Instructions

STATIC POWER CONVERSION EQUIPMENT for

AC ADJUSTABLE SPEED DRIVES

DC LINK COMMUTATED (DCLC) INDUCTION MOTOR DRIVE (IMD)

GEK-95647A

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WARNING	Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.
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NOTE	Commands attention to an essential operating or maintenance procedure, condition, or statement that must be highlighted.

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CHAPTER 1

SYSTEM DESCRIPTION

1-1. BASIC DRIVE CONTROL

The DC Link Commutated (DCLC) Induction Motor Drive (IMD) is a static, adjustable frequency drive system which controls an induction motor. Depending on size of the load capacitors, C_1 but typically from 50-70% to 100% speed, the load inverter is load commutated by leading KVAR load. From 0% to 50-70% speed the load inverter is forced commutated by shunting the DC link current into a shunt path in the DC link. For the current to transfer from the load inverter to the DC link shunt and thereby commutating the load inverter thyristors, the voltage that the DC shunt switches into must be less than the motor voltage reflected to the load side of the DC link at the instant of commutation. A simplified one-line diagram of a single channel IMD is given in Figure 1-1.

The transformer provides isolation from the ac system bus and provides the correct voltage at the terminals of the rectifier. Also, the internal impedance of the transformer limits the magnitude of any downstream bus faults.

The rectifier is a thyristor bridge whose gating is controlled to produce a variable dc voltage at its output. The output of the rectifier is fed through the dc link reactor, whose function is to smooth the current and keep it continuous over the operating range of the system.

The dc link reactor output is then fed into the DCLC and inverter bridge which produces variable frequency ac at the stator terminals of the motor. The inverter bridge , the DCLC, and the rectifier bridge use different power hardware and are controlled by microprocessor-based electronics. Although the bridges are labelled "rectifier" and "inverter" it is possible for their roles (when load has leading KVARS) and hence power flow to reverse. In this case the motor would be braked by pumping its energy back into the ac line. A more general nomenclature is to call the line side bridge the source converter, and to call the motor side bridge the load converter.

The filter provides the leading KVARS so that the drive may operate in the more efficient self-commutating mode (rather than forced comutated) when sufficient motor volts and speed are reached.

A more detailed diagram of a single channel IMD is shown in Figure 1-2. As shown in this figure, the electronic control receives the following signal inputs:

1. Attenuated line and load bus voltage signals.
2. Attenuated line and load current signals from current transformers.
3. Speed reference signal.
4. Process commands such as stop, start, etc.

The attenuated bus voltage signals are used by the control to synchronize thyristor firing with the line and load voltages, to provide voltage sensing across the thyristor cells, and to provide electronic over/under voltage detection. The attenuated current signals are

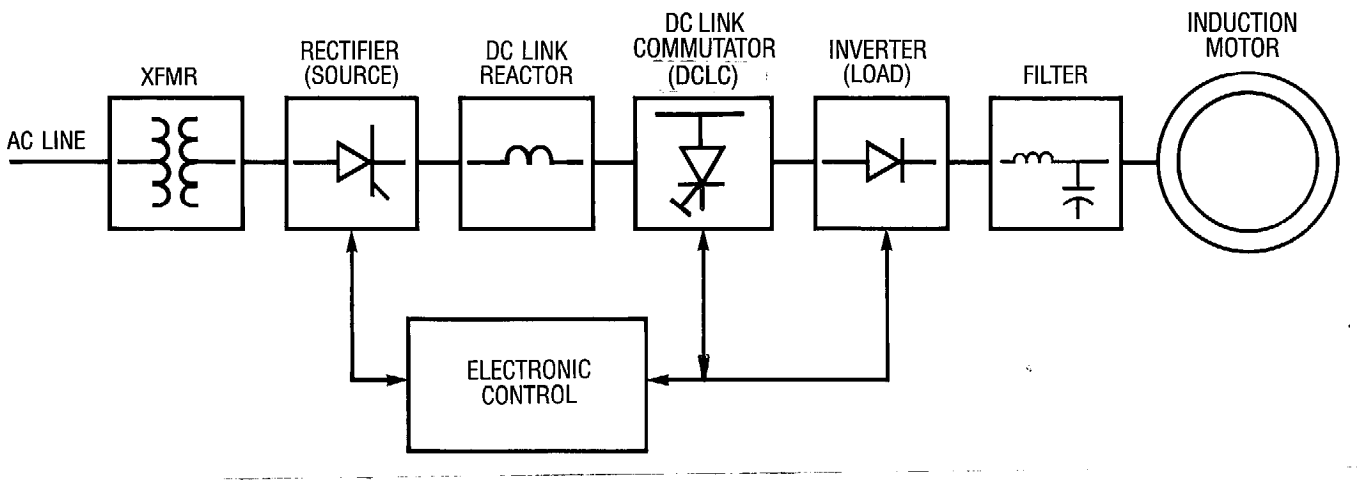


Figure 1-1. Simplified IMD One-Line Diagram

used for regulator current feedback, electronic overcurrent detection, and software-implemented fault detection.

The electronic control evaluates the process commands and internal status signals to determine whether the drive should be in a stopped, started, alarmed, or faulted condition. If in a started condition, the electronics provides gate signals to control the thyristor bridges. These signals are low level and are conditioned in the power bridge circuitry to provide the necessary isolation and power level.

The power bridges are of the six pulse, double way type and the thyristor bridge legs "fire" in the order that they are numbered in Figure 1-2 and as shown in Figure 1-3.

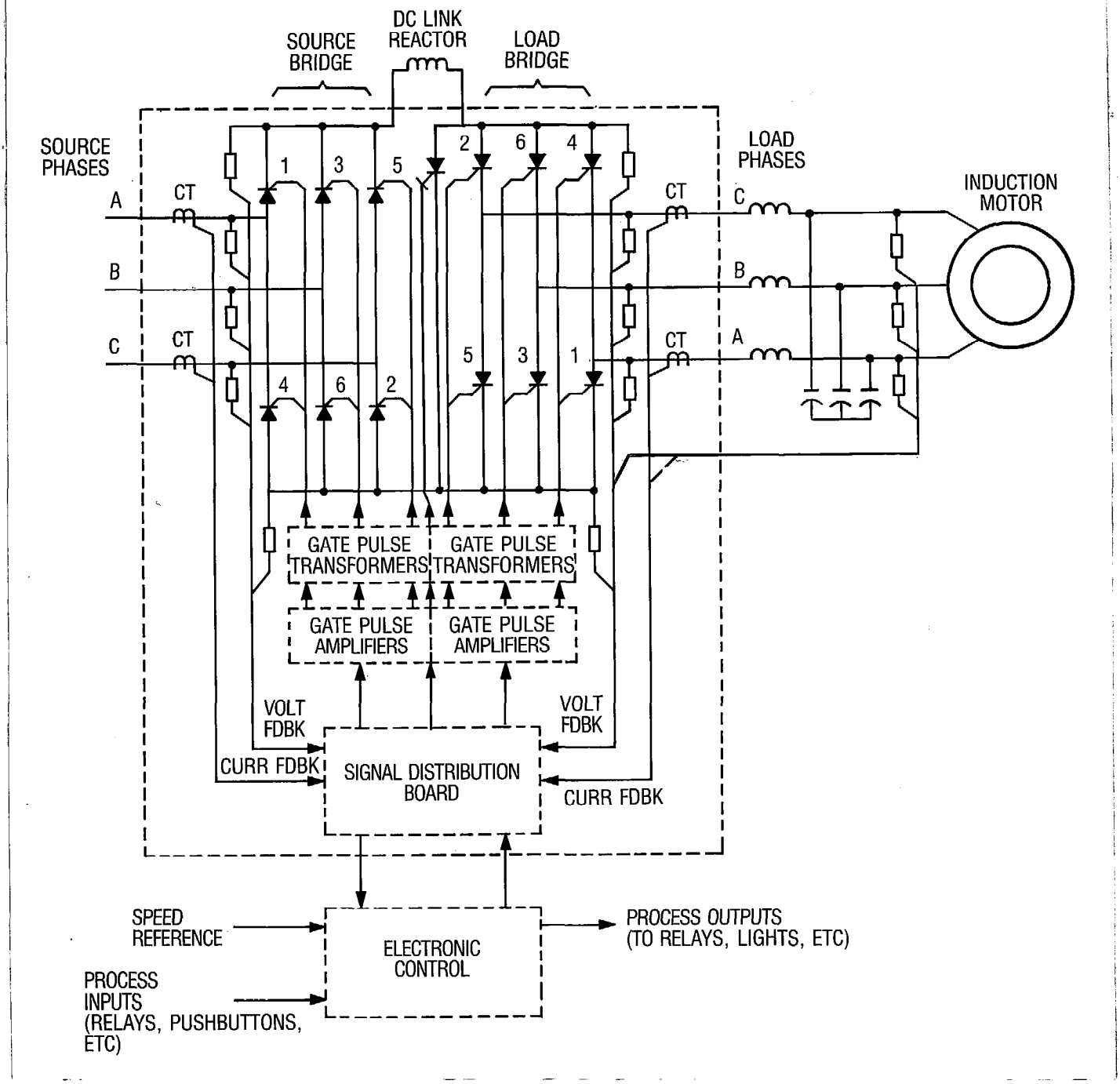


Figure 1-2. Single Channel IMD Drive

The source converter current is successfully transferred from one leg to the next by the AC line voltages. Similarly, the inverter current is successfully transferred by load's leading KVAR (75% SPD).

The process of switching bridge current from one leg to the next is illustrated in Figure 1-4. The principles illustrated here apply to a rectifier bridge as well as to an inverter bridge (when load has leading KVARS). This phase-controlled switching is accomplished by using the following two thyristor characteristics: (1) When the voltage across the thyristor is positive, it can be triggered into conduction, and (2) it will not permit current flow in the reverse direction - thus, in an alternating voltage circuit, thyristor conduction will cease and reverse voltage will begin to appear when the current becomes zero.

Current transfer must be completed before voltage cross-over, with a margin angle γ . This angle must be long enough to allow the previously conducting leg thyristors to recover to their blocking state before reverse voltage is applied. This is why the fundamental component of current must lead the voltage, from an inverter/load viewpoint. From the rectifier/source viewpoint, the fundamental component of current will always lag the voltage; but for successful commutation, the angle $\gamma = 180° - \beta = 180° - \mu - \gamma$ must always be less than 180° ; a practical limit is 155°. Correspondingly, a practical minimum value for β , for the inverter bridge, is 25°.

1-2. GENERAL OPERATIONAL THEORY

Startup modes and assumptions for GTO DC Link Commutation approach:

1. Commutation by phasing the source converter to inversion limit: (typically 0 to 15 Hz)..
2. Initially ther is no motor flux, so at .25 PU IDC the OPEN_LOOP frequency is ramped from 1 to 5 Hz.
3. At 5 Hz, motor flux has been established - thus the load PLL (phase-lock-loop), speed regulator, flux regulator, and load angle regulator are activated.
4. Now in CLOSED-LOOP mode, continue to ramp up motor speed to 15 Hz using source converter commutation. 15 Hz is about the limit of source converter commutation capability. At this speed there should be sufficient motor voltage to allow shifting to the GTO DC Link Commutation mode.
5. GTO DC Link Commutation mode: (typically 15 Hz to 50-70% speed)
6. From 15 hz to about 50-70% PU speed, the load thyristers are commutated by turning on the GTO shunt. GTO commutation is maintained as long as the commanded load thyrister firing angle (alpha) is greater than (180 degrees - margin) i.e. about 165 degrees. For succesful transfer of current from the load thyrister to the GTO, there must be sufficient commutating VOLT-SEC and this is accomplished by limiting the max firing angle (alpha) to about 225 deg. This prevents the vector control from being able to independently control flux and torque as could be done if alpha was unrestricted. The DC link voltage wa-

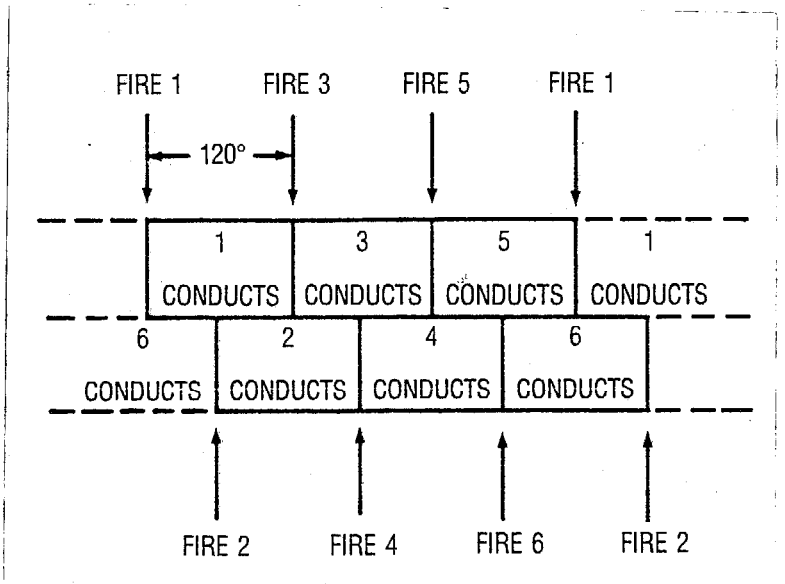


Figure 1-3. Thyristor Bridge Leg Conduction Sequence

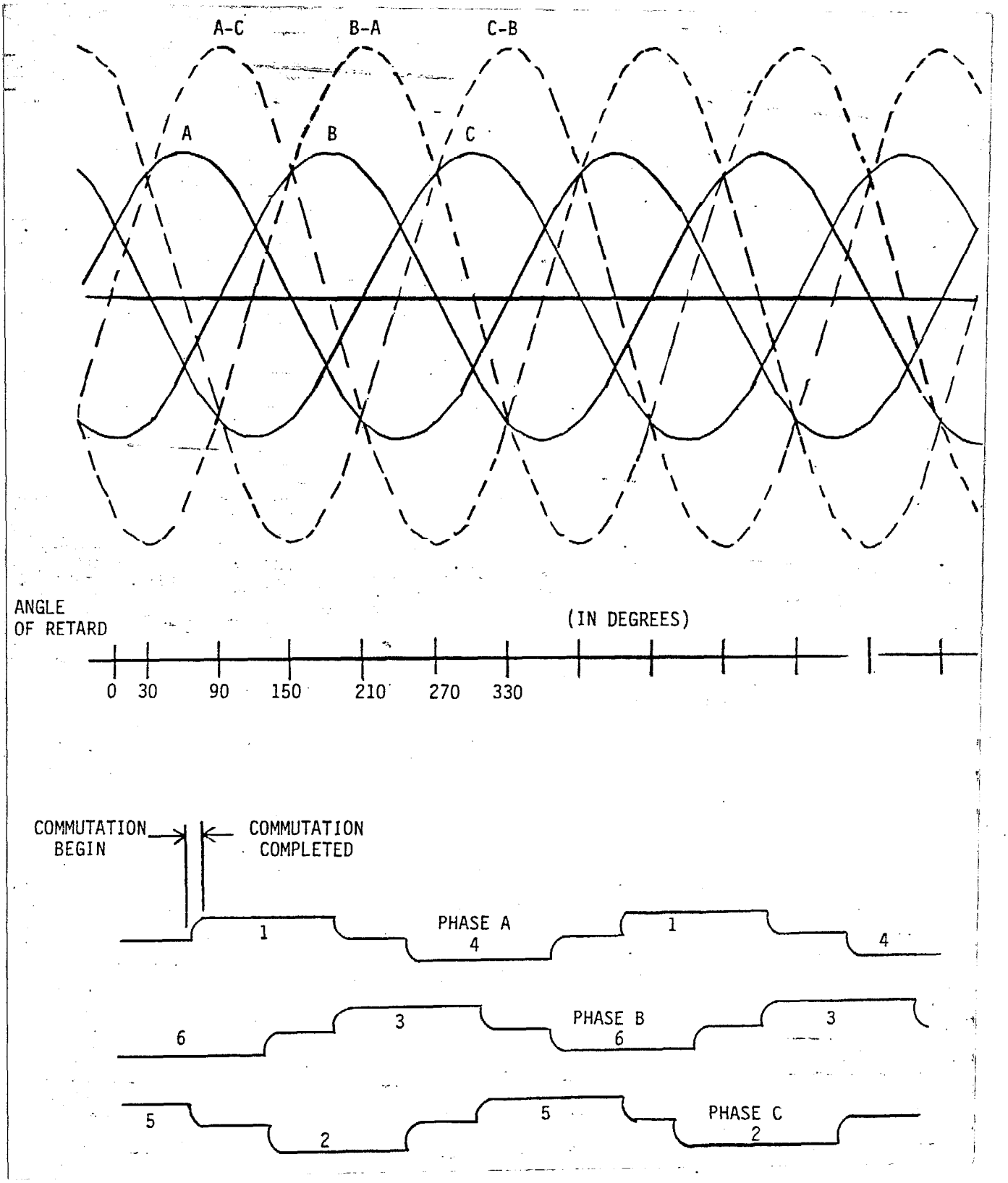


Figure 1-4A. Source Voltage and Current

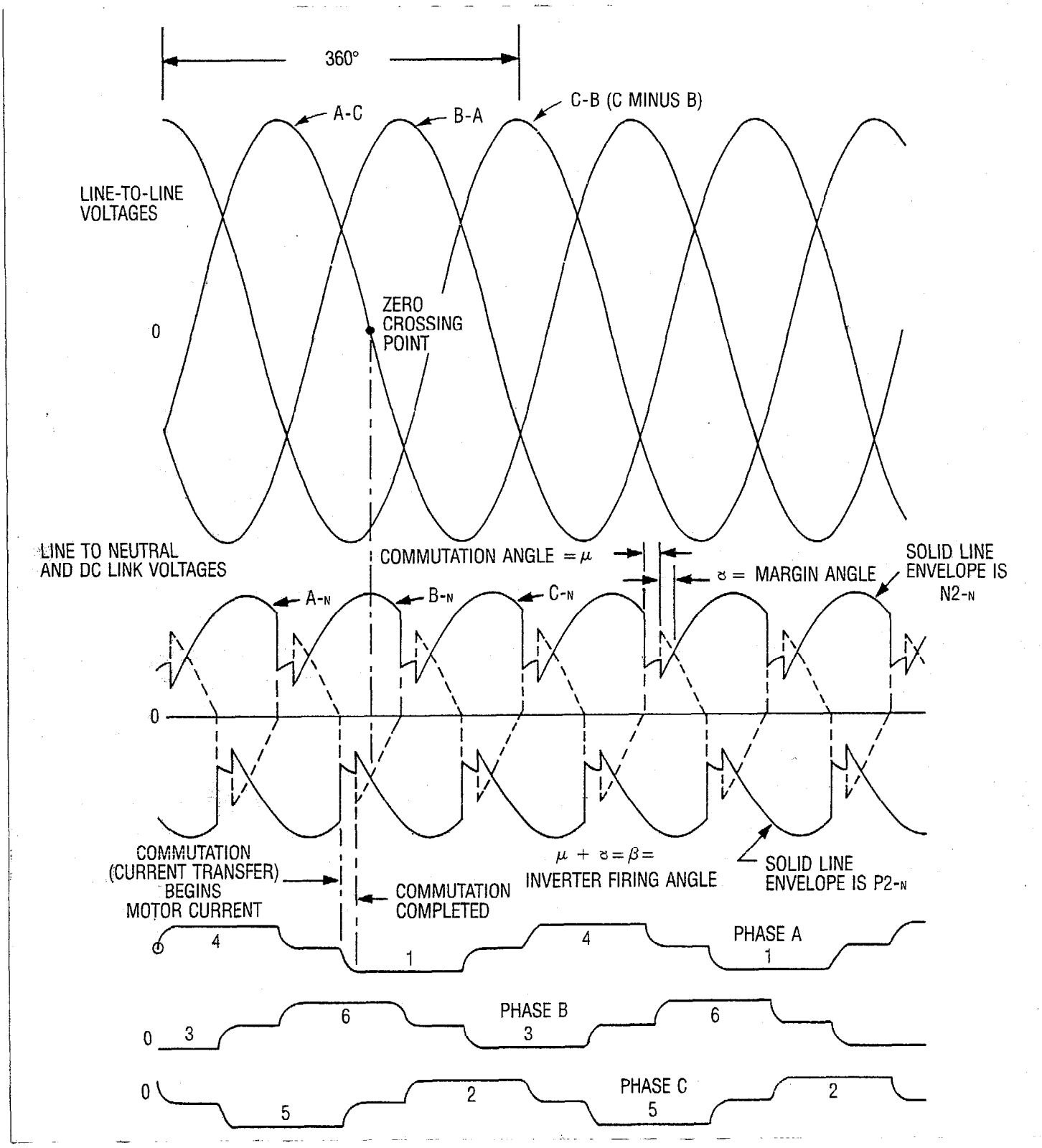


Figure 1-4B. Load Converter Voltage and Current (Self-Commutation Mode)

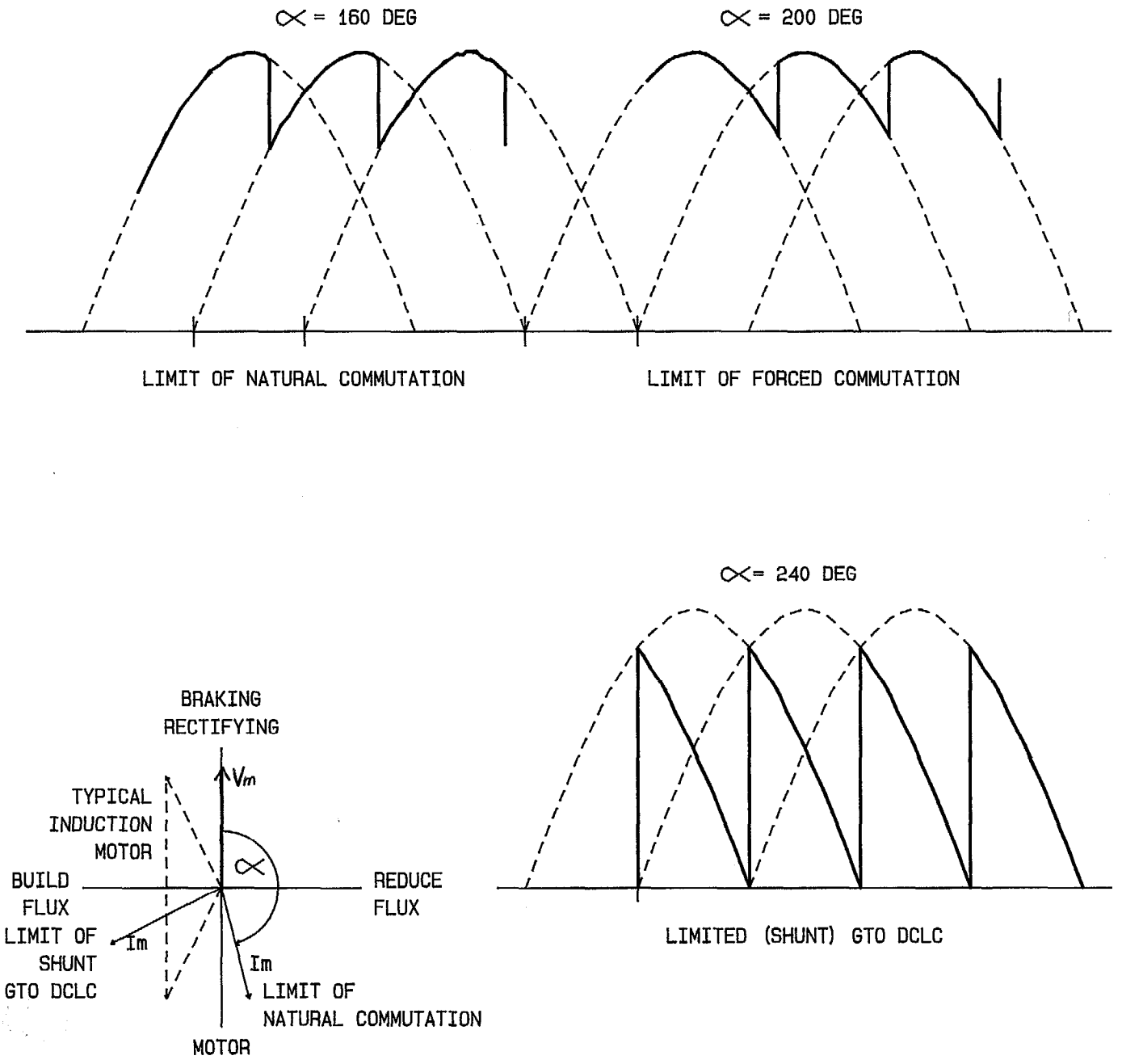

∝ = 160 DEG
∝ = 200 DEG
LIMIT OF NATURAL COMMUTATION
LIMIT OF FORCED COMMUTATION
∝= 240 DEG
BRAKING
RECTIFYING
Vm
TYPICAL
INDUCTION
MOTOR
∝
BUILD
FLUX
REDUCE
FLUX
LIMIT OF
SHUNT
GTO DCLC
Im
Im
LIMIT OF
NATURAL COMMUTATION
MOTOR
LIMITED (SHUNT) GTO DCLC

Figure 1-5.

veshape is illustrated in Figure 1-2 for various inverter firing angles. From this figure the firing angle alpha is limited to 240 degrees. For regenerative braking, the firing angle range is alpha between 270 to 360 degrees, thus thisinverter approach does not provide regenerative braking at low speed. As a means of maintaining speed regulator control down to light load-low speed conditions, when the commanded firing angle exceeds the max angle limit, link current is commanded to current-lower-limit and the angle spillover is used to stretch the GTO conduction interval - reducing torque and flux and increasing harmonic content in the inverter output. This all serves to make the control task of satisfying the speed regulator while ensuring successful load thyrister commutation in this low speed region very difficult - particularly because this is also the region of resonance of the load capacitor and motor leakage inductance with the 5th and 7th harmonics of the fundamental freguency. To reduce this "self-excitation", the current in the 8-18 Hz region is "chopped" to reduce the 5th and 7th harmonics and 'chopped' in the 18-28 Hz region to reduce the 5th harmonic. This essentially limits sustained drive operation to the 50-100% speed range although the drive will allow acceleration through the 0-50% speed range.

7. Load Commutation Mode: (typically 50-100% speed)
8. For (0 degrees < commanded firing angle (alpha) < 165 deg) the load thyristers are commutated by the load capacitor. In fact, at high speed the load inverter must 'eat' some of the excess leading KVAR to prevent motor overvoltage. In this region the GTO DC Link forced Commuation mode does not need to operate. The relationship between motor, inverter, and load capacitor currents as a fuction of inverter firing angle is illustrated in Figure 1-3. At high speed motoring the firing angle (alpha) range is 90 deg < alpha < 180 deg. For transient high speed regeneration the alpha range is 0 deg < alpha < 90 deg. For low speed motoring operation the alpha range is 180 deg < alpha < 270 deg. These three cases are illustrated in vector diagrams in Figures 3a, 3b, 3c respectively.
9. Assumptions in Power Circuit Design:
10. Fan load, IQ = K1*(SPD**2), IX = K2, IDC = K3*SQRT((IX**2)+(IQ**2))
11. Nominal motor volts = rated voltage * actual speed / rated speed
12. Diverter will not operate for alpha < 165 deg, which implies that the diverter will not operate for speed > .5-.7 PU. Nonlinear reactors (Lf) with 20% unsaturated impedance, 1.5% saturated impedance, and saturation at 10% rated current will buffer the inverter from the load capacitor. A voltage spike is experinced on turning off the GTO and picking up current in the load inverter. This voltage is proportional to Vp = IDC * SQRT(2*Lf/Csnub_eqv) . This voltage spike does not occur when the GTO is not operating but the equipment will be sized such that semiconductors will not be overvoltaged if a control failure occurs and the GTO inadvertantly fires at high speed, high IDC provided that all series cells are intact. The forced commutation circuit will allow nominal operation with one shorted GTO but will not be rated for two concurrent faults such as one cell shorted plus the control failure discussed above.
13. The drive will not provide regeneration in the GTO DC Link Commutation mode of operation.
14. The drive will be able to accelerate through the resonance region of the load capacitor with the motor leakage inductance excited by harmonics of the fundamental frequency, but need not operate continously in this region.
15. No Synchronization-and-Transfer-to-Bus feature.
16. Does have Start-on-the-Fly feature.
17. The GTO DC Link forced Commutation design uses either three or four series 4500 volt GTOs - depending on the voltage and horsepower requirements.
18. Nominal DI/DT in the GTO string is controlled by the load inductors, Lf. However, there could be parasitic capacitor discharge associated with the DC link inductor and its cabling The GTO leg is designed for DI/DT protection for a max of .01 microfarad parasitic capacitance.
19. Shunt-Series GTO DCLC:
20. For drives needing to use the 10-50% speed range during normal operation, it is possible to get successful commutation with less Volt-Secs by using a GTO string in the DC link between the shunt GTO string and the inverter bridge (hereby called Series GTO). The series GTO firing pattern is essentially the opposite of the Shunt GTO i.e. when turn-on the Shunt GTO then turn-off the Series GTO and vis-versa.

This allows the drive to essentailly operate over the entire speed range (10-100%) although the DCLC and harmonic chopping are still used in the 10-70% range.

1-3. AC INDUCTION MOTOR

AC motor torque is produced by a force or pull between two magnetic fields in the air gap. One being produced by current flowing in the conductors of the stator or stationary member and the other by current flowing in the conductors of the rotor or rotating member.

In the controlled inverter system, only the stator circuit is accessible therefore both magnetic fields in the air gap must be controlled by controlling the stator parameters (or stator current, volts, and frequency).

Torque is proportional to the "torque producing component" of stator current, $I_S \sin \Theta$ and motor flux,ψ

$$T_{AC} = K_4 I_S \sin \Theta (\psi) = K_5 I_S \sin \Theta (I_S \cos \Theta) = K_6 I_S^2 \sin \Theta \cos \Theta$$

AC motor flux is produced by the flux producing component of stator current $I_S \cos \Theta$. The motor angle Θ is described as the angle between the stator current and the motor flux produced in the air gap by the stator current. See Figure 1-8.

In the induction motor, constant torque can be maintained by controlling the torque producing component and the flux producing component of I_S which is done by regulating constant stator current and motor angle Θ.

When the motor takes on additional load, or torque is required to change, the speed drops back and both Θ and motor slip increase which, in turn increases the rotor current and draws corresponding torque producing stator current to maintain load.

Figure 1-9 shows the speed-torque characteristic of a typical induction motor. With a constant voltage applied and a frequency equal to B applied, the motor would accelerate to speed A where the torque curve equals the load torque at that speed. As the motor accelerates, the developed torque equals the characteristic shown. As long as the applied frequency, voltage and load remain constant, the drive will run at speed A and a slip equal to B-A. Under these conditions of constant volts/cycle and constant slip-stator current, motor angle Θ and flux are constant. To regenerate power back into the line, the motor speed exceeds the applied frequency B, the torque becomes negative, and the motor acts as a generator.

Constant torque is controlled by holding constant stator current and motor angle. Motor angle is increased (Figure 1-8) to reduce flux (and torque) for constant HP operation. Stator current and slip also change during this region.

In addition to torque control, motor speed can be controlled by being able to go from motoring to regenerating as required to hold the set speed. This can be done without changing speed, by changing the input frequency. See Figure 1-9. Assume the motor is running in the set speed mode of operation in the motoring quadrant at speed A. The load changes to require a regenerating torque equal to E to maintain speed A. The control would change the input frequency to D (motor angle Θ and slip change polarities).

To control motor torque and speed, the inverter current and stator voltage, signals are fed back to the control. These components are used to calculate flux, torque, capacitor current and motor current. The converter circuit is controlled to regulate current. The inverter circuit is controlled to regulate motor speed and volts.

At high load on the motor, the apparent power factor "seen" by the power source increases. This is because the source converter firing advances (is reduced) to obtain more current. The harmonics in the current and the resultant harmonics in the voltage, due to commutation "notching", will decrease.

The fundamental control strategy is: To increase motor current in response to a load torque increase, the D-C link voltage, on the source side, is increased, thereby raising motor current so as to keep motor speed constant.

1-4. SERIES TWELVE-PULSE OPERATION (OPTION)

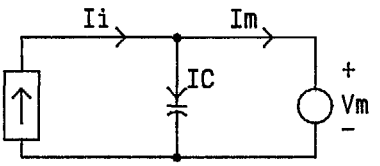
Series Twelve-pulse operation consists of two identical, separate, six-pulse bridges operating from a common source at the same current with the voltage and firing reference angles shifted 30 degrees by an incoming isolation transformer. But the two drives are combined into one by connecting the "+" bus of one to the "N" bus of the other .

The two bridges deliver equal power and the transformer design makes the source-side converter voltage equal, but also 30 degrees apart between the two channels. It is necessary that source-side and load-side voltages be phase-shifted. The transformer windings are shifted to reduce the harmonic distortion on the power system and to raise the harmonic frequencies, and to increase the power flow.

Normally the source master voltages (WYE) LEAD the source slave voltages (DELTA) .

1-5. BYPASS (ACROSS THE LINE) OPERATION - OPTION

On drives with a bypass mode, the motor can be connected directly to the line - "bypassing" the IMD



PER PHASE EQUIV. CIRCUIT

Ii = Ic + Im

Im = MOTOR CURRENT
Ic = LOAD CAPACITOR CURRENT
Ii = INVERTER LOAD CURRENT
Vm = MOTOR = CAPACITOR VOLTAGE

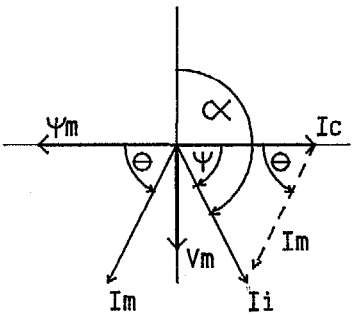


FIGURE 1-3A

90° < ∝ ≤ 180°
HIGH SPEED MOTORING
LOAD CAPACITOR COMMUTATION

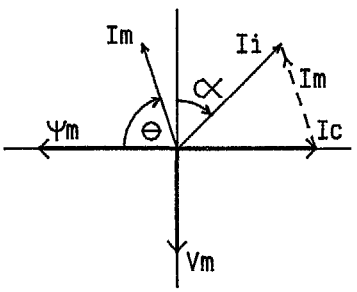


FIGURE 1-3B

0° < ∝ ≤ 90°
TRANSIENT HIGH SPEED REGENERATION
LOAD CAPCAITOR COMMUTATED

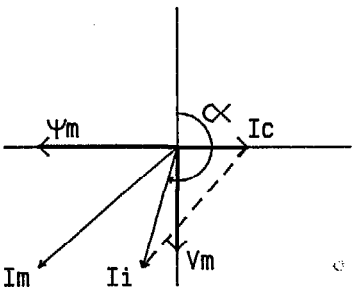


FIGURE 1-3C

180° < ∝ ≤ 240°
LOW SPEED MOTORING
DC LINK COMMUTATION

Figure 1-6.

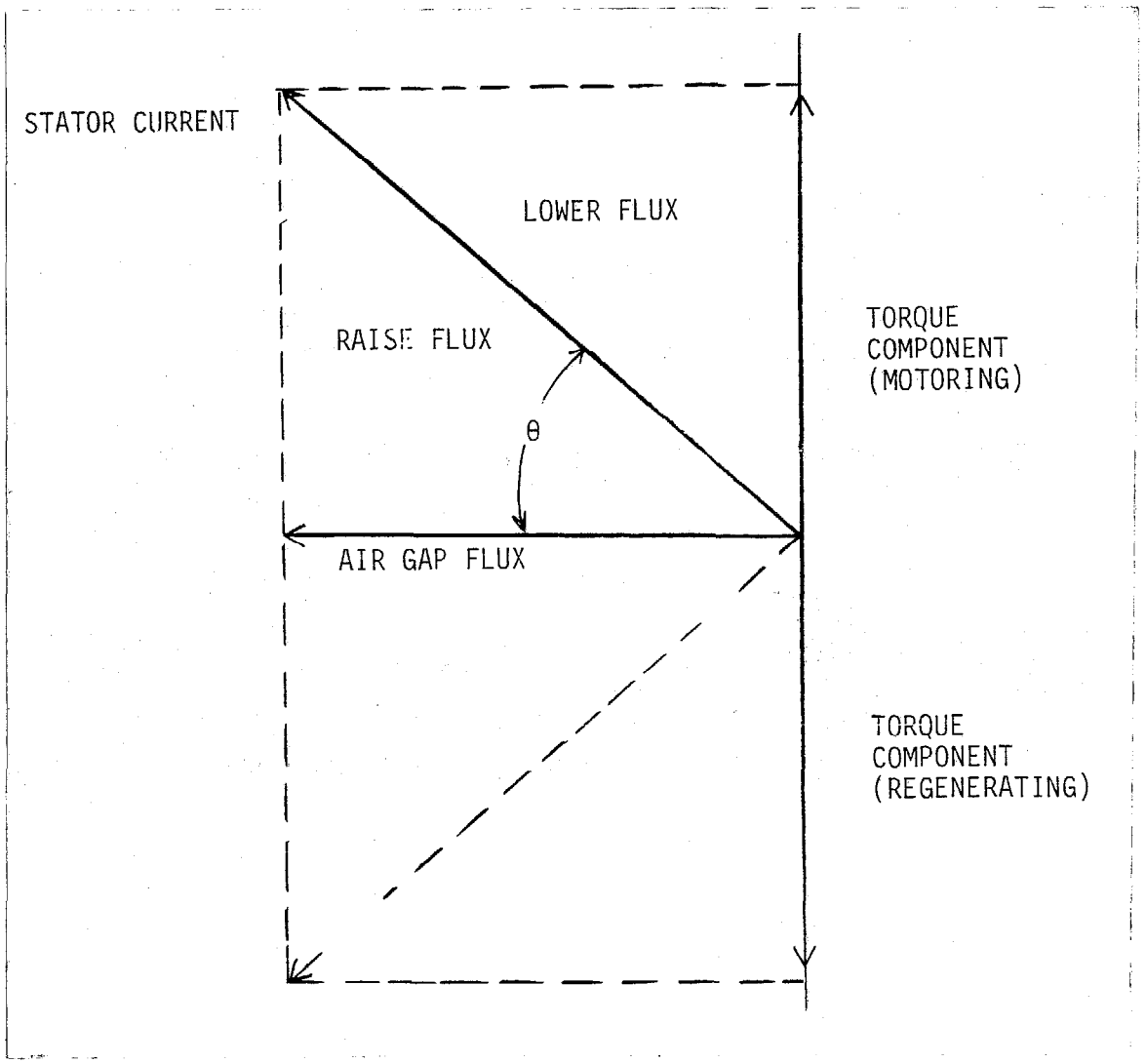


Figure 1-8. Motor Angle

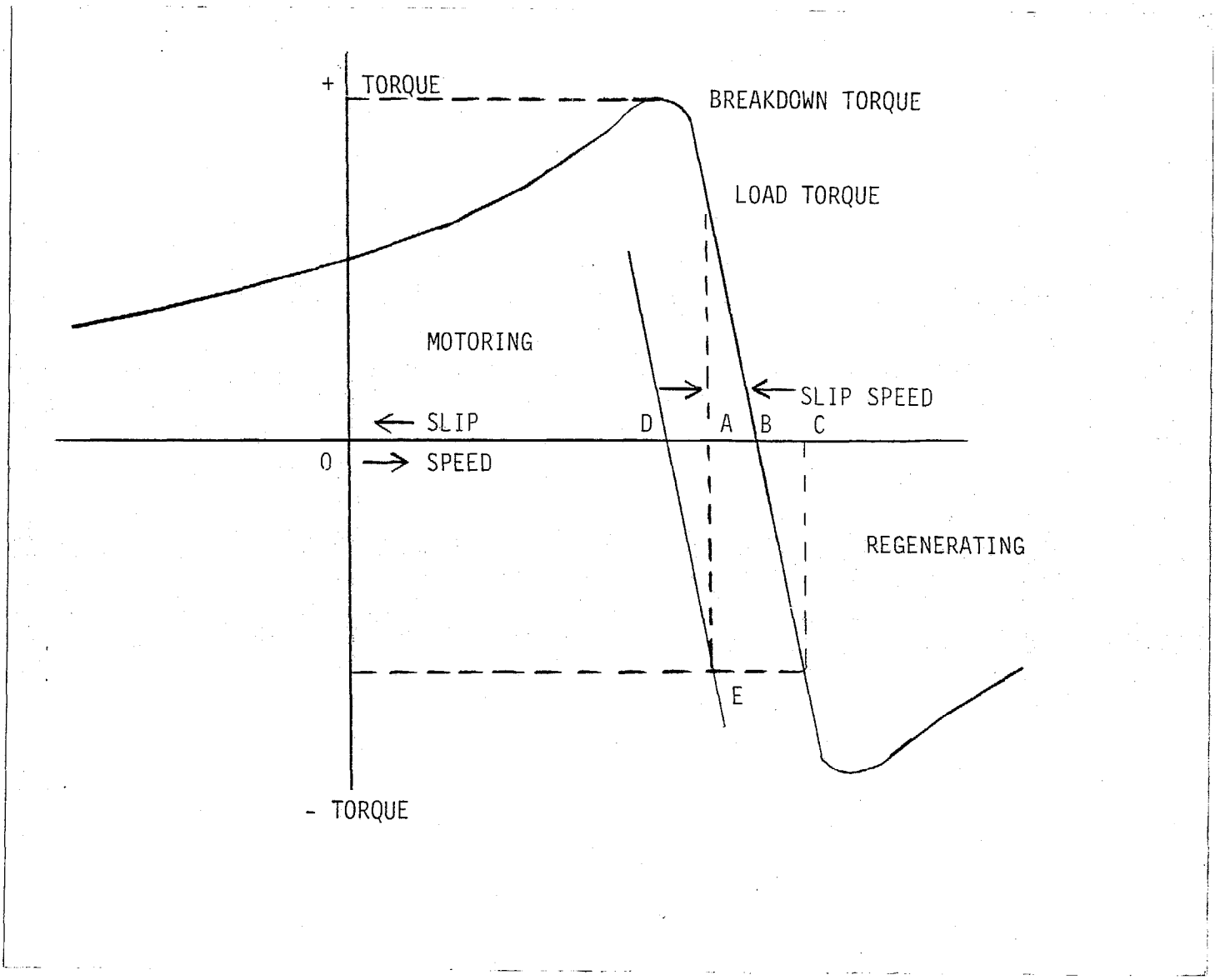


Figure 1-9. Speed Torque Characteristics of AC Induction Motor

drive. This feature gives "back-up" capability to the system, i.e. the motor can still be run if the drive is faulted.

CHAPTER 2

2-1. EQUIPMENT OPTIONS

Three types of bridges may be utilized for the power conversion equipment. One type uses three thyristors in series and will be designated the medium voltage bridge (MVB). Another type utilizes one or two thyristors in series and will be designated the SILCOMATIC® Plus power conversion bridge. The third uses three or four gate-turnoff thyristors (GTO) in series and is designated as a GTO leg.

2-2. 2KV MEDIUM VOLTAGE BRIDGE

The arrangement of the power converter SCR stack assemblies is shown in Figure M2-1. This figure and the elementary diagram show the major power components.

The SCRs are stacked horizontally; three in series, between four heat sinks. The 18 SCRs between the P-bus (P1, P2) and the AC lines, source and load, are on the left side in the bridge compartment; also CT1, CT3, FU1-4 and FU-6.

The N-bus SCRs are on the right side of the bridge compartment; also CT4, CT6, FU-5, and FU7-10.

The blower compartment contains the blowers, source and load AC and DC filters, MOVs, CT1A, CT3A, CT4A, and CT6A. The hinged base in this compartment contains the blower starters, the 70V gating power supply, the thyristor fan-out PWB XTFH, and bottom blower selection relay.

2-3. THYRISTOR MODULES, MEDIUM VOLTAGE BRIDGE (MVB)

Each power converter leg module consists of three thyristors arranged in series in a heatsink assembly.

The master leg module is mounted in an insulated box with associated printed wiring boards in front as shown in Figure M-2.

An R-C snubber network is also contained in the leg module box, above the cells.

Each cell stack assembly consists of three cells sandwiched between heatsinks as shown in Figure M-3. The stack assemblies are oriented horizontally in the module as shown. Air flow over the heatsinks is from front to back of the bridge enclosure and the cell modules.

The cells are selected, matched, and assembled at the factory under the correct amount of pressure. For these reasons, it is recommended that the entire stack be replaced if a cell failure occurs. The defective stack may then be sent to the factory for repair.

Each cell in the module is paralleled by a voltage-equalizing resistor and by a neon lamp and dropping resistor network which indicates when voltage is present across the cell. These lights are visible through the front doors of the equipment and can be used to detect a shorted cell. The cell light network is mounted on a small card (NTDA) located above each cell on the front of the thyristor modules. The location of these cards is as shown in Figure M-2. The NTDA card also contains thyristor gate interface circuitry which is illustrated in Figure M-4. Thyristor gating signals are routed from the gating transformers through the NTDA cards to the thyristor gates.

2-4. GATE DRIVER TRANSFORMERS

The cell gates are driven by current transformers mounted on the front of the module as shown in Figure M-2. The current transformers for each module are wired so that all their primary windings are in series. The primary winding is simply a conductor through the center hole of the transformer. Current pulses are applied in this conductor by the gate driver circuitry. This circuit insures that all thyristors in the string are gated simultaneously and it further provides voltage isolation between the thyristor gates and the electronic gate pulse generation circuitry.

2-5. GATE DRIVER CARD (HPTK)

Gate pulses of proper duration and timing are developed in an electronics module located in a control enclosure at the left end of the MVB. These signals do not possess the necessary drive and isolation to interface directly with the gate driver current transformers. This interface is provided by the HPTK gate driver card located in the thyristor module in each leg. The HPTK receives the low level signals and, after conditioning, passes them through a pulse transformer to provide isolation. The pulse transformer output drives a pulse amplifier circuit which is powered by a 70 VDC gating power supply located in the fan and filter compartment of the MVB. The pulse amplifier output drives all gate pulse transformers in the leg as described above. A red LED on the board glows to indicate when gate pulses are present.

A second function of the HPTK card is to detect an undervoltage condition of the 70 VDC supply. In the event that an undervoltage is detected, the HPTK outputs a logic signal to the electronics module indicating the fault. The electronic control then takes appropriate action to shut down the drive. Since there is an undervoltage detector on each gating card, wiring defects in the gating power will also be detected.

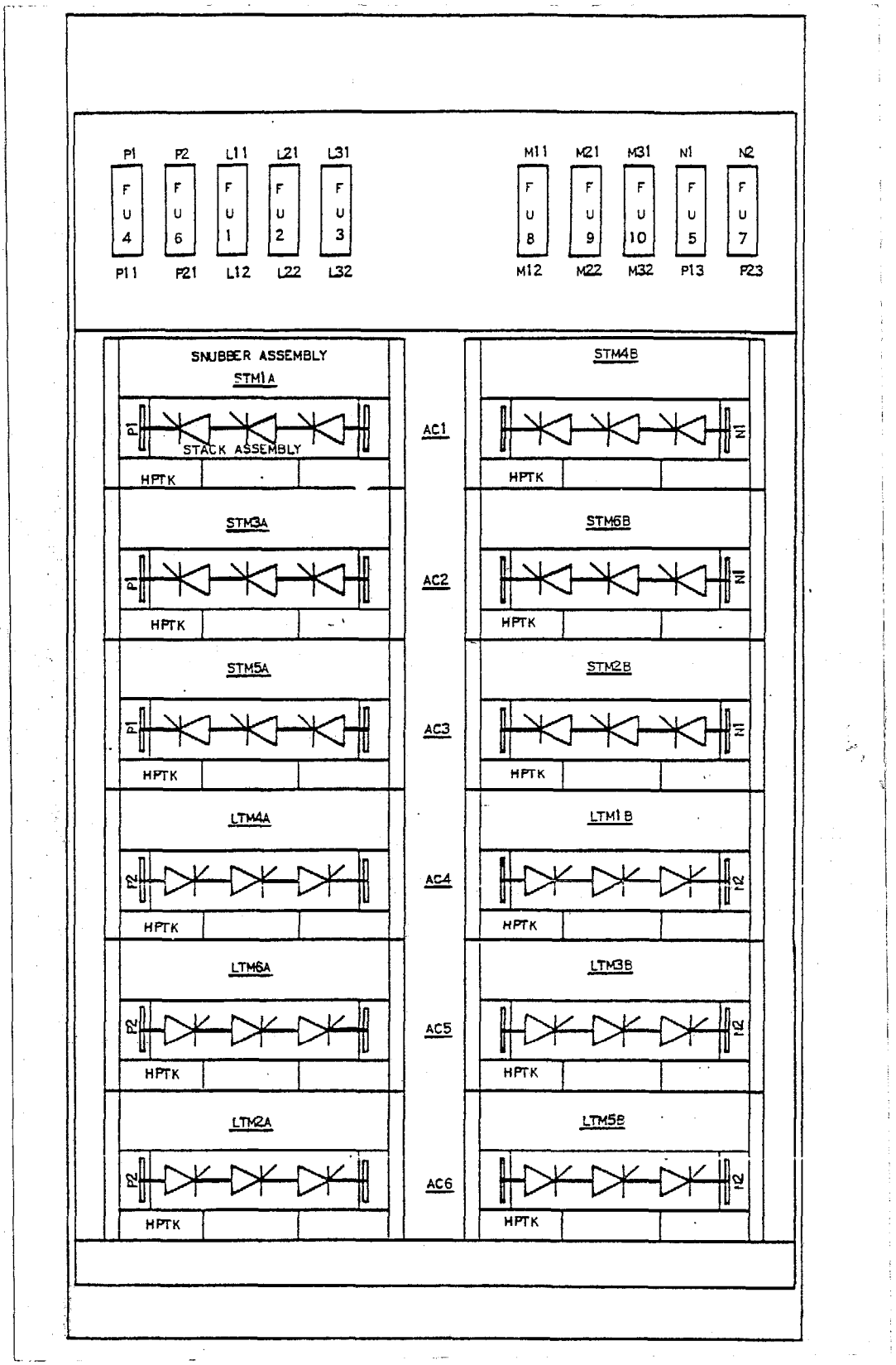


Figure M2-1. Medium Voltage Bridge Assembly

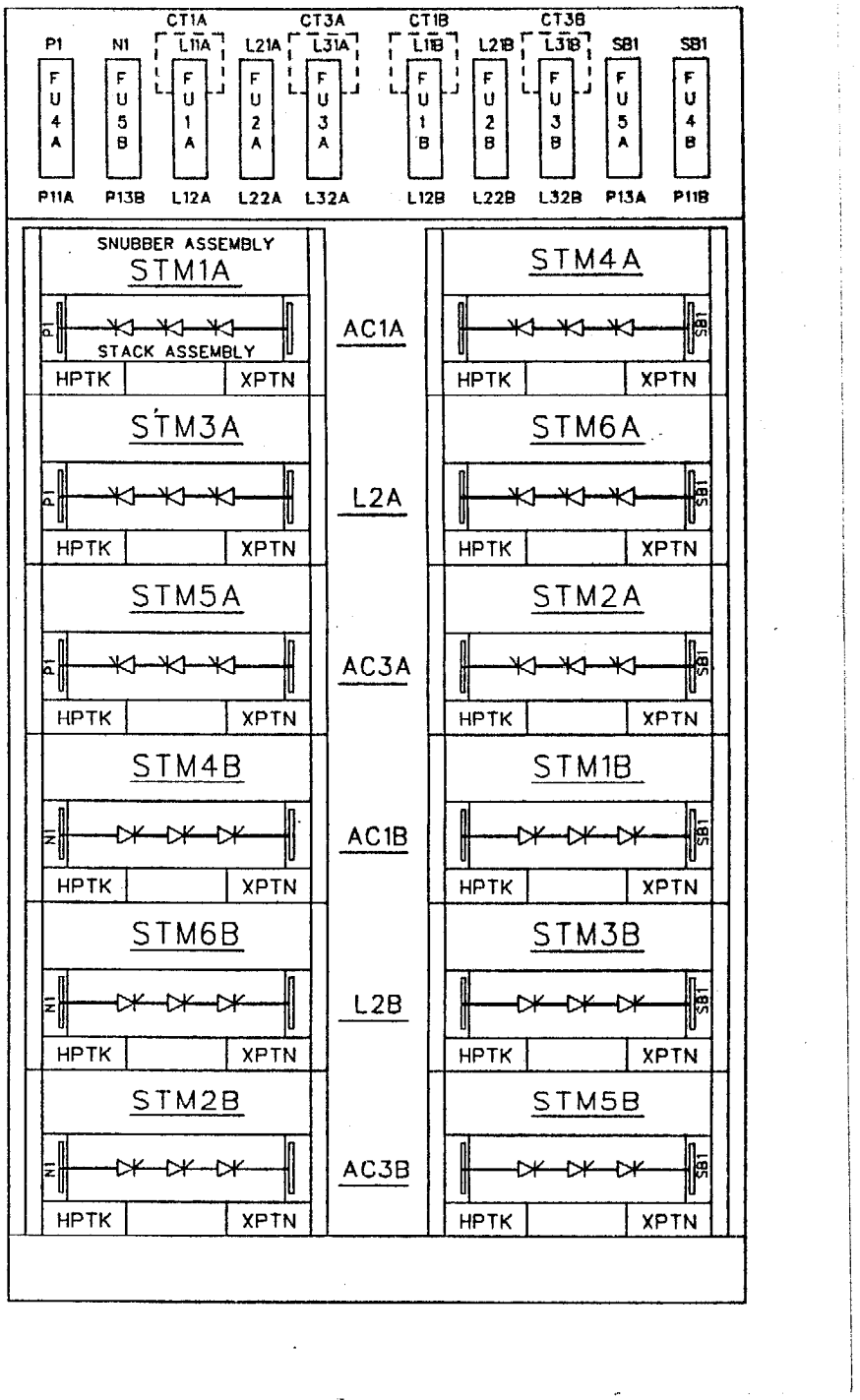


Figure M2-2. Medium Voltage Bridge Assembly

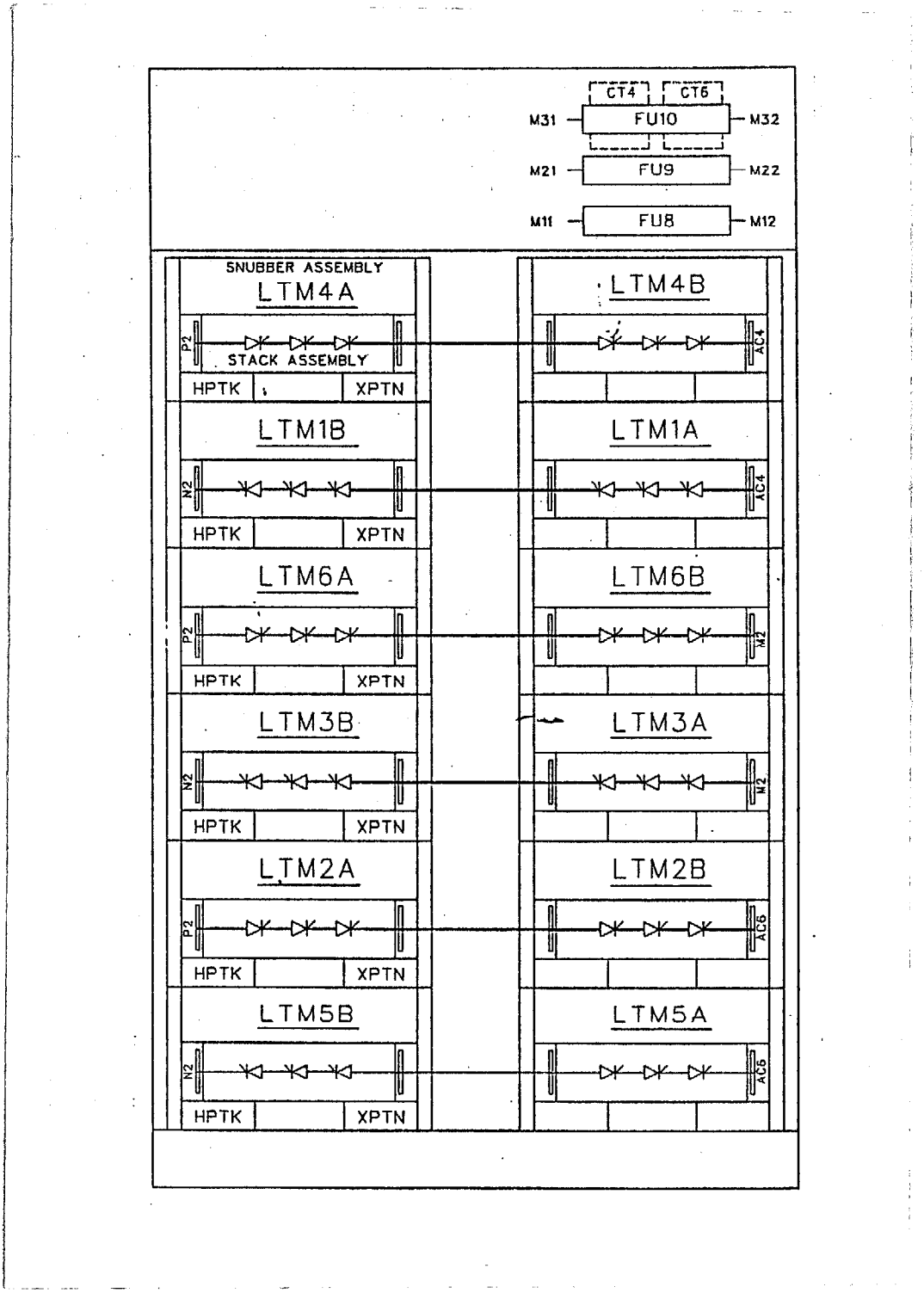


Figure M4-2. Medium Voltage Bridge Assembly, 4KV

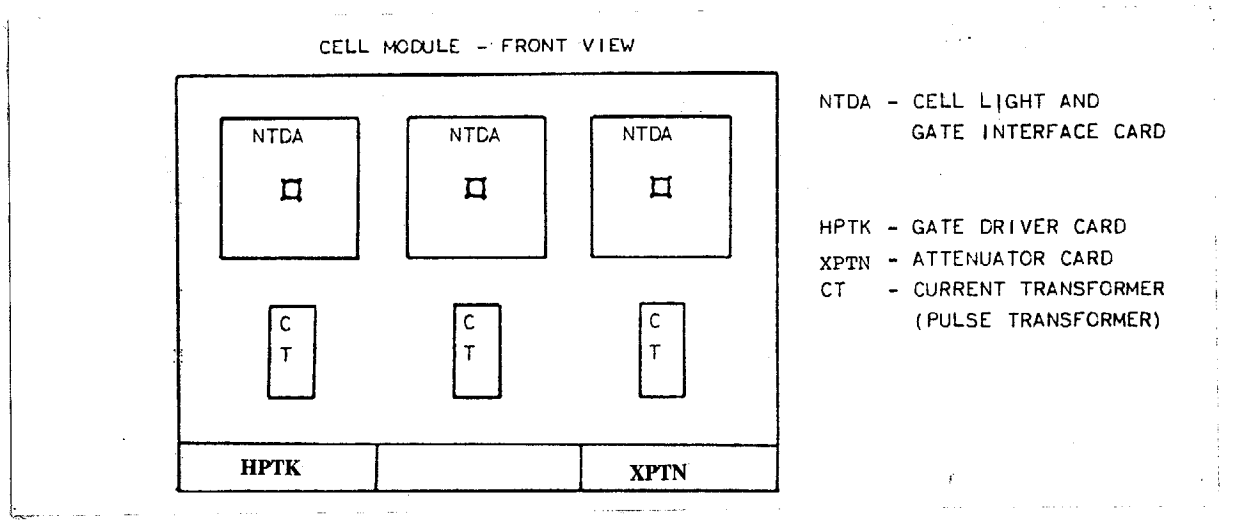


Figure M-2. MVB Thyristor Module

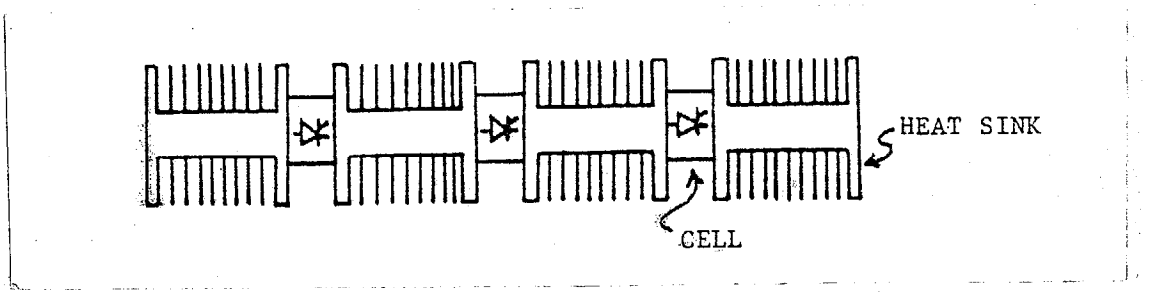


Figure M-3. MVB Cell Stack Assembly

A block diagram of the HPTK card is given in Figure M-5. The pulse driver delivers a current pulse simultaneously to all thyristors in the leg module.

2-6. VOLTAGE FEEDBACK ATTENUATOR CARD (XPTN)

In order to properly control the gating signals to the MVB, the electronics in the control enclosure must have available analogs of all AC and DC bus voltages. The XPTN attenuator cards in the thyristor modules provide bus-to-ground voltage signals which are combined in the electronics module to produce bus-to-bus signals for each converter bridge.

Figure M-6 illustrates the overall attenuator connections for developing bus voltage feedback signals for an entire bridge. Note that only five bus-to-ground signals are required. Consult MVB elementary DS5201 for complete details of attenuator connections.

Five signals are derived from the attenuator units as illustrated in Figrue M-6:

In Figure M-6, a series string of thyristors is represented by a single SCR symbol.

Note that the line-to-ground attenuator in leg module 4 is not used.

As explained in the paragraph discussing power converters, the voltage signals are used to control firing of the SCRs and to provide various protective functions.

The attenuation is variable via a Berg Jumper (BJ) for various system voltages: 1.75KV-STO; 2KV-LO; 4KV-HI.

2-7. SIGNAL DISTRIBUTION BOARD (XTFH)

All signal interchange between the drive electronic control and the medium voltage bridge is made through the XTFH signal distribution board located in the fan/filter compartment. A small ribbon cable from each HPTK gating card connects to the XTFH board. These ribbons each contain gating control signals, attenuated bus voltage signals, and 70 volt power

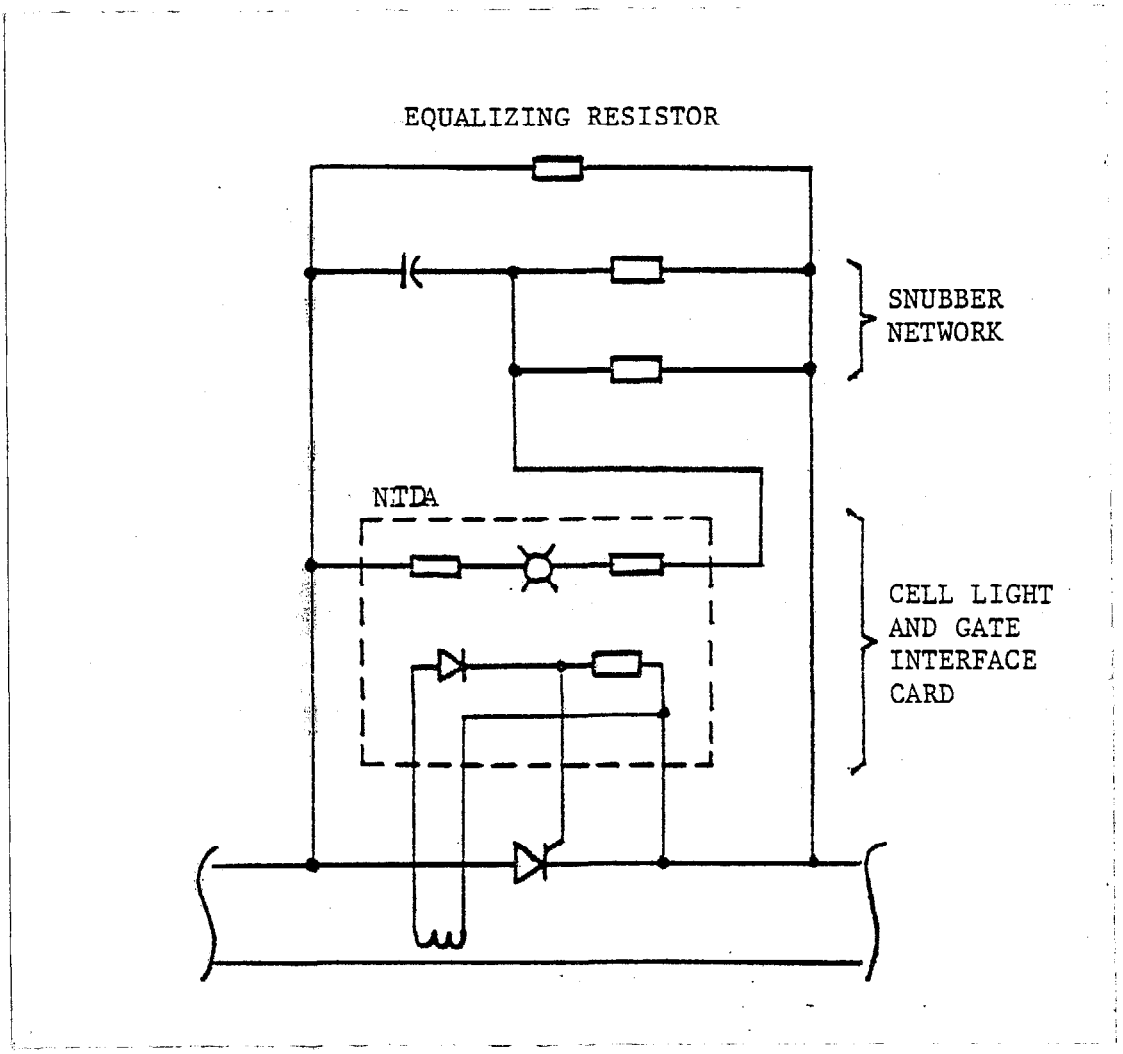


Figure M-4. Cell with Gating, Snubber, and Voltage Equalizing circuits

supply undervoltage signals. The undervoltage signals from each bridge are combined in the XTFH card to produce a composite undervoltage signal for each bridge. The gating signals originate in the electronics module in the control compartment, while the remaining signals originate in the medium voltage bridge and must terminate in the electronics module. Signal interchange to the electronics module is made via two large ribbons which connect the XTFH board and cards in the source and load electronics.

The XTFH contains the burden resistor circuits for the source and motor line current transformers. The auxiliary ct leads plug into the XTFH board and two small ribbons route these signals to the source and load electronics in the control cabinet.

2-8. PWB ADJUSTMENTS, TEST RINGS, LIGHTS, AND SWITCHES (MVB)

1. HPTK

TP1 - Gate pulse signals from SEM. Leg 1 same as SIG1, leg 2 same as SIG2, etc. 0 to 28VDC.

TP2 - 70V Gating power. Less than 10 volts DC indicates power undervoltage. Measure from PC1, PC2, or PC3 (Stabs on card).

TP3 - Gate Pulse signal common (reference for TP1)

CR -Light is on while gate pulses present.

2. NTDA

NE - Neon light. Normally illuminated. Light goes out if cell shorts.

TP1 - Cell cathode. Voltage across cell can be measured between adjacent test points.

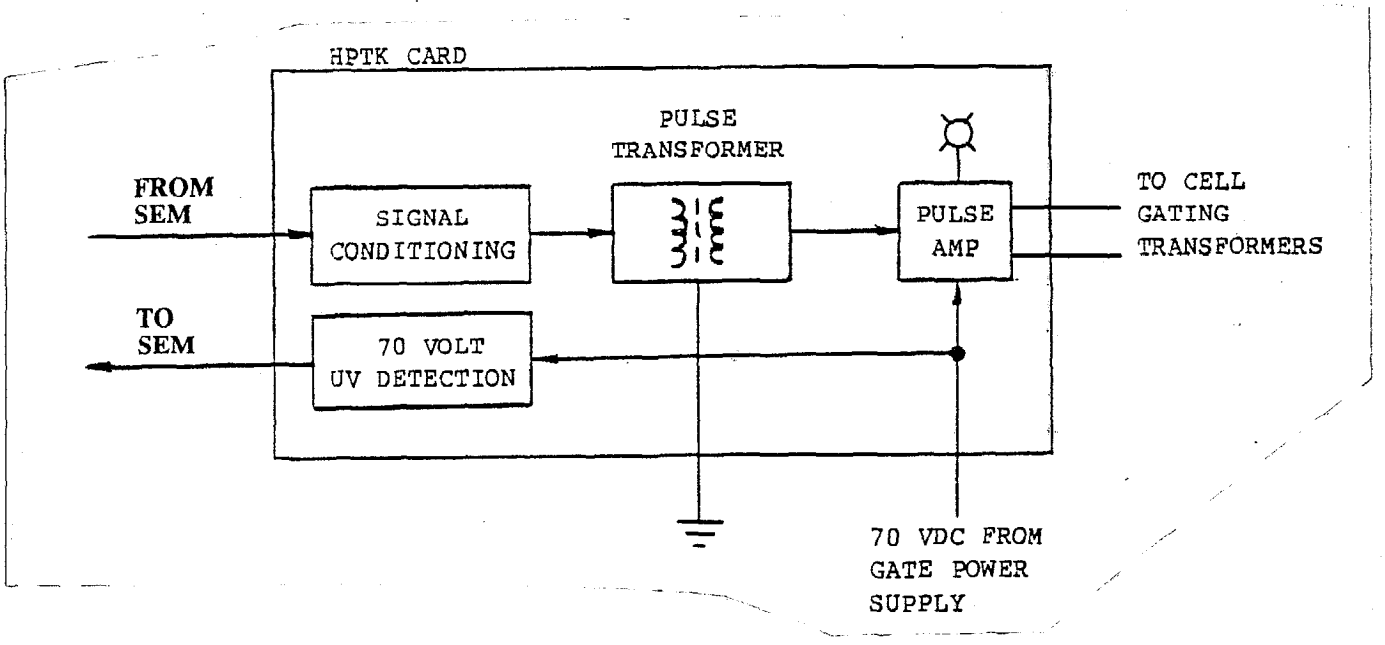


Figure M-5. HPTK Block Diagram

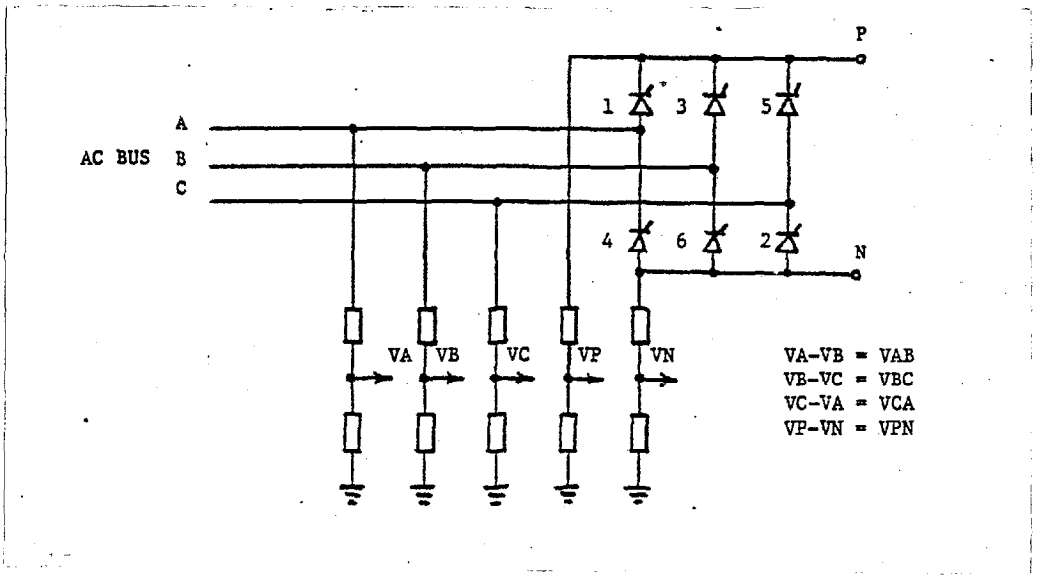


Figure M-6. Bridge Attenuator Circuits

CHAPTER 3

CONTROL ELECTRONICS PANEL

3-1. INTRODUCTION

The control electronics panel is made-up mainly by various modules interconnected thru ribbon cables and harnessed wiring. The main modules are:

DS3800IME	Main Control electronics
DS3820IOM	Inputs/Outputs (I/O)
DS3820SIM	Auxiliary interface electronics (When more than the normal signal are needed)
DS3820LDM	Display Module

The modules are typically configured as in Figure 3-1. Following are brief descriptions of the functions of the module. For more specific information the corresponding modules elementary and documentation should be consulted.

3-2. ELECTRONICS MODULE (IME)

The control electronics module consists of cards in 2-rows x 6-slots. Its primary purpose is to control the firing of the power converters and field exciter module SCRS for driving a motor. It also provides various protective functions for the motor and bridges. Interface between the IME to the customer is thru the IOM and SIM.

See Figure 3-2A for the physical layout of the module. Connection is made to the electronics module (SEM) through the plugs located at the top, and also through card front connectors.

From a functional standpoint, the electronics module processes signals as represented in Figure 3-2B.

The signal interfaces for the SEM are as follows: Attenuator signals - Attenuated bus to ground signals from each converter.

SIM module - See SIM section

Line/load current feedback - Signals from line/load bus current transformer burden resistors.

Power supply - The module requires +5, ±15, and +28 volts which are provided by a base-mounted power supply.

Thyristor gate signals - These are the electronic signals which drive the gate pulse circuits in the medium voltage bridge.

The hardware of the electronics module is represented from a block diagram standpoint in Figure 3-2C.

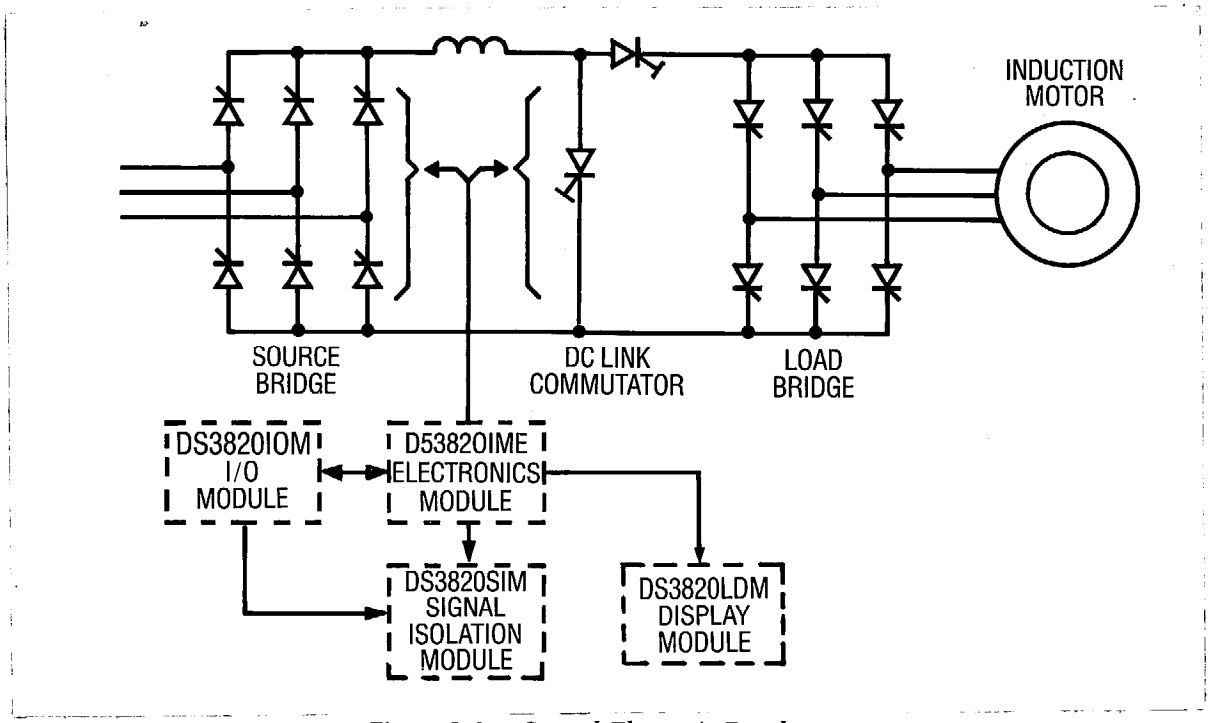


Figure 3-1. Control Electronic Panel

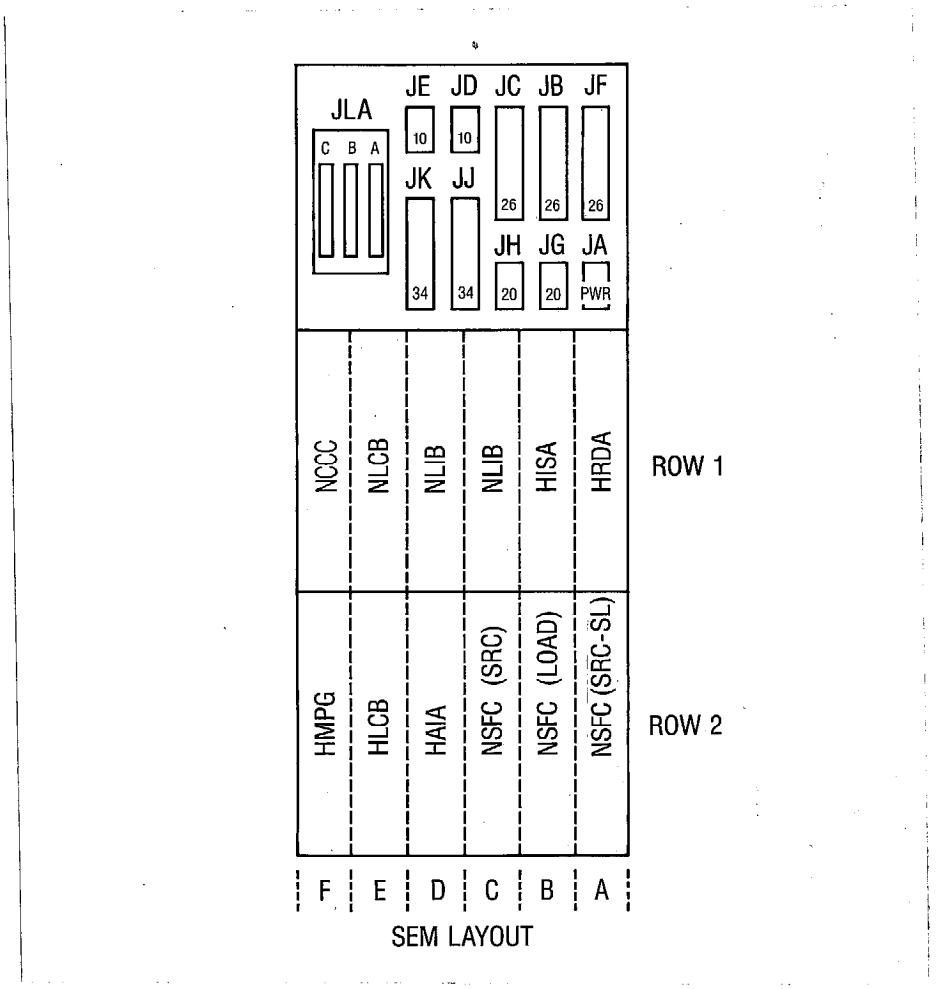


Figure 3-2A. Small Electronics Module (SEM) Card and Connector Layout

There is one microcomputer controlling the firing of both of the power converters and the field exciter.p

The microcomputer establishes a local bus (L-bus) structure on which the input/output for it is interfaced. In addition, it communicates to an 8k byte read/write (RAM) memory and a 2k byte electrically alterable memory (EAROM). This memory combines features of read/write memory (RAM) and read only memory (ROM) in that it is electrically alterable (although slow) and at the same time will not lose its contents on power loss. The primary use of this EAROM memory is to store parameters which must be tailored to each different system. These parameters are field adjustable for drive tuneup.

The local bus has a timer and interrupt card which provides the necessary hardware for control of interrupt priority and timing for the microcomputer.

There are input/output ports on the local bus which interface static logic signals to/from the microcomputer. As an example, thyristor gating commands are written out to one of these ports (on each side) and sent to an analog card to produce the gating signals to the converter bridge.

The electronic control contains a number of analog cards which process the voltage feedback, current feedback, meter signals, etc.

Analog to digital converter cards interface the speed reference, voltage feedback, current feedback signals, etc. to the microcomputer bus on each side.

I/O cards interface process command and status signals to the microcomputer. These inputs are combined logically by a sequencing program in the microcomputer to control the relays in the IOM. These outputs are used to control system devices such as panel lights, contactors, etc.

3-3. CONTROL OPERATIONAL DESCRIPTION

3-3.1. Basic Principles. Refer to the Control System block diagram in Figure 3-3.1A. The speed refer-

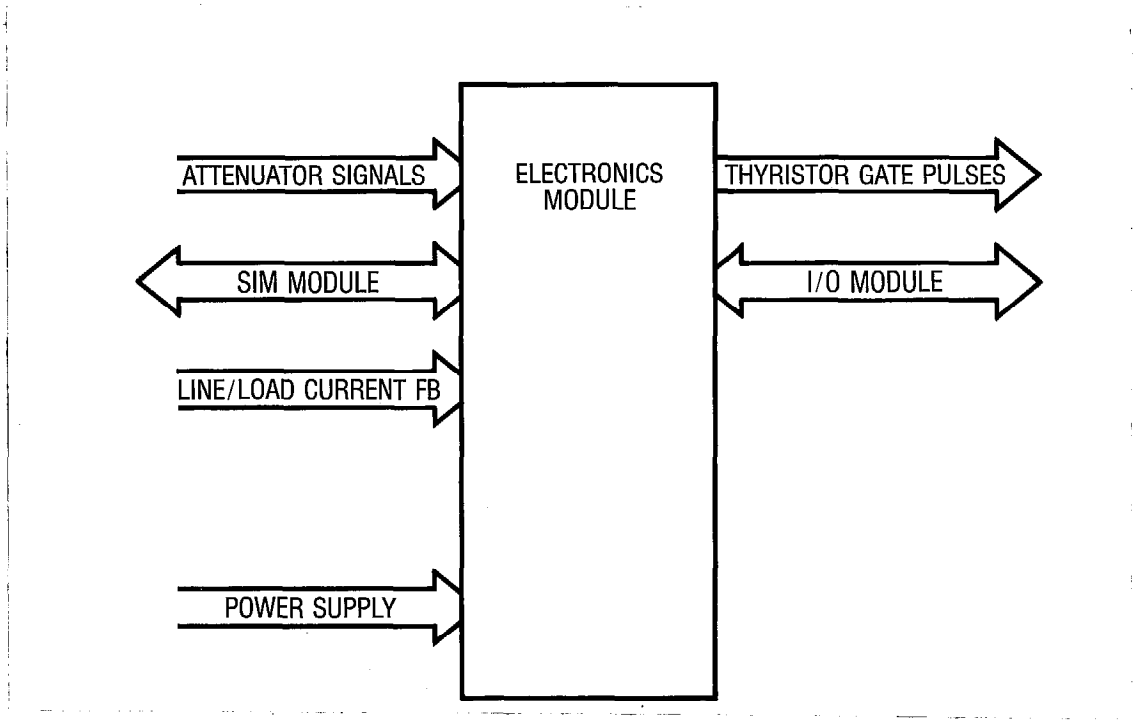


Figure 3-2B. SEM Input/Output

ence from the process controller is compared to a speed feedback derived from the integrated motor voltage to develop a speed error input to the speed regulator. The output of the speed regulator is a torque command. The speed regulator program resides in the load side processor.

The torque command signal is applied to both the source and load side control. As illustrated in Figures 1-6 and 1-7, motor torque is a function of voltage, current, and the angle between them. Therefore, torque can be controlled by adjusting current magnitude at a fixed angle or maintaining a constant current and varying the displacement angle. The current is controlled from the source side , whereas the angle is varied from the load side . At any instant in time, the torque is actively controlled by only one of these means.

The torque command to the source side control is applied to a maximum and minimum current limiter. The minimum current level is set to maintain continuous current in the DC link.

The current level control also affects the load side angle control. When the voltage spillover is not activated, the angle control is in one of its limits. If current is adjusted to control voltage, the angle control gain is decreased and displacement angle O is increased, which reduces developed torque to that required by the load.

The current command is compared with current feedback, I_{DC}, and the error is applied to the current regulator. The current regulator controls the firing of the thyristors in the source side converter (rectifier). Thus, the source side control adjusts the DC link voltage as required (within its limits) to produce the motor current needed to drive the load.

The load side control is designed to adjust firing angle so as to maintain sufficient flux, by controlling the angle between voltage and current .

Referring to Figure 1-4, it is shown that the angle of advance,B, is the sum of the commutation angle,U, and the margin angle,γ. For maximum efficiency, it is desired to keep the margin angle as small as possible without encountering commutation failure. The objective is to make the displacement between voltage and current as small as possible while maintaining torque and flux.

When the drive requires regeneration, the load-side thyristors are fired full advance (point "X" in Figure 3-3.1B), and the source side controls current by reversing the DC voltage to match the rectified motor voltage.

3-3.2. Phase Locked Loop Operation. Since the inverter thyristors are fired in synchronism with the integrated motor stator voltages, a phase locked loop

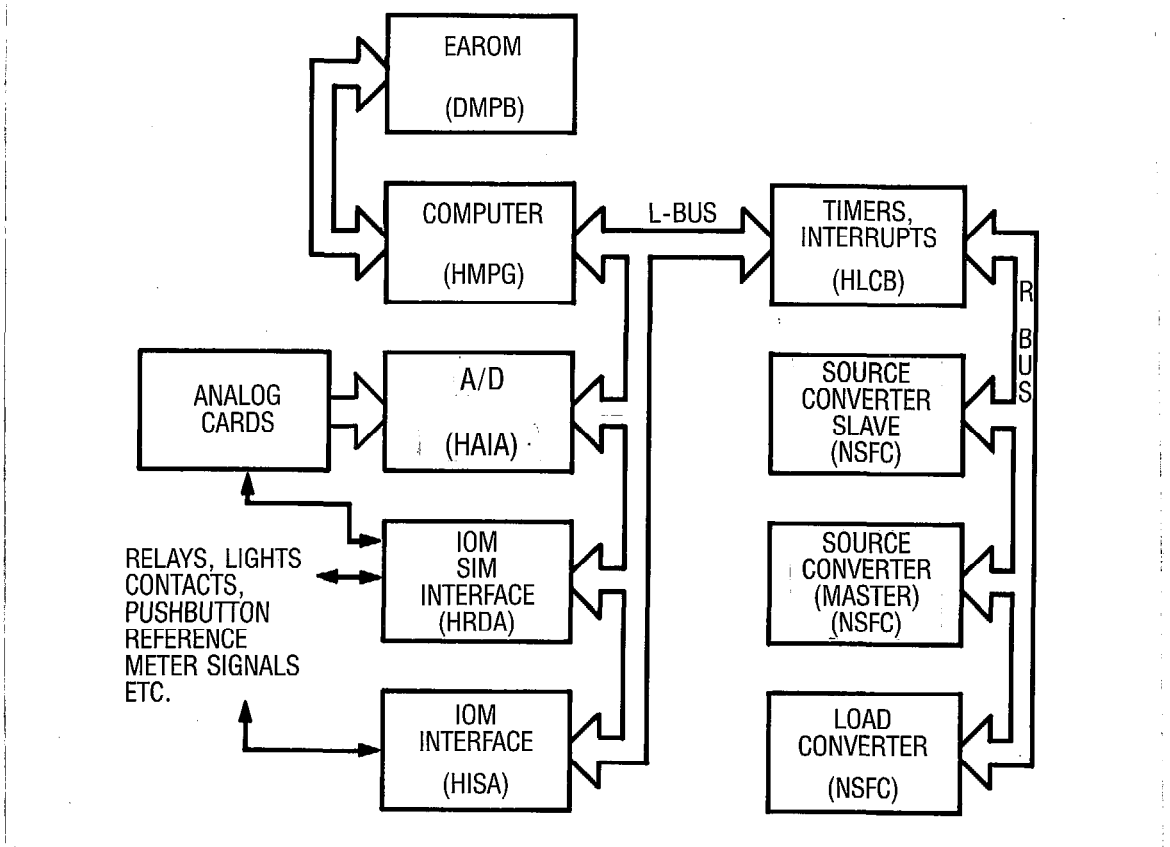


Figure 3-2C. Electronics Module Block Diagram

(PLL) counter is used to maintain correspondence between the integrated voltage wave zero crossings (Figure 3-3.1B) and the firing pulses.

When the PLL counter output is precisely in synchronism with the "flux" wave zero crossing, the loop is said to be "locked". If the counter output is lagging (slower than) the flux crossings, the PLL controlloer acts to increase the counter frequency (by decreasing PLLC LOAD). Similarly, if the counter frequency is leading (faster than) the flux crossings, the controller acts to decrease frequency (by increasing PLLC LOAD).

Although the output of a phase locked loop, when "locked", will have the same frequency as the reference signal, there will be a phase difference between the two due to the integrating action of the loop low-pass filters; thus, the name phase locked loop.

3-4. ELECTRONICS MODULE (IME) PRINTED WIRING BOARD (PWB)

Descriptions Following is a list of cards used in the electronics module. The function and condensed theory of operation of these cards follows the list. It is not the intent to describe in great detail the inner workings of each card, but to relate their function in the overall system. For more detail, refer to the appropriate DS3800 drawing, Section 2, for operating notes and Section 4 for elementary diagram. It is also generally helpful to refer to the card symbols in the SEM elementary diagram.

3-4.1. ALCI Processor extender board = DS3800HLCB . The HLCB board extends the functions provided by HMPG to include those required for an LCI system.

The following functions are provided.

1. Address decoding for on-board functions. NSFC, and NSFF.
2. Interrupt controller. - Source crossover
   - load crossover
   - source angle control
   - field excitation control
3. 5 bit purposes ionput port used for:
   - ground fault
   - I/O module input ribbon cable continuity

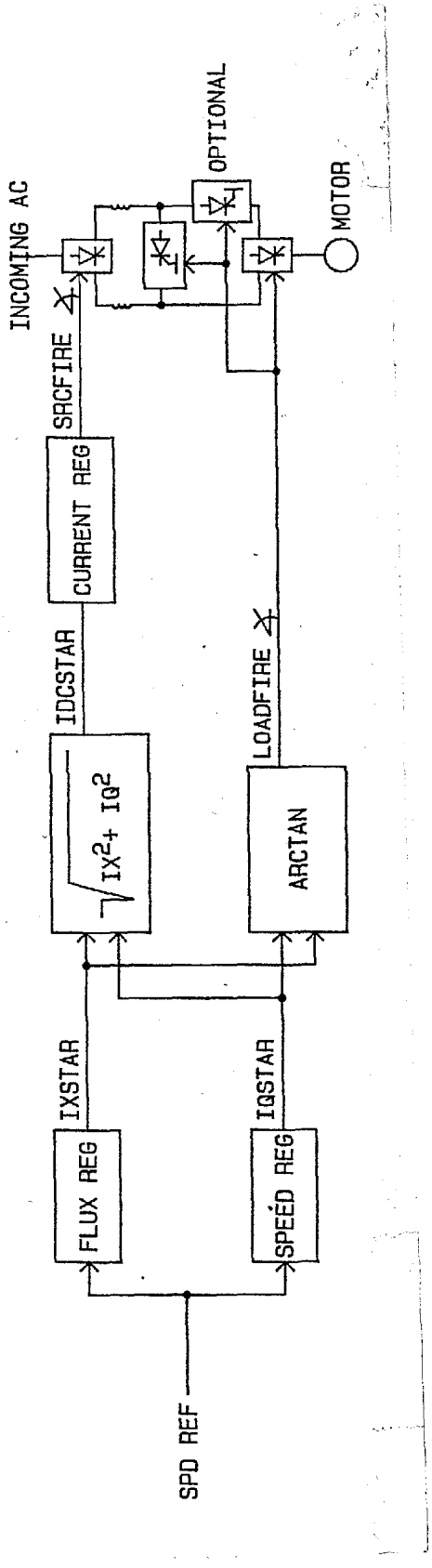


Figure 3-3.1A. DCLC-IMD System Digital Regulator Block Diagram

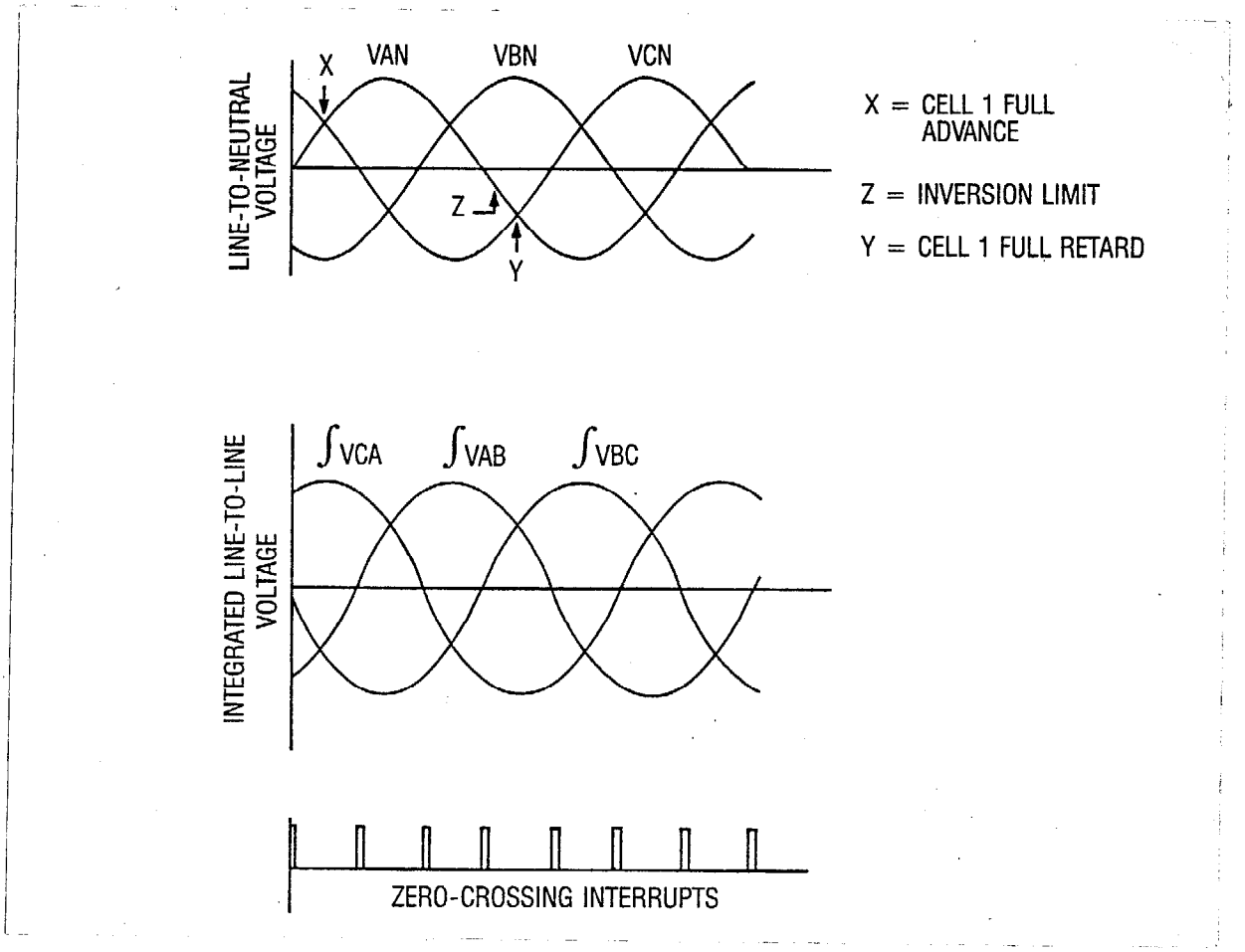


Figure 3-3.1B. Flux Wave Zero-Crossings

- I/O module signal cable continuity
- door module ribbon cable continuity

4. Zero crossing detection. Circuity is included to derive source and load crossover interrupts from square wave information originating on the MLIB boards. A single clock pulse interrupt is obtained at each crossover.
5. Two programmable timers provide the functions of - PLL clock (source and load)
   - TIME counter (source and load)
   - SPEED calculation
   - CLOCK for serial communications port
6. RS-232 compatible port. Only the transmitt function is implemented at the port is intended for driving a printer locaated in the door module. The control line, DSR (data set ready), is also monitored to determine the status of the remote printer.
7. Digital to analog converters. three 8-bit D/A converters are suplied. D/A 1 and D/A 2 are 8 bit mon-committed outputs which are available for transmission to the signal isolation module (SIM) if supplied. The 12 bit D/A is committed to supplying the torque reference between 6 pulse converter controls when used in a 12 pulse redundant system. System level documentation will define its use (if any) in other systems.

Berg jumpers

None used.

LED's

Nomenclature	Function
IMOK	Software controlled as a function or selftest. On normally, will be off if selftest is not passed.

Testpoints

Nomenclature	Function
ACOM	Analog common.
DA1	Digital to analog output 1. Nor-

	mally uncommitted.
DA2	Digital to analog output. Normally uncommitted.
DA3	Digital to analog output 3. Normally used for horsepower output.
DA4	Digital to analog output 4. Normally used for torque command to other channel (if supplied).
OTESTIN0	?
OTESTIN1	?
SFT1	Fxecution flag for source crossover and angle control routine.
SFT2	Execution flag for load crossover and angle control routine.
SFT3	Execution flag for field angle control interrupt.
SFT4	Execution flag for background program.
VA1	Phase A signal for source power.
VA2	Phase B signal for motor voltage.
DCOM	Digital common.

3-4.2. Microprocessor - HMPG/DMPB. The DS3800HMPG is a single board computer with the following features.

1. 80286 microprocessor
2. Programmable communications interface
3. Programmable interval timer
4. On-board RAM memory (16K bytes)
5. On-board PROM memory (64K bytes)
6. Mapped I/O
7. Bus hog timer
8. Stall timer

The configuration chosen gives a powerful 16 bit processor.

Connectors.

A 50 pin ribbon connector is included on the DMPG to interface the EAROMmemory (DMPB).

A 25 pin male connector is present to serve as an RS232C communications port. (See Tables 3-1A and 3-1B) A "null modem" is supplied to allow a standard data terminal interface cable to be used with either communications port.

3-4.3. Electrically-alterable . ROM = DS3800DMPB

This is a non-volatile memory board containing 8k bytes of memory used to store drive tuneup parameters which may be customized on a per drive basis. Physically, the board is configured as a daughter board and mounted on the HMPG.

Constants stored in the board's memory are normally changed by first enabling board writes with switch "WREN" and then supplying the data through the drive monitor.

In the Drive Systems documentation procedures, DMPB and the two PROMS in sockets U2 and U3 comprise module DS3815PLHA.

DS3815PLHA and HMPG are then combined to form module, DS3815PLGA. (See Table 3-2)

3-4.4. Analog/Digital Interface Card DS3800HAIA

The HAIA board accepts 16 analog input signals, multiplexes them to one line and coverts to a two byte digital word. The A/D provides 11 bits of magnitude resolution. Also provided are three, 8-bit digital input ports, three 8-bit output ports and one 4-bit digital port. Two boards are used, one for source microprocessor and one for the load. (See Table 3-3)

3-4.5. DC LINK COMMUTATOR CONTROL CARD - DS3800NCCC, 1F. This card sends out the gating signal (from the microprocessor control) for the shunt and series (if used) GTO to the XTFZ card (which fans it out via light pipes). It also receives the status (OK) of each of the GTO's. This information is available for use by the control program via the L-Bus.

Other functions include:

- Source slave bridge's current conditioning.
- A fault trip comparator of source slave current vs load current.
- A shorted motor filter capacitor detector.
- Motor current calculation circuitry which consists of the filter capacitor current and subtracting it from the load current.
- synchronizing function circuitry which includes PT match comparator (motor volts vs source (bypass) volts) and motor current as read from the bypass or motor isolator.
- DC link overvoltage detection

Test points and lights include:

TP1	OIA2	SRC slave phase A current
TP2	OIB2	SRC slave phase B current
TP3	OIC2	SRC slave phase C current
TP4	IFB2	SRC slave bridge current 5V=1PU
TP5	IERR	(SRC slave current)-(load current)
TP6	CIAB	Capacitor phase A-B current
TP7	CICA	Capacitor phase C-A current
TP8	MIAB	Motor phase A-B current
TP9	MIBC	Motor phase B-C current
TP10	MICA	Motor phase C-A current
TP11	MIFB	Motor current 5V=1PU
TP12	PTER	Motor PT(B) - Source PT(A)
TP13	GSH	Shunt GTO firing pulse
TP14	GSR	Series GTO firing pulse
TP15	Test	
LED	IMOK	On board op-amps not saturated

Daughter Board

Table 3-1A. Microprocessor Board Features - RS232C Connector Signals

Pin#	Name	Signal	Direction	Function
		RS232C CONNECTOR SIGNALS		
JB-2	BA	TXD	output	transmit data
-3	BB	RXD	input	receive data
-4	CA	RTS	output	request to send
-5	CB	CTS	input	clear to send
-6	CL	DSR	input	data set only
-7	AB	DCOM	input	signal ground
-9	CD	DTR	output	data terminal ready

Table 3-1B. Microprocessor - Berg Jumpers, Testpoints and LED's

Nomen.	Function	Position
	BERG JUMPERS	
J1	Selects type of PROM	8/16
J2	Selects size of PROM	16k
J3	Selects bootup method (on/off bd.)	DGN(not)
J4	Selects 8 or 16 address segments on L bus	8
J5	ground for RS232	GND
	TESTPOINTS	
RST	Reset jumper to common to reset processor	
DCOM	DCOM – Digital common	
I/S	Inhibit stall – Jumper to common to activate	
TNMI	Test nonmaskable interrupt – Jumper to common to activate	
	LED'S	
BHOG	Bushog – On when non-existent address is called by software	
TST	Software controlled and defined – On during initialization self test	

Table 3-2. Electrically-Alterable ROM - LED'S and Switches

(DMPB)

Nomen.	Function
	BERG JUMPERS
None Supplied	
	LED'S
IMOK	Software controlled as a function of self diagnostics — On normally, will be off if selftest is not passed
WREN	Write enable. On when write capabiltiy to board is enabled with switch, "WREN"
	SWITCHES
WREN	Activates the board to allow alteration of the drive tuneup parameters.

TP1/R1	Adjustment of current unbalance (IERR) trip point
TP2/R2	Adjustment of PT match point
TP3/R3	Adjustment of shorted capacitor detection point
R4	Gain adjustment of contactor motor current for metering
TP4	TP4 is motor current 5V = 1PU
TP5/R5	Adjustment of DC link overvoltage detection point

3-4.6. 4KV MEDIUM VOLTAGE BRIDGE-COMPACTED. The arrangement of the power converter SCR stack assemblies is shown in Figure M4-1. This figure and the elementary diagram sheets show the major power components.

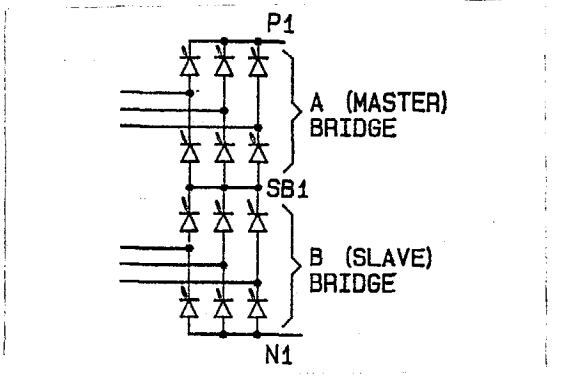
Thyristors (SCRs) are stacked horizontally, three in series, between four heat sinks. The AC buses are on the right side (consecutive stack pairs). The DC buses are on the left side (every other stack). The fuses (AC and DC filters) are located at the top of the component. The primary CTs are located behind the fuses on the bus work.

The blower compartment contains blowers, AC and DC filters, MOVs, secondary CTs and the terminal boards. Also, the hinged base in this compartment contains blower starters the 70V gating power supply, and thyristor fan-out PWB XTFH.

3-4.7. 2KV MEDIUM VOLTAGE BRIDGE-SERIES TWELVE PULSE. The arrangement of the power converter SCR stack assemblies is shown in Figure M2-2. This figure and the elementary diagram show the major power components.

SCRs are stacked horizontally; three in series, between four heat sinks. The top three rows are the master bridge, while the bottom three rows are the slave stacks. The AC lines come into the center of the rows respectively. The DC buses are on the outside edges of the rows. The fuses (AC and DC filters) are located at the top of the compartment. The primary CTs are located behind the fuses on the bus work.

The blower compartment contains the blowers, AC and DC filters, MOVs, secondary CTs, and terminal boards. The hinged base in this compartment contains the blower starters, the 70V gating power supply, the thyristor fan-out PWB XTFH, and bottom blower selection relay.



Typical 12-Pulse Series Bridge

3-4.8. GTO POWER CONVERSION SECTION.

The arrangement of the GTO power converter stack assemblies is as shown in figure G4-1. This figure and the elementary diagram show the major power components.

The GTO is mounted with its power diode in a heat sink assembly. The GTO may be mounted in series or in "center tap" arrangement depending on the GTO ratings and characteristics. Like the SCR's, it is recommended that the entire stack be replaced if a failure occurs and that the stack be sent back to the factory for repair.

Each GTO stack is mounted behind a GTO driven module which provides gating power, the gating signals (on, off) and status indication. Gating signals come in and status signals go out on fiber optic "pipes" to the XTFZ which converts these to signals the SEM uses.

Mounted to the side and behind the stack is the snubber network for the GTO diode, and the NTDA which contains a LED that goes out when the GTO shorts.

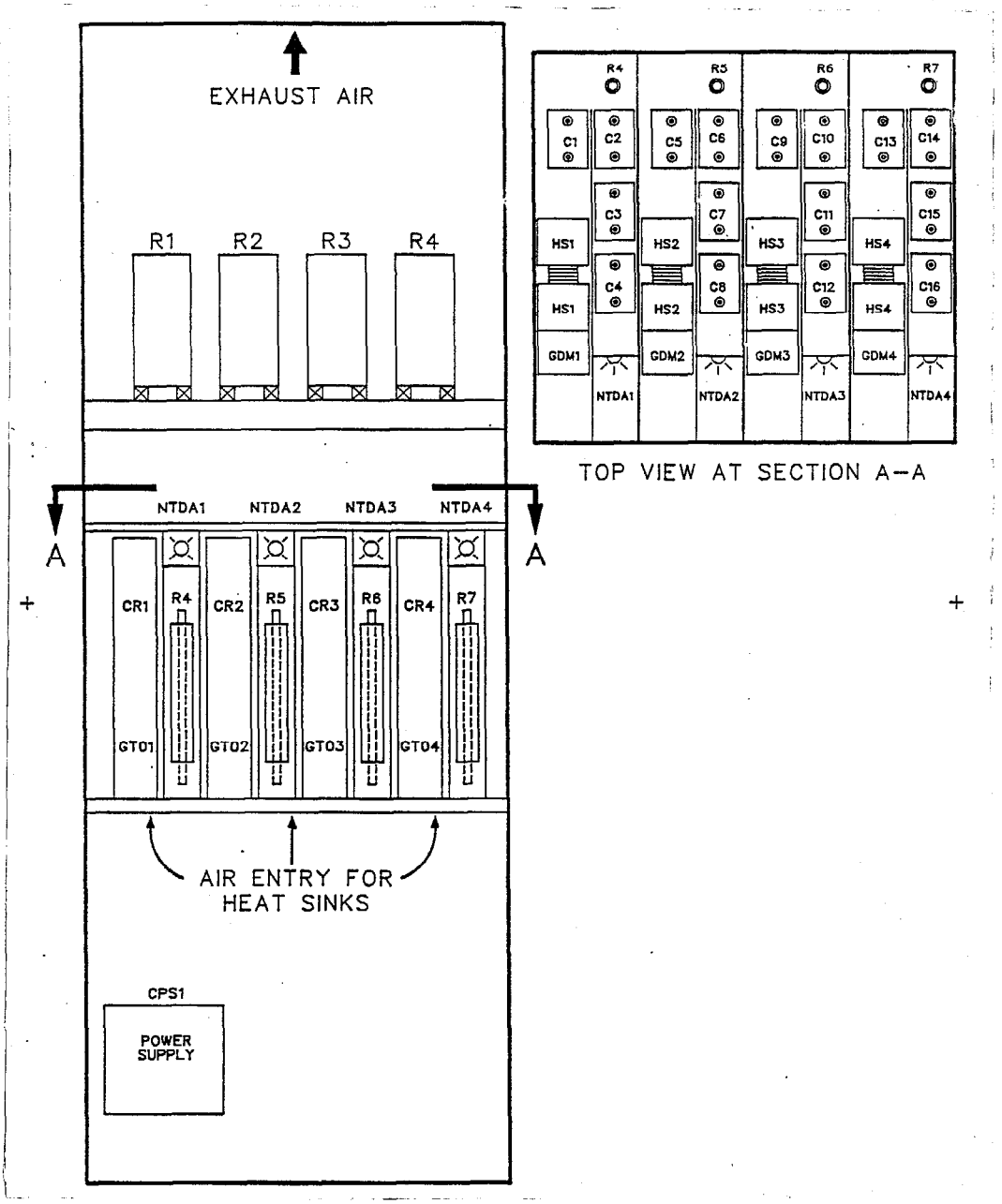


Figure G4-1

In the bottom of the compartment is located the GTO chopper power supply. It converts the 120 Vac, 60HZ into 120 Vac 400 HZ which is feed to the GT0 driven modules.

Mounted in the top of the compartment are some low ohm, high power resistors that are also used in the GTO's snubber network, also located above is the cooling blower.

3-4.9. Flux Integrators - DS3800NLIB. The NLIB contains various analog signal conditioning and protective functions. Two identical boards are used - one for the source side and one for the load. The following functions are provided:

1. Voltage integrators
2. Line/Motor current transformer receivers
3. Voltmeter driver
4. Overvoltage protection
5. Undervoltage sensing
6. Suppression
7. Variable reference comparator
8. Current meter driver

3-4.10. Voltage integrators.. A signal proportional to each line-to-line voltage is obtained from the NSFC board. This signal is applied to an analog integrator to obtain a replica of motor flux according to the equation:

$$V = K * dPhi/dt$$
or
$$Phi = V * dt/K$$

Components of line currents are used to compensate for drops due to motor resistance and commutating impedance so that the flux represents the motor internal counter-emf or the so called voltage behind the commutating reactance. The analog sine wave corresponding to flux is then applied to a zero crossing comparator where the digitized output is then available to the HITA for phasing information. The analog flux signals are also read by the microprocessor based software for use in the control algorithms.

Line/motor current transformer receivers.

Table 3-3. Analog/Digital Input Signals - Berg Jumpers and Testpoints

(HAIA)

Nomen.	Function	Position
	Berg Jumpers	
J1	Selects either a fixed +7.5v input to channel 15 or a variable input thru pin12. The variable input is selected.	A
J2	Selects either a fixed +7.5v input to channel 14 or a variable input thru pin12. The variable input is selected.	A
J3-J8	Selects L bus address for board.	F
J9	Selects filter for analog side of A/Dconverter. Filter is selected.	O
	Testpoints	
ACOM	Power supply analog common.	
FOG	Test input. Tie to common to latch internalcard data bus.	
AN	Analog signal at input to A/D conversion	
DB	Data buffer direction. High transmits to microprocessor and low receives.	
DV	"O" data valid signal	
BSY	"1" busy signal	
ES	"1" enable switch	
CLK	card clock - 1.152 mhz	

Two signal conditioning differential receivers are included for line currents A and C. Line current B is calculated from A and C assuming a balanced load. Delta currents are also determined for use in the flux reconstruction.

3-4.11. Voltmeter driver.. The magnitude of the line-to-line voltage is determined by a 3 phase absolute value network. The absolute value is then scaled by an operational amplifier for use as a line voltage signal.

3-4.12. Overvoltage protection.. The absolute value of line/motor voltage obtained above is compared with an adjustable reference for a time determined by daughter board component, C37, then the overvoltage latch is set. The signal is wired to an I/O port for fault logging by the control software.

3-4.13. Undervoltage sensing.. The absolute value of line voltage is compared with an adjustable reference. If the voltage is less than the reference for time delay set by daughter board component, C17, then signal "1LV" becomes true. This signal is used by the software system sequencing to determine low source volts condition. The function is not used on the load side.

3-4.14. Suppression.. The absolute value sum of the line currents is taken and applied to a comparator. If the rectified current exceeds the reference threshold for 26ms, a fault latch is set. The time delay is included since the primary protection is in software which includes an on-line recovery. The function implemented here is strictly backup.

3-4.15. Variable reference comparator.. "COMP" presently not used

3-4.16. Current meter driver.. The absolute value of current is scaled and available from either source or load for meter driving. Scaling is 5v = 1pu/ (See Table 3-4)

3-5. GATING REFERENCE BOARD - DS3800NSFC

The NSFC's primary task is to convert information from the microprocessor based control to cell pair and timing for firing 6 thyristors configured into a 3 phase - double way bridge. In performing this task, the following functions are used:

1. R bus interface
2. Firing mask latch
3. Cell state sensors
4. Line-to-line voltage signal conditioners
5. Pulse train generation
6. Gate drivers
7. Attenuator buffers
8. D-C voltmeter driver

Quantity 4 NSFC's are used, assigned as follows:

1. Master source converter
2. Master load converter
3. Slave source converter
4. Slave load converter

R bus interface.Control logic is provided for decoding read and write strobes. Eight data lines (either upper or lower byte) are connected. All information exchange with host processor takes place through the R bus interface.

Firing mask latch.The particular cell pair to be fired is selected in software and communicated to the NSFC over the R bus. The mask information is temporarily stored in the primary firing mask latch. Also obtained from the software is time delay corresponding to the number of degrees the cell pair firing is to be delayed. This information is routed to a timer. At the end of the delay an interrupt is generated and on the load side the cell mask is clocked into the final latch. On the source side, the generated interrupt calls an "execute firing" program which (if all permissives are met) signals the NSFC to move the firing mask to the final latch.

3-5.1. Gate drivers.. Six darlington transistors shift the gating information from logic level to analog levels sufficient to drive the remotely mounted gate pulse amplifiers. These drivers operate at 28 VDC levels.

Pulse train generation.The gate pulse reference driver signals consist of a "O" true pulse train nominal 120 degrees long during the conduction period. The reference pulse train is generated from a crystal clock and shaped for a duty cycle of 15 microseconds on and 30 microseconds off by a timer and control logic.

3-6. BUFFERS.

Attenuator buffers are provided to receive each attenuator string connected to phase voltages 1,2,3 and also the positive and negative bus. These buffered signals are then used for analog line-to-line voltage signals as well as cell state sensor inputs.

3-6.1. Line-to-line voltage signal conditioners.. The buffered phase to neutral converter voltage signals are differentially combined to recreate line-to-line voltages. These are used after further processing (on NLIB) as information for the software control algorithms.

Table 3-4. Flux Integrators - Jumpers, Adjustments, LEDs and Testpoints

(NLIB)

Nomen.	Function	Position
	BERG JUMPERS	
BJ1-BJ3	Selects voltage integrator scaling for 60Hz or 120Hz	60Hz or 120Hz (var)
	Adjustments	
FAB, FBC, FCA	Scale flux integrators which calculate flux. Scaling is 4.75 volts RMS corresponds to 1 pu.	
OV	Set overvoltage protection trip point.	
UV	Set undervoltage protection trip point.	
SUPP	Set trip point for overcurrent backup fault.	
COMP	Sets field loss level on source side and field overcurrent level of load side.	
VDC	Not used.	
	LED's	
OV	On when overvoltage fault latch set.	
SUPP	On when overcurrent fault latch set.	
IMOK	On if microprocessor based self test is passed and no saturation condition are present.	
	Testpoints	
OVR	Analog D-C voltage proportional to setpoint of over-voltage fault latch.	
UVR	Analog D-C voltage proportional to setpoint of source undervoltage sensor.	
SUPP	Analog D-C voltage proportional to setpoint of overcurrent fault latch.	
FAB,FAC FCA	Analog voltage proportional to motor/line flux reconstructed. See fig. 3-6 for typical wave forms.	
FBAR, FCBR, FACR	Analog voltage proportional to motor/line flux reconstructed for IX drop. See fig. 3-6 for typical wave form.	
OIA,OIB OIC	Analog waveform representative of inverted line current. See fig. 3-6 for typical waveforms.	

3-6.2. Cell state sensors. . From a cursory check of the standard three phase double way bridge configuration, it can be seen that the voltage across each cell is a combination of one phase voltage and one D-C bus voltage. Since each of these voltages is available from an attenuator, the voltage across each cell or "on state" maybe uniquely determined. Circuitry is provided for reading the "state" of all cells at a particular instant and for making the information available to the microprocessor via the L bus.

3-6.3. D-C Link volts. . The positive and negative D-C link bus voltages are combined differentially to obtain a D-C link volts signal suitable for driving a remote analog meter with approximately 10 VDC maximum scale (See Table 3-5).

3-7. ANALOG LCI BOARD = DS3800NLCB

The NLCB provides various miscellaneous signal conditioning and test functions required for the structured LCI. These features include:

1. Speed feedback generation from motor voltage zero crossing signals with overspeed protection and indication

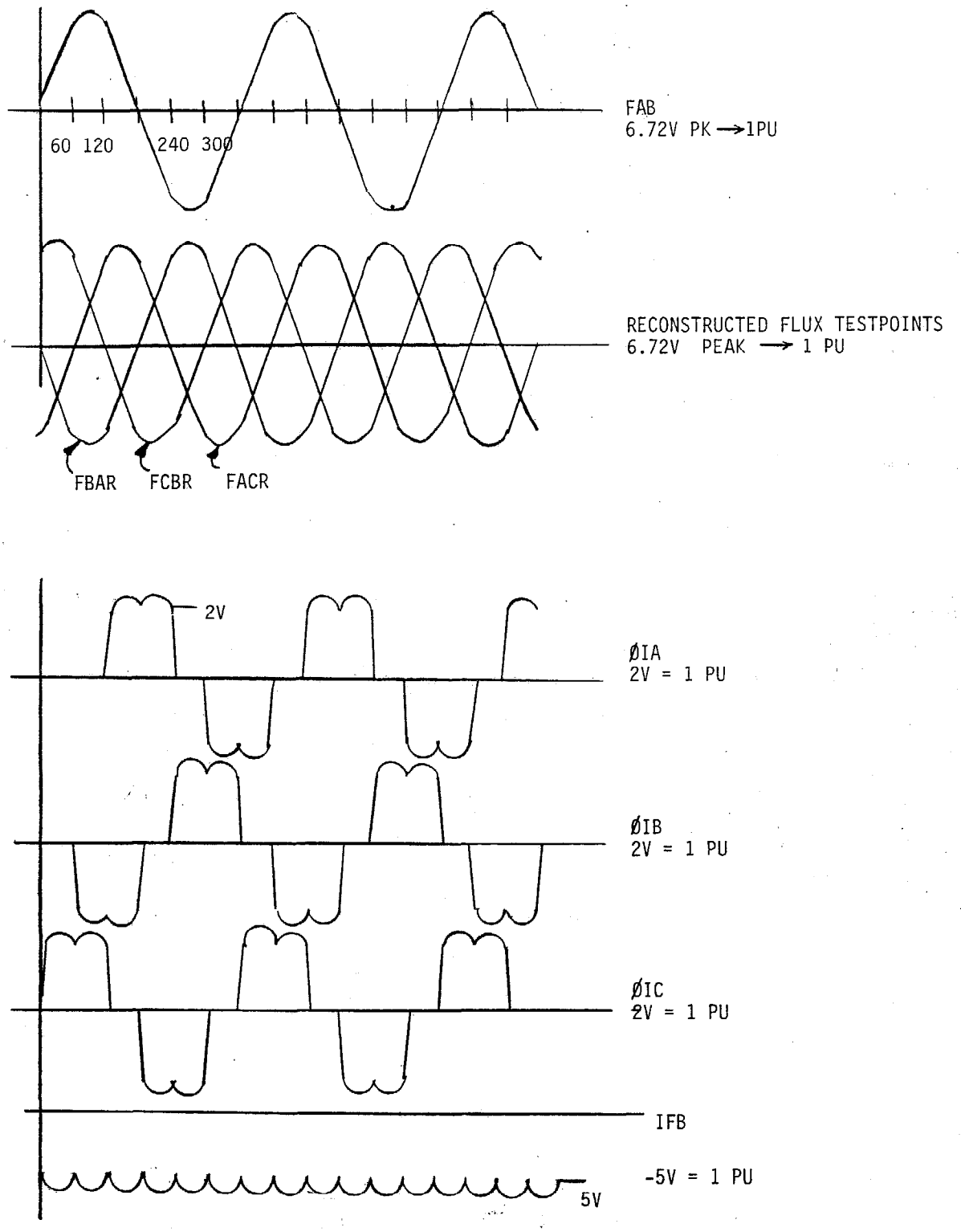


Figure 3-6. Analog Typical Waveforms

Table 3-5. Gating Reference - Berg Jumpers, Adjustments, LED's and Testpoints (NSFC)

Nomen.	Function	Position
	Berg jumpers	
J1	Determines if new mask should be clocked into final latch at termination of count down timer.	NINT(1d) INT(src)
	Adjustments	
VDC	Sets remote meter full scale to corresponding D-C link volts.	
	LED's	
IMOK	Board hardware status indication. On normally. Off if self diagnostics determines a board hardware failure.	
	Testpoints	
SIG 1-SIG 6	Gate pulse command. "O" true. Steady state length of 120 degrees relative to operating frequency. Each gate pulse is shifted 60 degrees relative to lower numbered ones. SIG numbers correspond to thyristor leg being gated.	
VDC	Output of D-C link voltage meter driver.	
VAC,VCB	Analog output proportional to motor/source line-to-line voltage.	
ACOM	Power supply analog common.	

2. External reference signal conditioning
3. Test reference signal conditioning
4. Torque reference signal conditioning
5. Selftest features
6. 4-20 MA signal drivers
7. Ground fault detector
8. Power applied reset
9. Fault relay drivers
10. Horsepower meter driver
11. Continuity detection circuits
12. Test switch
13. Restart upon power loss

3-7.1. Speed feedback:. A pulse rate signal proportional to drive speed is obtained from the zero crossing detection circuitry on the HLCB. A pulse rate to analog converter then produces a corresponding D-C level proportional to speed suitable for driving a meter and protective functions.

The overspeed function is implemented by comparing an overspeed setting reference to the analog speed voltage attained above. A latch insures that transient conditions will be sealed. A red LED indicates the overspeed condition until reset.

A differential amplifier committed to accepting the remote speed reference is supplied. Berg jumpers are used to configure the circuit to accept a 0 to 10 volt signal or a 4 -20 MA signal. In either case, the output voltage is transmitted to channel of the HAIA analog to digital converter for reading by the system processor.

3-7.2. Test speed reference.. A differential amplifier is included to condition a voltage source signal from a potentiometer located on the I/O module. This is also known as the local speed reference.

3-7.3. Torque reference signal conditioning. A differential amplifier for receiving the master torque reference from the other six pulse drive (in a 12 pulse system) is also supplied.

3-7.4. Selftest features . An array of static switches and voltage sources is included for use in testing the analog circuitry of this and other boards in the module during the module selftest mode. A sinewave oscillator is used to simulate line and motor voltage during "gate test mode". This is to allow the power converter electronics to be tested without the necessity of applying high voltage to the associated bridges.

3-7.5. 4 - 20 MA signal drivers . Two voltage to current converters provide capability of remote driving system information to customer interfaces. The signals selected for transmission are motor current and motor speed. In each case, 4 ma corresponds to 0.

3-7.6. Ground fault detector . The ground fault detector sums line-to-neutral voltages Va, Vb, and Vc. It operates on the principal that when all impedances from phase to neutral are balanced that the summation of the phase-to-neutral voltages will be zero. Any unbalanced impedance will cause an error to be detected and a ground fault noted by the system processor.

3-7.7. Power applied reset . A power applied reset is generated here and distributed throughout the electronics module. It is used to prevent false operations of relay drivers and fault latches during the process of powering up the module.

3-7.8. Fault relay drivers . Relay driver no. 1 is generally used as a method of tripping the source side drive switchgear from both hardware and software sensed fault conditions. It is inhibited from false operations during the power up phase by the power applied reset.

Relay driver no. 2 is generally used as a method of tripping the motor side isolator (if used) from both hardware and software sensed fault conditions. It is also inhibited from false operations during the power up phase.

3-7.9. Horsepower meter driver . An adjustable meter driver for powering a door mounted meter indicating horsepower is also included. The horsepower signal is calculated in software and converted to analog on the HLCB board.

3-7.10. Continuity detection circuits . Four circuits are included to monitor the continuity of various ribbon cables connected to the electronics module when it is integrated in to a system control panel. These signals are only monitored during the panel selftest. An appropriate diagnostic message is printed out if a cable is not in the correct location.

3-7.11. Test switch . The test switch on the board front edge is used to initiate the panel selftest mode. When the switch is operated, then the electronics module will begin to execute a selftest routine which extends to other components of the control panel. Upon completion of the selftest, the control will enter gate-test-mode using the three phase test oscillator as a source of synchronous waveforms.

3-7.12. Restart upon power loss . If the drive has been previously running and the control power is lost for a time (determined by setting of RTD potentiometer), then the drive will be automatically restarted when power returns by a pulse which appears at signal 1RSRT. Note this is an option and the I/O module must be properly configured. (See Table 3-7)

3-8. RELAY DRIVER BOARD = DS3800HRDA

The HRDA is relay driver board with 16 open-collector outputs. Output voltages can range from 5 to 50 volts. In the LCI system, it is applied at a nominal voltage of 28 volt D-C. The board may be connected to the computer R-bus for accessibility by the system control.

The relay driver board is used principally as an interface between the system software control and the power devices located in the I/O module. (See Table 3-8)

3-9. INPUT SIGNAL CONDITIONING = DS3800HISA

HISA is designed to work in conjunction with HRIA (in I/O module) to condition thirty high level (105VDC) signals to logic levels and allow access to them through the processor R bus. Each signal is separately isolated by a photocoupler and indicated by a bar LED. Two relay drivers are also included which are only utilized in the hardware selftest of the I/O module. (See Table 3-9)

3-10. DISPLAY MODULE (LDM)

The DS3820LDM is the display module for the structured LCI control panel. The display provides fault, alarm and diagnostic information. The display module comes in two basic layouts. Layout Option 1 provides the following features:

- 3 indicator lights (RUN, ALARM and FAULT)
- 2 push buttons (RESET AND DRIVE STATUS)
- 3 1/2 digit motor speed meter (RPM)
-24 column printer with real-time clock

and Layout Option 2 provides the following features:

- 4 push-to-test indicator lights (RUN, ALARM, FAULT and READY)
- 2 push buttons (RESET and DRIVE STATUS)
- 4 DB40 type meters (motor speed, motor current, motor voltage and horsepower)

All lights and push buttons are on a 28v circuit. The functions and color codes for each light are:

Red for the motor RUN light

Red for a system FAULT light

Amber for a system ALARM light

Green for the system READY light

Table 3-7. Analog LCI Board - Berg Jumpers, Adjustments, Testpoints and LEDs
(NLCB)

Nomen.	Function	Position
	BERG JUMPERS	
LOAD	Selects remote speed reference for either current input or voltage input. Should be coordinated with GAIN.	variable
GAIN	Selects gain necessary for remote speed when used as current input or voltage input.	variable
	Adjustments	
SPD	Adjusts analog speed feedback circuit for 10 voltsat 60 Hz.	
SPM	Adjusts door mounted speed meter for proper fullscale reading	
OSP	Hardware overspeed reference.	
RTD	Adjusts allowable time for auto restart after loss of control power.	
	Testpoints	
SPD	Output of analog speed feedback calculation	
OSP	Reference to hardware overspeed detection circuit. Scaled same as speed i.e. 10 volts = 60Hz.	
IOA	Output of channel A current source.	
IOB	Output of channel B current source.	
REF	Output of remote speed reference. With a voltage input will be 0 to 10 volts. With a current source input, will be 2 to 10 volts.	
TREF	Output of local test reference. Varies from 0 to 8.3 volts.	
12P	Output of torque reference obtained from other channel.	
NTL	Neutral. Summation of line to neutral voltages.Normally near 0 volts.	
RTD	Run time delay. Sets allowable time for loss of control power while still allowing an auto restart. Scaled 0 to 10 volts.	
PHA	Test oscillator, phase A. Shows a 55Hz. sine wave.	
PHB	Test oscillator, phase B. Shows a 55Hz. sine wave,lagging phase "A" by 120 degrees.	
	LED's	
OSP	Red LED indicating that the hardware overspeed latch is set.	
RSRT	Green LED which is pulsed when the auto restart circuit is activated.	
FT1	Green LED indicating the status of fault relay driver No. 1. Will be on when the relay driver is energized.	
FT2	Green LED indicating the status of fault relay driver No. 2. Will be on when the relay driver is energized.	
TEST	Green LED which luminous when the test switches closed and the system software indicates gate test mode.	
IMOK	Yellow indicating light will be on normally. An off condition indicates a possible hardware problem with the board. It can be turned off either locally or from the processor as a function of selftest.	

Table 3-8. Relay Driver - Testpoints and LED's

(HRDA)

Nomen.	Function
	BERG JUMPER
None supplied	
	ADJUSTMENTS
None supplied	
	TESTPOINT
TP1	Reset. Connecting testpoint 1 to testpoint 2 resets only this board. This function isn't useful in the LCI system.
TP2	DCOM
	LED's
O - F	A red LED for each of the 16 relay drivers indicates the energized status of each circuit.
ADR	Yellow LED. Turns on when the board is addressed; stays on if the board is addressed at least every 200 microseconds.
WPOK	Parity check. Not used in this system will always be on.

The push buttons are connected to a 28v circuit and direct a 28v signal to the output when the button is depressed. The RESET push button is used to reset latched faults in either software or hardware (NLIB and NLCB latched faults - overspeed, suppression, overvoltage, etc.). The DRIVE STATUS push button; when depressed, provides information on the motor voltage, current, speed, etc. via the printer.

The digital meter is a DC voltage measurement panel type. The conversion factor for the meter is 2 volts correspond to 1999 cnts. The panel meter is powered by the 5 volt bus via the 26-pin ribbon cable.

The four DB40 type analog meters are 10 volt movement meters with ± 10% adjustment rheostats (back).

The printer included in the display module is a Digitec Model 6610 Panel mount printer. The 24 column printer inputs data and control signals via a serial RS232 connection and uses an impact print head mechanism to output the data. The printer also supplies a real-time clock that will be used to log the time of status information and system failures.

The LDM interfaces to the LCG thru a 26 pin ribbon cable JK.

3-11. SIGNAL ISOLATION MODULE (SIM)

The DS3820SIM is used to provide optional functions to an LCI control panel. In particular, it provides four impedance isolated signal outputs for use in a purchaser's external control system. The four signals may be selected as 0 to 10 volts or 4 to 20ma outputs by header connection jumpers at JM. Each circuit may be selected independantly. Inputs to the module (TEM) are received from the LCI control SEM via 26-pin ribbon cable (JM). The four inputs to the isolation circuits may be chosen by header connector jumpers at JB from the following list of signals: Source volts, source current, SPD ref., motor SPD, horsepower, DA1, DA2, exciter current, motor volts.

A DS3800NVAA provides the functions of voltage to current source transformation and isolation. A differential amplifier accepts the signal from the remote control SEM. The signal is isolated from the NVAA on-board ground by approximately 100 K ohms of resistance. The isolated signal is then applied as a reference to a current source with the gains weighted such that a 0 to 10 volt signal provides a delta output of 16 milliamperes. An additonal bias is provided ot give the 4 milliamperes at 0 volts input. A chopper isolated power supply is present on the NVAA board to power the output stages of the signal isolators. The

output of each of the four circuits is referenced to XCOM, the on-board power supply common. It is intended that the XCOM connection be tied to the customer's external circuit common. While it is not necessary for the signals used as inputs to the isolators be referenced to the customer's common or XCOM, the difference between ACOM and XCOM must not exceed 3 volts steady-state or transiently. The voltage developed across any external impedance in series with a current source must a not exceed 10 volts.

The output signals are also available as voltage followers (0 to 10v). Maximum output load is 1 ma.

An operational amplifier monitor (or saturation detector) is provided for monitoring the output stages of the current driver circuits an LED, "IMOK", is luminous as long as no fault condition exists. In additon, if FUSE 1 in the on-board power supply is blown, then the LED will be non-luminous. (See Table 4-0)

3-12. I/O MODULE (IOM)

The purpose fo the DS3820IOMA (I/O) module is to provide signal conditioning and connection for interpanel and purchaser wiring in a load commutated inverter (LCI) control panel. The following functions are provided.

1. Signal conditioning and isolation for interface wiring at the 105vdc level.
2. Isolated signal interlocks (form C) to remote locations as a function of programmed system requirements. Rated .25A, 115vac (Resistive).
3. Fused 105vdc power supply to provide a source of high level power for interface.
4. Power relay contro for interface with a motor-side isolator as well as interface with a line-side switchgear device. Rated 10A, 600vdc (Inductive).
5. Local operator's station for running the drive during maintenance modes. (Channel selector, local/remote, start/stop, local speed reference).
6. Termianl boards for terminating interconnecting wiring.
7. Diagnostic functions.

The IOMA module is designed to be intergrated into a system diagnostic test function residing in the system electronics module. Therefore, no local diagnostic messages are displayed. However, the necessary circuits aand feedbacks are provided to allow the system function resident in the electronics module to isolate failed components to the replaceable board level.

Table 3-9. Input Signal Conditioning - Berg Jumpers, Adjustments, Testpoints and LED's

(HISA)

Nomen.	Function
	BERG JUMPERS
None Supplied	
	ADJUSTMENTS
None supplied	
TESTPOINTS	
None Supplied	
	LED's
I0O - I1D	Green bar LED's. Each indicates the status of the associated input, luminous when the input is energized.
IMOK	Yellow LED. On during normal operation. will be off only as the result of module selftest.

Table 4-0. Signal Isolation Module (SIM)

Nomen.	Function
	TESTPOINTS
TP1	Voltage output of channel 1 relative to XCOM.
TP2	Voltage output of channel 2 relative to XCOM.
TP3	-2.5 VDC bias for V-I circuits.
TP4	Voltage output of circuit 3 relative to XCOM.
TP5	-2.5 VDC bias voltage for I-V circuit.
TP6	XCOM, isolated supply ground.
TP7	Voltage output of channel 4 relative to XCOM.
TP8	Voltage output of 4-20ma to voltage converter.
TP9	N15X, isolated supply.
TP10	P15X, isolated supply.
TP11	ACOM, analog power supply ground.

CHAPTER 4

4-1. SECTION A: EPROM MEMORY

4-2. EEPROM MEMORY

EEPROM Memory is a non-volatile memory space used to hold critical drive constants that may be system dependant. The use of EEPROM Memory allows the drive constants to be changed in the field in order to fine tune the system performance. EEPROM Memory space resides on the DMPB card (HMPG daughter board) and is configured to contain 1024 bytes (1K X 8).

Default values for the drive constants are burned into the ROMs residing on the HMPG (ROMs also contain the executable software). These default values are intended to provide a starting base to work from when tuning the drive. They may be loaded into the EEPROM Memory by turning the DMPB write enable switch 'on', jumpering testpoint TIN0 on the HLCB pwb to DCOM, and then executing a hardware SEM reset. In order to change a value in the EEPROM memory, turn the DMPB write enable switch "on" and use the monitor ZAP command.

After power up or hardware reset, the microprocessor on the HMPG makes a local copy (shadow) in RAM of the EEPROM Memory. During normal running, the microprocessor accesses the drive constants from this local RAM shadow. The addressing for the drive constants is shown graphically:

4-3. EEPROM DEFINITIONS

The number in parenthesis following the variable name is the address offset for that variable.

ADVANC LINE E(:4A)
ADVANC LINE E Sets the maximum clampfor the source dc link voltage commane. The scaling is:

256 counts = 1 pu dc link voltage command

The recommended seting is OFFH.

ALPH LD IMP MARG E (:340)
ALPH LD IMP MARG E is presently not used.

ANG SPILL GAIN E (:0C2)

ANG SPILL GAIN E is gain used to contol the spillover of load-side firing angle into GTO DCLC "on" time. When the load-side firing angle runs into retard clamp, the difference between commanded angle and actual firing angle spills over and increases the GTO on time.

ANG SPILL GAIN E = 16 * desired gain

The recommended setting is a gain of 1 (10H).

BACKGROUND OPTION E (:33C)
BACKGROUND OPTION E's individual bits are used to enable/disable various functions in the background

MEMORY MAP

DMPB EEPROM		HMPG RAM	HMPG ROM
3000:3FF	RESERVED	20:3FF	FFC0:3FF
3000:3F0		20:3F0	FFC0:3F0
3000:3EF		20:3EF	FFCO:3EF
	SELF TEST EEPROM MEMORY		
3000:300		20:300	FFC0:300
3000:2FF		20:2FF	FFC0:2FF
	FOREGROUND EEPROM MEMORY		
3000:000		20:000	FFC0:000

(application) program. See the software logic ladder in the system elementary for specific usage. Typically a "1" in the bit position indicated below forces indicated state.

bit	function
0	no motor overtemperature
1	no bridge differential pressure alarm
2	blower 1 starter energized
3	blower 2 starter energized
4	blower 3 starter energized
5	blower 4 starter energized
6	no user defined alarm
7	no transformer alarm
8	no DC link reactor alarm
9	no PCR alarm
A	no blown bridge filter fuse
B	run permissive energized
C	start permissive energized
D	blower fault is not system fault
E	no master-slave current unbalance fault
F	fan door closed

BACKGROUND TIMER1 E (:330)
BACKGROUND TIMER2 E (:332)
BACKGROUND TIMER3 E (:334)
BACKGROUND TIMER4 E (:336)
BACKGROUND TIMER5 E (:338)
BACKGROUND TIMER6 E (:33A)

BACKGROUND TIMER? E is the time delay in counts used in various timing functions in the background (application) program. See the software logic ladder in the system elementary for specific usage.

1 count = 1 / (source line frequency)

BAUD 4800 E (:308)

BAUD 4800 E controls the baud rate of the RS232 port. The selections are:

= 0 - 300 baud
= 1 - 4800 baud

BRAKE TCOM E (:09E)

BRAKE TCOM E sets the level of dc current applied to the motor during braking. The scaling is:

1024 counts = 1 pu dc link current

The recommended setting is 0.5 pu current (200H).

BRAKE TMR LMT E (:0A2)

BRAKE TMR LMT E controls the lenght of time that dc current is applied to the motor during braking. The scaling is:

1 count = 1 second

The recommended setting is 8 seconds (8H).

CRNT ZERO E (:084)

CRNT ZERO E sets the threshold below which the load-side dc link current is assumed to be zero. It is used in the load-side overcurrent detection/protection routine. The scaling is:

1024 counts = 1 pu dc link current

The recommended setting is 0.06 pu or (40H).

CROSS TIE MARG E (:048)

CROSS TIE MARG E sets the level of source dc link voltage command at which the crosstie regulator becomes active. The crosstie regulator works to lower the flux setpoint in order to keep the source dc link voltage command from going any higher. CROSS TIE MARG E is calculated as a percentage of the 1 pu source dc link voltage command. The scaling is:

CROSS TIE MARG E = 256 * desired pu margin

The recommended setting is 6.25% (10H).

CROWBAR TEST MODE E (:33E)

CROWBAR TEST MODE E, when set to 1, causes the load-side bridge to fire rotating diametric pairs. Used during the installation set-up and check-out testing

CRSST INT GAIN E (:0E2)

CRSST INT GAIN E is the integral gain of the crosstie regulataor. The recommended setting is 012H

CRSST MAX LMT E (0CE)

CRSST MAX LMT E set the clamp for the maximum contribution from the cross tie regulator on the flux setpoint. The scaling is:

8192 counts = pu flux setpoint

The recommended setting is 0.4 pu (OCCDH).

CRSST PRP GAIN E (:0E0)

CRSST PRP GAIN E is the proportional gain of the crosstie regulataor. The recommended setting is 8H.

CRW BR TIME E (:080)

CRW BR TIME E controls the duration of the load-side crowbar (diametric firing) when the crowbar flag is set. The scaling (sec/count) is dependent upon the frequency at which the load side code cycles in the idle mode. Thus the scaling is:

CRW BR TIME E = Tcr * 61440 / LD FIX PERIOD E

where

LD FIX PERIOD E = EEPROM variable controlling load-side code cycle period
Tcr = desired crow bar time (sec)

The recommended setting is 2.5 seconds (382H for the typical load cycle period of ABH).

CURLL SRC E (:044)

CURLL SRC E clamps the minimum commanded value of dc-link current. It is chosen to maintain continuous dc-link current. The scaling is:

1024 counts = 1 pu current

For a system using standard scaling, set for 0.13 pu (85H).

CURRENT MTCH TIME E (:102)

CURRENT MTCH TIME E specifies the length of time that the motor current and inverter current must be matched during de-synchronization before the by-pass contactor is opened. The scaling is:

1 count = 16.7msec

The recommended setting is 130msec (8H).

DA1 SCL E (:02E)
DA2 SCL E (:030)

DA? SCL E controls the scaling of the variable being addressed by VAR DA? PTR E for output through the d/a converters on the HLCB board. This variable is calculated based on the maximum absolute value of the variable chosen to be output through the d/a:

DA? SCL E = log(base 2) [max abs value / 127]

NOTE: DA3 is usually used for Horse Power meter and DA3SC6 must be offset by +7.

DCLC CHOPA THETA E (:186)
DCLC CHOPB 2THETA E (:188)

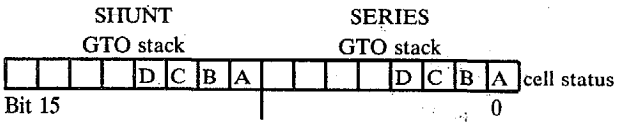
These variables control the harmonic elimination chopping patterns. The scaling is:

1 count = 0.3516 degrees

The recommended setting are:
DCLC CHOPA THETA E = 1CH
DCLC CHOPB 2THETA E = 22H

DCLC CONFIG E (:342)

DCLC CONFIG E's individual bits are used to mask-out GTO status for the various stack configurations used (3 or 4 cells per stack) (series and/or shunt stacks). A one in the bit(cell) location forces that cell status to "OK".



DCLC IDC COMP E (0) (:19E)
DCLC IDC COMP E (1) (:1A0)
DCLC IDC COMP E (2) (:1A2)
DCLC IDC COMP E (3) (:1A4)

DCLC IDC COMP E is an array of gains which are applied to the dc link current command. The gains are intended to compensate for dc link current that is shunted through the dc link commutator instead of going to the motor. The scaling is:

DCLC IDC COMP E (?) = 32 * (desired gain)

The recommended settings are:

DCLC IDC COMP E (0) = 20H
(gain = 1 for self com)
DCLC IDC COMP E (1) = 39H
(gain = 1.78 for dclc com elim 5th,7th)
DCLC IDC COMP E (2) = 33H
(gain = 1.59 for dclc com elim 5th)
DCLC IDC COMP E (3) = 20H
(gain = 1 for dclc com no harm elim)

DCLC TEST E (:196)

DCLC TEST E places the control into a test mode intended to be used to verify correct dc link commutator operation. In this mode, the load side interrupts are not activated and thus there is no inverter gating. The power circuit must be modified to place the series dclc arm in parallel with the shunt dclc arm. If there is

no series dclc arm, some resistive impedance (5 to 50 ohms) of appropriate wattage must be placed in parallel with the shunt arm. In this test mode, the source current regulator controls current and also selects the path for dc link current (shunt arm or series arm/resistive impedance). The current level is set by CURLL SRC E (:044). The conduction times of the shunt dclc and series dclc are controlled by TEST OFF CNT E (:198) and TEST ON CNT E (:19A). Setting DCLC TEST E to 1 and performing a hardware sem reset will place the control in the test mode. Setting DCLC TEST E to 0 and performing a hardware reset will return the control to normal operating mode.

DIAG LINE WIDTH E (:01E)

DIAG LINE WIDTH E defines the width of the error message test printed out by the diagnostic list function. The terminal line width should be entered directly (in Hex). The recommended setting is 050H.

DIAG LIST ENABLE E (:002)

The variable arms the diagnostic list. The options are

DIAG LIST ENABLE E = 00H disable the list
= 01H arms the list

DIAG LIST INVLIM MASK E (:01C)

DIAG LIST INVLIM MASK E controls the triggering of the diagnostic list based on the INVLIM word. Setting a bit to a "1" will cause a list to trigger when the corresponding bit in INVLIM gets set by software. Setting a bit to "0" will inhibit that invlim bit from triggering the list. Following is a list of the invlim bits.

bit	hex	flag name
0	1	zero current request
1	2	unused
2	4	unused
3	8	unused
4	10	load idle request
5	20	unused
6	40	source over current
7	80	unused
8	100	drive idle request
9	200	loss of hardware speed
A	400	load over current
B	800	source phase-lock-loop error
C	1000	load phase-lock-loop error
D	2000	crow bar
E	4000	unused
F	8000	stack overflow

DIAG LIST SITE E (:018)

DIAG LIST SITE E selects the software routines in which samples are taken for the diagnostic list. The variable is bit coded, setting the appropriate bit to a "1" selects that site. Multiple sites may be selected by setting more than one bit. These sites are available for diagnostic list sampling:

BIT 0 - after SOURCE CROSSOVER/PLL REG
1 - after FLUX REG
2 - after TCOM COND
3 - after SYNCHRONIZING REG

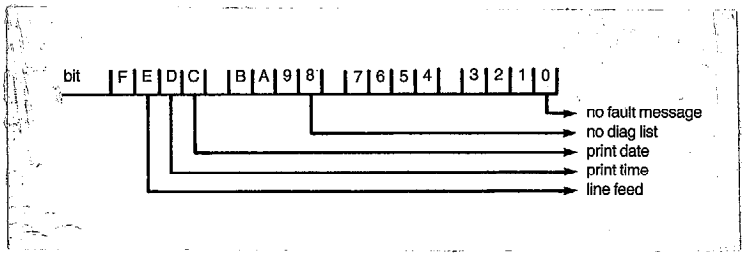
4 - after SOURCE IDLE (SACM00)
5 - after CURRENT REG (SACM10)
6 - unused
7 - unused

8 - after LOAD CROSSOVER/PLL REG
9 - after LOAD IDLE (LACM00)
A - after LOAD FIX FIRE (LACM10)
B - after LOAD RUN FIRE (LACM20)

C - unused
D - unused
E - unused
F - unused

DIAG PRINT STATUS E (:020)

DIAG PRINT STATUS E controls the automatic printout of the English language fault messages and the printout of the diagnostic list. The following options are available:



A one in the appropriate bit enables the function. (Note: Line feed refers to the insertion of a line feed at the end of each printed line. Some printers require this, others do not. Print time and date apply only to specially equipped printers.)

DIAG VAR PTR1 E (:004) (offset)
1 (:006) (base segment)
DIAG VAR PTR2 E (:008) (o)
2 (:00A) (b)
DIAG VAR PTR3 E (:00C) (o)
3 (:00E) (b)
DIAG VAR PTR4 E (:010) (o)
4 (:012) (b)
DIAG VAR PTR5 E (:014) (o)
5 (:016) (b)

DIAG VAR PTR? E is used with the diagnostic list function. The address of the variable to be monitored is loaded into PTR?. For further information see section Diagnostics and Troubleshooting.

DIAG VAR2 MAX LMT E (:01A)

DIAG VAR2 MAX LMT E is used with the diagnostic list function. It's value is essentially a trigger level. When the value of the variable monitord by DIAG VAR PTR2 E exceeds this value, the diagnostic list is triggered. Setting the level to FFFFH defeats this triggering method.

DSYNC IRATE LIM E (:100)

DSYNC IRATE LIM E controls the rate limit on current command as the control tries to match inverter current to motor current during de-synchronization. The scaling is:

DSYNC IRATE LIM E = rate limit (pu/sec) * 136

The recommended setting is 0.11pu/sec (0FH).

EEPROM RESERVED E (:000)

EEPROM RESERVED E is a reserved location marking the start of the eeprom variable memory space. It should be set to 0H.

FAEAP INT GAIN E (: 0FE)

FEAP GAIN E is the integral gain of the fire as early as possible regulator. The recommended setting is 0H.

FAEAP MIN LMT E (:17E)

FAEAP MIN LMT E sets the clamp for the maximum contribution from the fire as early regulator on the flux setpoint. The scaling is:

8192 counts = 1 pu flux command

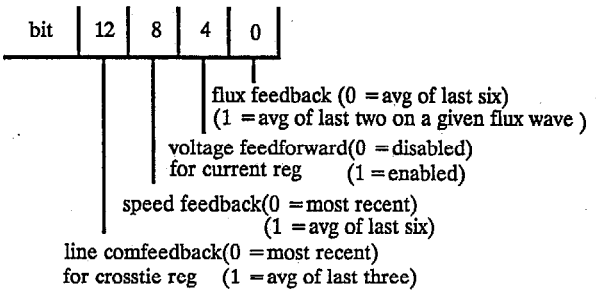
The recommended setting is 0h.

FAEAP NCLMP E (:1B0)

FAEAP NCLMP E ...

FBK MODE E (: 1AA)

FBK MODE E is a bit coded variable which controls the feedback sampling format for several of the regulators. The bit assignments are:



The recommended setting is 1110H.

FLUX MAG TOL E (:08C)

FLUX MAG TOL E sets the minimum limit to which flux is allowed to dip in a regenerative condition before the drive is shut down. The scaling is

1024 counts = 1 pu flux

The recommended setting is at 0.25pu (100H).

FLX BRK1 E (:0D0)
FLX BRK2 E (:0D2)
FLX BRK3 E (:0D4)

FLX BRK? E controls the break points on the flux program vs speed command profile (flux vs speed profile - sh.1EA of system elem.). The scaling is:

683 counts = 1 pu speed command

The recommanded setting are 0.0pu (0H), 0.8pu (222H), and 1.pu (2ABH).

FLX DELAY TIME E (:1AE)

FLX DELAY TIME E defines the time allowed for the load side flux integrators to settle before an attemt is made to run the load side phase lock loop. The scaling is:

1 count = LD FIX PERIOD E / 61440 (sec)

The recommended setting is 0.7 sec (0FFH).

FLX DROOP MAX E (:094)

FLX DROOP MAX E sets the maximum amount that the flux setpoint will get depressed during a light load condition. The scaling is:

512 counts = 1 pu depression

The recommended setting is 0.3 pu (9AH).

FLX DROOP TCOM E (:096)
FLX DROOP TCOM E sets the light load torque command threshold below which the flux setpoint gets depressed. The scaling is:

1024 counts = 1 pu torque command

The recommended setting is 0.5 pu torque command (200H)

FLX ERR FIL GAIN E (:1B6)
FLX ERR FIL GAIN E set the break frequency on a lowpass filter applied to flux error. This variable is calculatedusing the following equation:

FLX ERR FIL GAIN E = 32 / (1 + 120/Wc)

where Wc = desired break freq (rad) (max = 50 radians)). The suggested setting is 0H.

FLX INT GAIN E (:ODC)
FLX INT GAIN E is the gain of the integral portion of the flux regulator. It is calculated by:

$$\text{FLX INT GAIN E} = 2.133 * \left[\frac{\omega}{1 + j\omega \frac{(L1 + L2)}{R2}} \right] \quad \omega = Wc$$

where L1, L2, and R2 are the motor parameters and Wc is the desired crossover of the regulator. The recommended crossover is 5 radians.

FLX LEV1 E (:0D6)
FLX LEV2 E (:0D8)
FLX LEV3 E (:0DA)

These variables define flux levels in the flux program vs torque or speed command profile (see Fig 1). The scaling for flux is:

1024 counts = 1 pu flux

The recommended settings when using the speed command profile are 0.6pu(266H), 1.15pu(499H), and 1.0pu(400H) (note maximum (max-min) =0.99pu)

FLX PRP GAIN E (:0DE)
FLX PRP GAIN E is the gain of the proportional portion of the flux regulator. It is calculated by:

FLX PRP GAIN E = 1.875 * FLX INT GAIN E * T

where T = motor time constant (Lm+L2)/R2

FLX RATE LIM E (:OE4)
FLX RATE LIM E controls the rate at which the flux setpoint is ramped when trying to match motor voltage to line voltage during synchronization. The scaling is:

FLX RATE LIM E = pu flux per second * 68.2

The recommended setting is .23 pu/sec (10H).

GTO COM ANGA E (:0B0)
GTO COM ANGA E clamps the minimum shunt DCLC conduction angle while chopping to eliminate the 5th and 7th harmonic. The scaling is:

1 count = 0.3516 degrees

The recommended setting is 5 degrees (0EH).

GTO COM TIME BC E (:18E)
GTO COM TIME BC E clamps the minimum shunt DCLC conduction time while chopping to eliminate the 5th and chopping just to commutate. The scaling is:

1 count = 0.25 mSec

The recommended setting is 1 mSec (0F9AH).

GTO E (:0AE)
GTO E is presently not used. Set to 0H.

GTO MIN ON CLMP E (:0CC)
GTO MIN ON CLMP E is a minimum clamp placed on the shunt dclc conduction time. The scaling is

1 count = 0.25 mSec

The recommended setting is 1 mSec (0F9AH).

GTO TUF ANGA E (:19C)
GTO TUF ANGA E sets the shunt dclc conduction width in degrees for one of the chops made in the 5th/7th harmonic elimination chopping mode. The scaling is:

1 count = 0.3516 degrees

The recommended setting is 9.8 degrees (1CH).

HI SCALE LIMIT E (:064)
HI SCALE LIMIT E is the minimum frequency (maximum pllc load) clamp on the output of the load-side phase-lock-loop regulator. The scaling is:

HI SCALE LIMIT E = 57600 / minimum pll freq (hz)

The recommended minimum frequency is approximately 1.2Hz (07BB8DH).

ICAP FLTR GAIN E(:0F4)
ICAP FLTR GAIN E controls the break frequency of the low pass filter on the calculated capacitor current. This variable is calculated using the following equation:

ICAP FLTR GAIN E = 32 / (1 + 120/Wc)

where Wc = desired break freq (rad) (max = 50 radians)Suggested break frequency is 12 radians (3H).

IFB MODE E (:1B2)
IFB MODE E is presenty not used.
The recommended setting is 0H.

I LINK ZERO E (:042)
I LINK ZERO E is the value below which the source dc link current is assumed to be zero. This variable is used in the protectives and also when the drive is being turned off. The scaling is:

1024 counts = 1 PU dc link current

Recommended setting is 0.094 pu (60H).

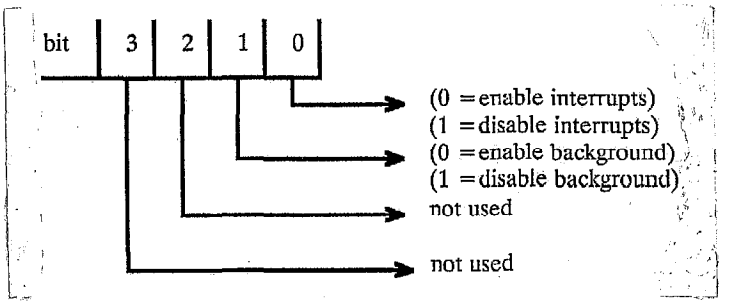
I MOTOR SCL E (:104)
I MOTOR SCL E is a scale factor applied to the motor current signal read through the a/d converter. This signal is used by the synchronizing funtion. The scaling is:

I MOTOR SCL E = 16 * Im' / 5

where

Im' = scaled motor current signal(volts) @ 1pu motor current

IO TEST MODE E (:034)
IO TEST LD E controls the enabling of the interrupts. It is useful when testing the I/O ports, etc. The available options are:



IQ COMP SPD E (:192)
IQ COMP SPD E set the motor speed above which iq star is calculated based on flux feedback instead of flux setpoint. The scaling is:

683 counts = 1 pu speed (60Hz)

The recommended setting is 0Hz (0H).

IREG FIX ALPHA1 E (:04C)
IREG FIX ALPHA2 E (:04E)
IREG FIX ALPHA? E are used to control the first two source-side firings out of a forced commutation. These first two firings are basically open loop with a fixed firing angle being commanded. The recommended settings for these variables are

IREG FIX ALPHA1 E = 800H
IREG FIX ALPHA2 E = 600H

I REG G1 E (:050)
I REG G1 E is the proportional gain for the current regulator. It is calculated using:

I REG G1 E = 2.7 * I REG G2 E

I REG G2 E (:052)
I REG G2 E is the integral gain for the current regulator. It is calculated using:

I REG G2 E = 23.7 * Idc * Wc * L / Vll

where

Idc = rated inverter dc link current
Wc = desired regulator crossover (radians)
L = inductance in commutating path (henries)
= dc link reactor + 2*(src xfrmer comm react)

(4*(src xfrmer comm react) if 12-pulse series)
Vll = rate line to line source voltage rms (2*Vll if 12-pulse series)

The recommended crossover is 170 radians.

ISTAR CLMP E (:046)
ISTAR CLMP E is a clamp applied to the dc link current command formed from the real and reactive components of current command. Scaling is:

1024 counts = 1 pu dc link current

This clamp is normally set to 1 pu (400H) unless overload performance is required.

ISTAR CLMP1 E (:1A8)
ISTAR CLMP1 E is the clamp applied to the dc link current command during dc link commuation. The scaling is:

1024 counts = 1 pu dc link current

This clamp is normally set lower than ISTAR CLMP E. The recommended setting is 0.5 pu (200H).

IX MIN CLMP E (:0EE)
IX MIN CLMP E sets the minimum clamp on the flux producing component of current while in light load operation. The scaling is

1024 counts = 1 pu command

The recommended setting is 0.12pu (80H).

LD ANG MAX ADV E (:0B8)
LD ANG MAX ADV E is a hard clamp on the maximum advance of the load-side firing angle. It is calculated by:

LD ANG MAX ADV E = (max adv alpha - 180) / 0.3516

where max adv is measured from the alpha=0 angle. The recommended setting is alpha = 15 degrees (-1D5H = FE2BH)

LD ANG MAX RET E (:0BA)
LD ANG MAX RET E is a hard clamp on the maximum retard of the load-side firing angle when chopping to eliminate the 5th harmonic. It is calculated by:

LD ANG MAX RET E = (max ret alpha - 180) / 0.3516

where max ret ang is measured from the alpha=0 angle. The recommended angle is 216 degrees (068H) for the shunt configuration. For the series/shunt configuration, the recommended angle is 270 degrees (0FFH).

LD ANG MAX RETA E (:0CA)
LD ANG MAX RETA E is a hard clamp on the maximum retard of the load-side firing angle when in the chopping mode to eliminate 5th and 7th harmonics. It is calculated by:

LD ANG MAX RET E = (max ret alpha - 180) / 0.3516

where max ret ang is measured from the alpha=0 angle. The recommended angle is 216 degrees (068H) for the shunt configuration. For the series/shunt configuration, the recommended angle is 270 degrees (0FFH).

LD DIVERT ANG E (:0B4)
LD DIVERT ANG E defines the load-side firing angle where the diverter is activated to accomplish commutation. The diverter is activated when the load-side firing angle is greater than the angle defined here. The scaling is:

LD DIVERT ANG E = (divert ang - 180) / 0.3516

where divert ang is measured from the alpha=0 angle. The recommended angle is 165 degrees (-02AH) = FFD6.

LD DIVERT HYST E (:0B6)
LD DIVERT HYST E sets the amount of hysterysis in the decision to use the dc link commutation circuit. The scaling is:

1 count = 0.3516 degrees

The recommended setting is 10 degrees (1CH).

LD FAST RATE LIM E (:0BE)
LD FAST RATE LIM E is a fast load angle rate limit which is normally in effect. The scaling is:

1 count = 0.3516 degrees / firing

The recommended setting is 30 degrees per firing (55H).

LD FIX PERIOD E (:07C)

LD FIX PERIOD E controls the repetition rate of the load-side firing interrupt while in idle and fixfire modes. The scaling is:

1 count = 0.3516 degrees

The recommended setting is every 60 degrees (0ABH).

LD INIT ANG E (:OB2)

LD INIT ANG E controls the angle of the first firing in runfire. The scaling is

LD INIT ANG E = (alpha - 180) / 0.3516

where alpha is the desired firing angle measured from alpha=0. The recommended angle is 216 degrees (067H).

LD IX COMP E (:1B8)

LD IX COMP E tells the selftest software if load side ix flux reconstruction is in use:

= 0H if no ix comp
= 1H if ix comp

LD SLOW RATE LIM E (:0BC)

LD SLOW RATE LIM E is a slower load angle rate limit which is imposed when a load firing angle quadrant change takes place. The scaling is:

1 count = 0.3516 degrees / firing

The recommended setting is 5 degrees per firing (0EH).

LD TRANS SFLAG LIM E (:07E)

LD TRANS SFLAG LIM E sets the maximum number of out-of-tolerance passes of the load-side phase-lock-loop during transitions from open to closed loop load-side angle control. The load pll invlim bit is set if this number is exceeded. The recommended setting is -30H(= FFD0H).

LEAD LAG ADJUST E (:134)

LEAD LAG ADJUST E is used to select the time constant of the lead/lag filter applied to the speed feedback signal. This time constant is coordinated with the speed regulator crossover in order to achieve critical damping. The following options are available:

spd reg crossover	lead Tc	LEAD LAG ADJUST E
--	filter bypassed	0H
80.00	0.0125	1H
34.25	0.0292	2H
16.00	0.0625	3H
7.74	0.1292	4H
3.80	0.2629	5H
1.89	0.5300	6H
0.94	1.0640	7H
0.47	2.1300	8H

Typically use 07H for a 1 rad speed regulator.

LEAD LAG ADJUST ALT E (:132)

LEAD LAG ADJUST ALT E is used to select the time constant of the lead/lag filter applied to the speed feedback signal when the control is trying to synchronize. This time constant is coordinated with the speed regulator crossover when trying to synchronize (ie. SPD G1 ALT E and SPD G2 ALT E) in order to achieve critical damping. It is selected from the same table as LEAD LAG ADJUST E. Typically use 04H.

LINE IMP E (:054)

LINE IMP E is used in the calculation of the source-side firing angle retard limit as a function of dc-link current. It is dependent on the source impedance and is calculated:

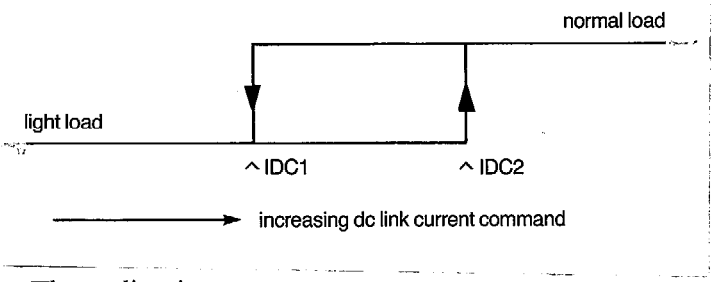
LINE IMP E = (63.75 * X)/VLL

where

X = pu impedance of the source
VLL = minimum pu source line voltage (typically 0.7pu)

LIT LOD IDC1 E (:0E6)
LIT LOD IDC2 E (:0E8)

LIT LOD IDC? E sets the level and hysterisis for the determination of light load conditions based on the level of commanded dc link current.

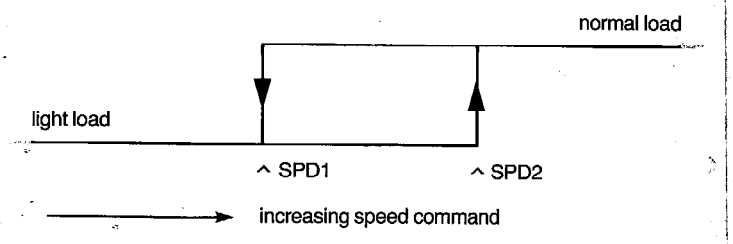


The scaling is:

1024 counts = 1 pu dc link current command

The recommended settings are:
IDC1 = 0.13pu (82H)
IDC2 = 0.22pu (E0H)

LIT LOD SPD1 E (:OEA)
LIT LOD SPD2 E (:0EC)
LIT LOD SPD? E sets the level and hysterisis for the determination of light load conditions based on the level of commanded speed.



The scaling is:

683 counts = 1 pu speed command (60Hz)

The recommended settings are:

SPD1 = 40Hz (1C7H)
SPD2 = 42Hz (1DEH)

LO SCALE LIMIT E (:066)
LO SCALE LIMIT E is the maximum frequency (minimum pllc load) clamp on the output of the load-side phase-lock-loop regulator. The scaling is:

LO SCALE LIMIT E = 57600 / maximum pll freq (hz)

The recommended maximum frequency is 87hz (298H).

L PLL LIM E (:068)
L PLL LIM E controls the allowed number of passes through the load phase-lock-loop with the phase-lock-loop error out-of-tolerance. It can be calculated by

L PLL LIM E = (N - 1) * 6

where N is the number of consecutive passes with the pll error out of tolerance. The recommended setting is N=86 or L PLL LIM E = 200H.

MANUAL FIX FIRE E (:098)
MANUAL FIX FIRE E places the control in a manually controlled fixed freqeuncy firing mode instead of the normal fix fire starting sequence. The inverter frequency and current level are manually controlled by the eeprom variables MANUAL FREQ E and MANUAL TCOM E. The options are:

= 00H - normal fix fire startup sequence
= 01H - manual frequency and current control

The recommended setting is for normal startup. For testing, the manual mode may be selected.

MANUAL FREQ E (:09C)
MANUAL FREQ E sets the inverter frequency when the control is in fix fire and the open loop mode has been set to 02H (manual mode). The scaling is:

MANUAL FREQ E = 6 * frequency (hz)

This option is usefull in tuning up the motor starting parameters and should normally be set to 2 hz (0CH).

MANUAL TCOM E (:09A)
MANUAL ICOM E sets the motor current level when the control is in fix fire and the OPEN LOOP MODE E has been set to 02H (manual mode). The scaling is:

1024 counts = 1 pu current

This option is useful in tuning up the motor starting parameters and should normally be set to 0.2 pu (0CDH).

MAX GTO ON ANGA E (:0C8)
MAX GTO ON ANG A E clamps the maximum shunt DCLC conduction angle in degrees while chopping to eliminate 5th and 7th harmonics. The scaling is:

1 count = 0.3516 degrees

The recommended setting is 14 degrees (28H).

MAX GTO ON ANGB E (:0C6)
MAX GTO ON ANGB E clamps the maximum shunt DCLC conduction angle in degrees while chopping to eliminate the 5th harmonic. The scaling is:

1 count = 0.3516 degrees

The recommended setting is 40 degrees (72H).

MAX GTO ON ANGC E (:190)
MAX GTO ON ANGC E clamps the maximum shunt DCLC conduction angle in degrees while chopping only to commutate. The scaling is:

1 count = 0.3516 degrees

The recommended setting is 50 degrees (8EH).

MAX OPEN LOOP FLX E (:08E)
MAX OPEN LOOP FLX E is presently not used. Set to 7FFFH

MIN PLL HZ LD E (:06A)
MIN PLL HZ LD E controls the frequency at which the load-side phase-lock-loop transitions from open loop prediction mode to closed lock mode. The scaling is:

MIN PLL HZ LD E = 7200 / freq (hz)

The recommended frequency is 3.75 Hz (780H).

MIN PRTC HZ E (:082)
MIN PRTC HZ E sets the minimum frequency at which the load-side cell state sensors are monitored. The scaling is:

683 counts = 1 pu speed

The recommended setting is 0.42 pu speed (122H).

MIN START FLUX E (:090)
MIN START FLUX E is a flux threshold used by the control when trying to start the motor from standstill. The control fires the inverter at some low frequency until the flux level exceeds MIN START FLUX E. At this point the control assumes the motor is spinning at some slip freqency so it begins to ramp the frequency up. The scaling is:

1024 counts = 1 pu flux

The recommended setting is 0.375 pu (180H).

MIN TRANS FREQ E (:092)
MIN TRANS FREQ E sets then minimum frequency at which the control can transition from fix fire to run fire when starting from standstill. The scaling is:

MIN TRANS FREQ E = 6 * freq (Hz)

For shunt only configurations the setting is 5 Hz (1EH). For series/shunt configurations the setting is 8 Hz (30H).

MOT CAP E (:0F2)
MOT CAP E defines the size of the motor capacitors. The scaling is:

MOT CAP E = 13558.4 * C * Zb * (Idcm / Idci)

where

C = motor capacitor size (μF)
Zb = base ohms for motor
(if wye cap use wye Zb, if delta cap use delta Zb)
(= KVA(per phase)/El-n**2)
Idcm = rated motor amps dc
Idci = rated inverter amps dc

NO FLUX THRESH E (:08A)
NO FLUX THRESH E sets the flux level below which the control no longer tries to run the load-side phase-lock-loop when the motor is not running (ie. idle mode). The scaling is:

1024 counts = 1 pu flux

The recommended setting is 0.1 pu flux (66H).

OVER CUR LD E (:086)
OVER CUR LD E sets the trip threshold for the load software detected overcurrent function. The scaling is:

1024 counts = 1 pu dc link current

The recommended setting is 1.5 pu current (600H).

OVER CURRENT CNT L E (:088)
OVER CURRENT CNT L E controls the number of consecutive load side software detected overcurrents that are allowed before the overcurrent fault is set and the drive is tripped. It can be calculated:

OVER CURRENT CNT L E = (N - 1) * 6

where N is the number of consecutive overcurrents. Presently OVER CURRENT CNT L E is set at N=2 (5H).

OVER CURRENT CNT S E (:058)
OVER CURRENT CNT S E controls the number of consecutive source side software detected overcurrents that are allowed before the overcurrent fault is set and the drive is tripped. It can be calculated:

OVER CURRENT CNT S E = (N - 1) * 6

where N is the number of consecutive overcurrents. Presently OVER CURRENT CNT S E is set at N=4 (10H).

OVER CUR SRC E (:05A)
OVER CUR SRC E sets the trip level for the source software dectected overcurrent function. The scaling for current is:

1024 counts = 1 pu dc link current

The recommended setting is 1.5 pu current (600H).

OVER CUR SRC1 E (:1A6)
OVER CUR SRC1 E sets the trip level for the source software dectected overcurrent function when the drive is operating in the dc link commutated mode. The scaling for current is:

1024 counts = 1 pu dc link current

The recommended setting is 0.75 pu (300H).

PHS PRP GAIN E (:106)
PHS PRP GAIN E is the gain of the synchronizing phase angle regulator. The recommended setting is 0AH.

PHS REG CLMP E (:108)
PHS REG CLMP E clamps the output of the phase angle regulator which feeds into the speed regulator. The scaling is for this clamp is:

11,656 counts = 1Hz

It is recommended that the phase regulator contribution to the speed regulator be clamped to approximately 0.46Hz (1500H).

PI CLMP RATE LIM E (:0F0)
PI CLMP RATE LIM E controls the rate limit on the ramp of the minimum clamp on the output of the proportional + integral portion of the flux during light load operation. The scaling is:

1 count = 0.0293 pu/sec

The recommended setting is 1pu/sec (20H).

PLL ERR LMT E (:06C)
PLL ERR LMT E is the bipolar clamp placed on the load-side phase-lock-loop error signal. The scaling is:

1 count = 0.3516 degrees

The recommended setting is 11.25 degrees (1FH).

PLL ERROR TOL L E (:06E)
PLL ERROR TOL L E sets the allowable tolerance for the load side phase-locked-loop error. load-side SFLAG gets set whenever the load-side PLL error exceeds this tolerance. The scaling for PLL error is:

1 count = 0.3516 degrees

The load PLL error tolerance should be set for 22.5 degrees (3FH).

PLL ERROR TOL S E (:03A)
PLL ERROR TOL S E sets the allowable tolerance for the source side phase locked loop error. source-side SFLAG gets set whenever the source pll error exceeds this tolerance. The scaling for pll error is:

1 count = 0.3516 degrees

The source pll error tolerance should be set for 22.5 degrees (3FH).

PLL G1 SRC E (:036)
PLL G1 SRC E sets the integral gain for the source-side phase locked loop regulator. It should be set to 1EH.

PLL G2 SRC E (:038)
PLL G2 SRC E sets the proportional gain for the source-side phase locked loop regulator. It should be

set to 1DH.

PLL MODE ERR TOL E (:070)

PLL MODE ERR TOL E is an error threshold used in the load-side phase-lock-loop. The phase-lock-loop error must be less than this threshold before a transition from one frequency calculating region to another may be made (refer to R0 UP E). The scaling is:

1 count = 0.3516 degrees

The recommended setting is 45 degrees (7FH).

PROT DIS SRC E (:05C)

PROT DIS SRC E controls the source-side protectives. The options are:

= 0FFH disables protectives
= 000H enables protectives (normal operation)

QC CUR BALENCE TOL E (:302)

QC CUR BALENCE TOL E is not used.

QC CUR LOWER LIMIT E (:306)

QC CUR LOWER LIMIT E is used during the GTO OFF test. The test man is requested to enter the 1 pu dc link durrent of the drive. The entered value is compared to QC CUR LOWER LIMIT E. If it is less, the value is not accepted and the test man is requested to enter a new value. QC CUR LOWER LIMIT E is in amps with a default value of 100.

QC CUR UPPER LIMIT E (:304)

QC CUR UPPER LIMIT E is used during the GTO OFF test. The test man is requested to enter the 1 pu dc link current of the drive. The entered value is compared to QC CUR UPPER LIMIT E. If it is greater, the value is not accepted and the test man is requested to enter a new value. QC CUR UPPER LIMIT E is in amps with a default value of 1250.

QC CUR VALUE TOL E (:300)

QC CUR VALUE TOL E is used during GTO OFF test. This variable is the allowable tolerance of the actual current reading as compared to the current commanded. The value is stored in percent (i.e. if a 5% tolerance is desired, QC CUR VALUE TOL E is set to 5). The following formula is used to calculate the actual tolerance in amps:

current tolerance (amps) = (commanded current(amps) * QC CURRENT VALUE TOL E) / 100

The current reading (IDC SRC) is compared to the commanded current value (QC IDC STAR) plus or minus the calculated current tolerance. QC CUR VALUE TOL E has a default value of 5.

RAMP UP E (:0A6)

RAMP UP E sets the rate at which the frequency ramps up from low frequency when the control is starting the motor from standstill. The scaling is:

RAMP UP E = 1 / accel rate (pu/sec)

The recommended acceleration rate is 0.0156pu/sec (40H). Note that this acceleration rate can be increased if starting current (RMP UP TCOM E) is also increased in direct proportion.

RTARD MARG SRC E (:056)

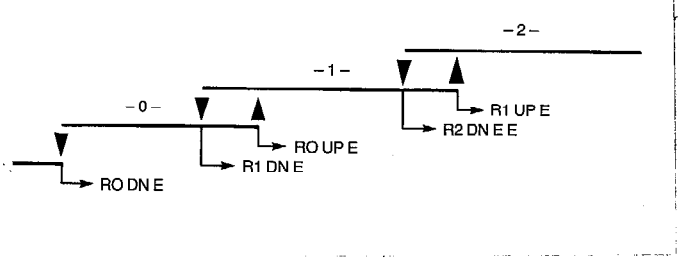
RTART MARG SRC E sets the fixed retard margin for the source side when its inverting. It is calculated:

RTARD MARG SRC E = 256 * [1 + cos(180 - ALPHA)]

where ALPHA is the desired retard margin in degrees. Normal margin is 17 degrees (0CH).

RO DN E (:072)
R1 DN E (:074)
R2 DN E (:076)
R0 UP E (:078)
R1 UP E (:07A)

R? UP E / R? DN E are used by the load-side phase-lock-loop to establish 3 different regions of pll operation. The boundaries for these regions are based on motor frequency. Which region the control is in dictates the number of times per cycle that the pll regulator calculation is run. The regions of operation can be graphicly illustrated:



region 0 - calc 6 times/cycle
region 1 - calc 3 times/cycle

region 2 - calc 2 times/cycle

The scaling for the frequency thresholds is:

frequency (hz) = 7200 / counts

The recommended settings are:

R0 DN E - 2hz (0E10H)

R1 DN E - 14hz (202H)
R0 UP E - 16hz (1C2H)

R2 DN E - 80hz (5AH)
R1 UP E - 85hz (55H)

SERIES SHUNT E (:1B4)
SERIES SHUNT E configures the software to match the dc link commutator configuration:

= 0 for shunt only
= 1 for series/shunt combination

SLIP FILTER PERIOD E (:136)
SLIP FILTER PERIOD E is presently not used.

Set to 0H

SLIP GAIN E (:138)
SLIP GAIN E is used to calibrate the slip calculation. The scaling is:

SLIP GAIN E = 683 * full load slip

SOURCE SLAVE BRIDGE E (:32A)
SOURCE SLAVE BRIDGE E determines whether the source slave firing leads or lags the source master bridge.
= 1 - slave lags master by 30 degrees
= 0 - slave leads master by 30 degrees

SPD BRK HYST E (:182)
SPD BRK HYST E places the hysteresis on the speed at which the control transitions from chopping to eliminate the 5th and 7th harmonic to chopping to eliminate only the 5th harmonic to chopping soley for commutation purposes. The scaling is:

683 counts = 1pu speed

The recommended setting is 2Hz (17H). This would place one of the thresholds 2Hz above the threshold set by SPD BRK CHOPA H and SPD BRK CHOPBC H.

SPD BRK CLOSE REG E (:184)
SPD BRK CLOSE REG E sets the speed above which the speed and flux regulators are closed. The scaling is:

683 counts = 1pu speed

For shunt configurations the setting is 7 hz (4FH). For series/shunt configurations the setting is 10 hz (71H).

SPD BRK SRC COM E (:0C0)
SPD BRK SRC COM E sets the speed at which the transition from source commutation to DCLC commutation is made during a start up sequence in shunt configuration. The scaling is:

683 counts = 1pu speed

The recommended setting is 8Hz (5BH).

SPD BRK CHOPAB E (:0C4)
SPD BRK CHOPAB E sets the speed at which the control transitions from chopping to eliminate the 5th and 7th harmonic to chopping to eliminate only the 5th harmonic. The scaling is:

683 counts = 1 pu speed

The recommended setting is 22Hz (0FAH).

SPD BRK CHOPBC E (:18C)
SPD BRK CHOPBC E sets the speed at which the control transitions from chopping to eliminate the 5th harmonic to chopping only to commutate. The scaling is:

683 counts = 1 pu speed

The recommended setting is 28Hz (13EH).

SPD BRK CHOPCA E (:1B2)
SPD BRK CHOPCA E sets the low speed at which the control trasnitions from 120 degree square output to chopping to eliminate the 5th and 7th harmonic. The scaling is:

683 counts = 1 pu speed

The recommended setting is 14Hz (09FH)

SPD ERR FILTER E (:13A)

SPD ERR FILTER E controls the filter applied to the speed error signal. The options are:

1 = 00H - Filter is bypassed
2 = 01H - Filter with double break at 17 radians
3 = 02H - Filter with double break at 40 radians

The recommended setting is 02H.

SPD G1 E (:13C)

SPD G1 E is the proportional gain of the speed regulator. It be calculated:

SPD G1 E = 3.32e-6 * N^2 * WK^2 / (HP * Tr)

where

N = rated motor rpm
WK^2 = total drive WK^2 at the motor shaft
HP = total rated drive Horsepower
Tr = desired speed regulator response time

It is recommended that the drive be tuned for 1 radian response (Tr = 1 sec).

SPD G1 ALT E (:140)

SPD G1 ALT E is the proportional gain of the speed regulator when the control is trying to synchronize. It is calculated the same as SPD G1 E. It is recommended that the drive be tuned for 4 radian response when trying to synchronize (Tr = 0.25 sec).

SPD G2 E (:13E)

G2 E is the integral gain of the speed regulator. It can be calculated:

SPD G2 E =3.54e-6 * N^2 * WK^2 / (HP * Tr^2)

where

N = rated motor rpm
WK^2 = total drive WK^2 at the motor shaft
HP = total rated drive horsepower
Tr = desired speed regulator response time

It is recommended that the drive be tuned for 1 radian response (Tr = 1 sec).

SPD G2 ALT E (:142)

SPD G2 ALT E is the integral gain of the speed regulator when the control is trying to synchronize. It is calculated the same as SPD G2 E. It is recommended that the drive be tuned for 4 radian response when trying to synchronize (Tr = 0.25 sec).

SPD RATE ACCL 6P E (0) (:144)
SPD RATE ACCL 6P E (1) (:146)
SPD RATE ACCL 6P E (2) (:148)
SPD RATE ACCL 6P E (3) (:14A)
SPD RATE ACCL 6P E (4) (:14C)
SPD RATE ACCL 6P E (5) (:14E)

SPD RATE ACCL 6P E (?) is an array of variables that set the acceleration rate limit on the speed regulator setpoint while in six pulse operation. Six different rate limits are available and are selected by the background program. The selection is based on the speed of the motor. The scaling is:

SPD RATE ACCL 6P E (?) = rate limit (pu/sec) * 186,499

SPD RATE ACCL 12P E (0) (:150)
SPD RATE ACCL 12P E (1) (:152)
SPD RATE ACCL 12P E (2) (:154)
SPD RATE ACCL 12P E (3) (:156)
SPD RATE ACCL 12P E (4) (:158)
SPD RATE ACCL 12P E (5) (:15A)

SPD RATE ACCL 12P E (?) is an array of variables that set the acceleleration rate limit on the speed regulator setpoint while in twelve pulse operation. Six different rate limits are available and are selected by the background program. The selection is based on the speed of the motor. The scaling is:

SPD RATE ACCL 12P E (?) = rate limit (pu/sec) * 186,499

SPD RATE DECEL 6P E (0) (:15C)
SPD RATE DECEL 6P E (1) (:15E)
SPD RATE DECEL 6P E (2) (:160)
SPD RATE DECEL 6P E (3) (:162)
SPD RATE DECEL 6P E (4) (:164)
SPD RATE DECEL 6P E (5) (:166)

SPD RATE DECEL 6P E (?) is an array of variables that set the deceleration rate limit on the speed regulator setpoint while in six pulse operation. Six different rate limits are available and are selected by the background program. The selection is based on the speed of the motor. The scaling is:

SPD RATE DECEL 6P E (?) = rate limit (pu/sec)

*186,499

SPD RATE DECEL 12P E (0) (:168)
SPD RATE DECEL 12P E (1) (:16A)
SPD RATE DECEL 12P E (2) (:16C)
SPD RATE DECEL 12P E (3) (:16E)
SPD RATE DECEL 12P E (4) (:170)
SPD RATE DECEL 12P E (5) (:172)

SPD RATE DECEL 12P E (?) is an array of variables that set the deceleration rate limit on the speed regulator setpoint while in twelve pulse operation. Six different rate limits are available and are selected by the background program. The selection is based on the speed of the motor. The scaling is:

SPD RATE DECEL 12P E (?) = rate limit (pu/sec) * 186,499

SPD REF BREAK POINT E (0) (:11E)
SPD REF BREAK POINT E (1) (:120)
SPD REF BREAK POINT E (2) (:122)
SPD REF BREAK POINT E (3) (:124)
SPD REF BREAK POINT E (4) (:126)

SPD REG BREAK POINT E (?) sets the breaks in the speed rate profile where SPD REG BREAK POINT E (0) sets the lowest speed break and SPD REG BREAK POINT E (4) sets the highest speed break. The scaling is:

683 counts = 1 pu speed (60 hz)

SPD REF MAX LMT E (:12A)

SPD REF MAX LMT E sets the maximum scaled speed reference in speed setpoint mode 01H (0-10 volt reference). The scaling is

683 counts = 1 pu speed (60 hz)

SPD REF MIN LMT E (:12C)

SPD REF MIN LMT E sets the minimum scaled speed reference in speed setpoint mode 01H (0-10 volt reference). The scaling is:

683 counts = 1 pu speed (60 hz)

SPD REF MODE E (:128)

SPD REF MODE E defines which speed reference is used to determine the speed setpoint:

SPD REF MODE E = 0 : 0-10 volt from customer
1 : 4-20 mA from customer
2 : -10 to +10 volt from customer
3 : 0-10 volt local reference
4 : raise and lower spd request
5 : turning speed mode

SPD REF PROCESS MAX E (:12E)

SPD REF PROCESS MAX E controls the maximum scaled speed reference in speed setpoint mode 00 (4-20 mA reference). It also sets the minimum speed when in speed setpoint mode 05 (down-contact reference). The scaling is:

683 counts = 1 pu speed (60 hz)

SPD REF PROCESS MIN E (:130)

SPD REF PROCESS MIN E controls the minimum scaled speed reference in speed setpoint mode 00 (4-20 mA reference). It also sets the minimum speed when in speed setpoint mode 05 (down-contact reference). The scaling is:

683 counts = 1 pu speed (60 hz)

SPD REF TURNING GEAR E (:344)

SPD REF TURNING GEAR E is the variable setting the fixed speed when spd ref mode is set to 05H. The scaling is:

683 counts = 1 pu speed (60 hz)

SPD REG ERR TOL E (:174)

SPD REG ERR TOL E defines the maximum speed error allowed before a loss of speed control alarm is declared. The scaling is:

683 counts = 1 pu speed (60 hz)

It is recommended the error tolerance be set to 0.2 pu (88H).

SPD REG REF MAX LMT E (:32C)

SPD REG REF MAX LMT E is the macimum speed reference clamp. The scaling is:

683 counts = 1 pu speed (60 hz)

SPD REG REF MIN LMT E (:32E)
SPD REG REF MIN LMT E is the minimum speed reference clamp. The scaling is:

683 counts = 1 pu speed (60 hz)

SRC CONFIG E (:194)
SRC CONFIG E matches the software to the configuration of the source side bridge(s). The available choices are:

00H - 6 pulse channel A
01H - 6 pulse channel B (qc factory test only)
02H - 12 pulse channel B lags
03H - 12 pulse channel B leads

SRC IDLE PERIOD E (:03E)
SRC IDLE PERIOD E controls the repetition rate of the source-side firing interrupt while in idle mode. The scaling is:

1 count = 0.3516 degrees

The recommended setting is 25 degrees (047H).

SRC PLL LIM E (:03C)
SRC PLL LIM E controls the number of consecutive source-side phase-lock-loop regulator calculations with the pll error out-of-tolerance that are allowed before a source phase look loop fault is declared. It is calculated using:

SRC PLL LIM E = (N - 1) * 6

where N is the number of consecutive out-of-tolerance calculations. It is recommended to set N = 65 or SRC PLL LIM E = 180H.

START DELAY LMT E (:0A4)
START DELAY LMT E is a delay imposed on the inititiation of a dead start attempt. The scaling is:

1 count = LD FIX PERIOD E / 614440 (sec)

The recommended setting is 0.5 sec (0B4H).

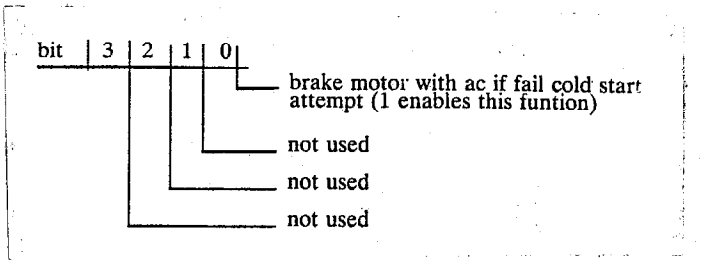
START FREQ E (:0AA)
START FREQ E sets the frequency at which the control starts ramping the firing frequency at the RAMP UP E rate when trying to start the motor from a standstill. The scaling is:

START FREQ E = 6 * start freq (hz)

The recommended setting is 2 hz (0CH).

START OPTION E (:0A0)
START OPTION E enables/disables the various start up options avaivlable:

.lm + 5
.lt



The recommended setting is 1H

START TCOM E (:0A8)
RMP UP ICOM E sets the level of current when the control is trying to start the motor from standstill. The scaling is

1024 counts = 1 pu dc link current

The recommended starting current is 0.35 pu (16AH). Under no condition should the starting current be set above 0.85 pu of the pcm continuous rating.

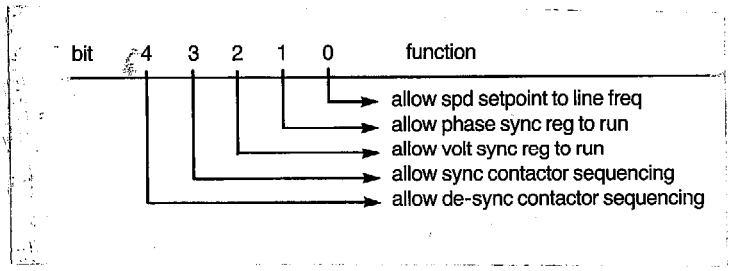
START TMR LMT E (:0AC)
START TMR LMT E sets the maximum lenght of time that the control will attempt to start the motor from a standstill. The scaling is:

1 count = 1 second

The recommended setting is 6 seconds (6H).

SYNC CONTROL E (:0F8)
SYNC CONTROL E contols the synchronizing func-

tions. The options available by setting various bits in the word are:



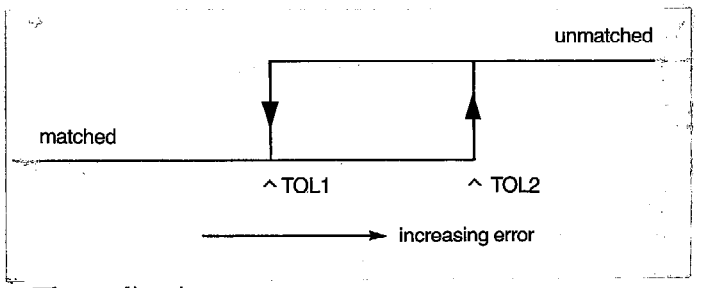
Setting the a bit to a "1" enables the appropriate function. To allow full synchronization, set to 1FH.

SYNC CURLL SRC E (:0FC)

SYNC CURLL SRC E is the minimum current allowed during synchronizing and de-synchronizing operations. It should be set approximately 0.25PU (100H) (never lower than CURLLSRCE).

SYNC CURRENT TOL1 E (:10C)
SYNC CURRENT TOL2 E (:10E)

SYNC CURRENT TOL? E sets the hysterysis limits for the determination of whether the motor current is magnitude matched to the inverter current when the control is trying to de-synchronize.



The scaling is:

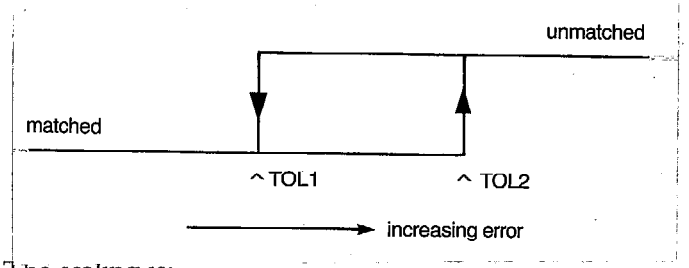
1024 counts = 1 pu current

The recommended settings are

TOL1 = 0.05 pu (33H)
TOL2 = 0.10 pu (66H)

SYNC PHASE TOL1 E (:110)
SYNC PHASE TOL2 E (:112)

SYNC PHASE TOL? E sets the hysterysis limits for the determination of whether the motor voltage is in phase with the source line voltage when the control is trying to synchronize.



The scaling is:

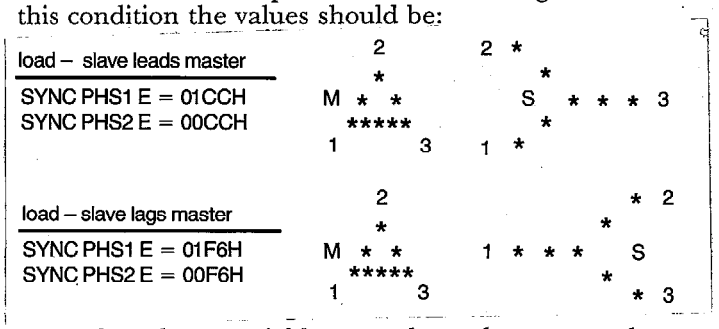
1 count = 0.3516 degrees

The recommended settings are:

TOL1 = 11 degrees (1FH)
TOL2 = 15 degrees (2AH)

SYNC PHS1 E (:114)
SYNC PHS2 E (:116)

SYNC PHS? E control the desired phase shift between motor volts and source volts during synchronizing. The normal desired phase shift is zero degrees. For this condition the values should be:



Note that these variables are dependent upon the relative phasing of the load inverters.

SYNC PT TIME E (:118)

SYNC PT TIME E defines the length of time that the pt signals must be matched before the control believes that they are truely matched. The scaling is:

1 count = 2.7msec

The recommended setting is 0.5 sec (0B4H).

SYNC SPD ERR TOL E (:10A)

SYNC SPD ERR TOL E defines a speed error tolerance used by the synchronizing function. When initially commanded to synchronize, the control ramps the speed setpoint up to sync spd stp E. When the speed setpoint reaches sync spd stp E and the speed error falls below SYNC SPD ERR TOL E, the phase and voltage matching regulators are enabled. The scaling for this variable is:

683 counts = 1 pu speed

The recommended setting is 0.007 pu speed (5H).

SYNC VOLTS GAIN E (:0FA)

SYNC VOLTS GAIN E is a scale factor used in synchronizing by the synchronizing function to place the source and load voltage signals on the same scale. The scaling is:

SYNC VOLTS GAIN E = 1.911 * Vf / Vs

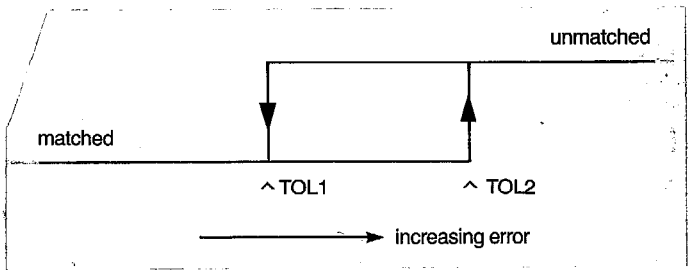
where

Vf = scaled(in the mem) motor flux (rms)
Vs = scaled(in the mem) source Vl-l (rms)

Note that standard scaling is Vf=4.75vrms and Vs=3.464vrms yielding SYNC VOLTS GAIN E = 03H.

SYNC VOLTS TOL1 E (:11A)
SYNC VOLTS TOL2 E (:11C)

SYNC VOLTS TOL? E sets the hysterysis limits for the determination of whether the motor voltage is magnitude matched to the source line voltage when the control is trying to synchronize.



The scaling is:

1024 counts = 1 pu volts

The recommended settings are:
TOL1 = 0.06 pu (40H)
TOL2 = 0.11 pu (70H)

SYN FAULT SRC E (:05E)

SYN FAULT SRC E is used in testing to generate faults in the source converter bridge. The available options are:

1. =000H normal operation (function disabled)
2. =001H induce source commutation failure
3. =0A0H induce source shoot-through (diametric)

TCOM DEFAULT E (:176)

TCOM DEFAULT E is the default torque command level when the speed regulator is opened. The scaling is:

1024 counts = 1 pu torque

This function is used primarily for testing and should normally be set at 0.25 pu (100H).

TCOM HI CLMP E (:178)

TCOM HI CLMP E is the maximum clamp placed on the torque command from the speed regulator. The scaling for torque is:

1024 counts = 1 pu torque command

The recommended setting is 1 pu current (400H).

TCOM LO CLMP E (:17A)

TCOM LO CLMP E is the minimum clamp for the torque command clamp that is calculated based on motor speed. The scaling for torque is:

1024 counts = 1 pu torque command

The recommended setting is 0.2 pu torque (0CCH).

TCOM PRECOND E (:1AC)

TCOM PRECOND E is a variable gain placed on the speed regulatoar precondition calculated from mmotor flux and open loop tcom. The scaling is:

unity gain = 8 counts

The recommended setting is 8H.

TCOM REGEN CLMP E (:17C)

TCOM REGEN CLMP E provides a variable clamp on the amount of regenerative torque the speed regulator may ask for. The scaling is

-1024 counts = 1 pu regenerative torque command

The recommended setting is 0.25 pu regen (-100H).

=

(FF00H)

TEST OFF CNT E (:198)
TEST ON CNT E (:19A)

TEST OFF CNT E and TEST ON CNT E control the conduction periods of the shunt arm of the dclc during the dclc test (selected by dclc test e). The counters are based on source side firings, thus:

1 count = 2.78 mSec

Setting test off cnt e to 0H will force the shunt arm to conduct all the time. Setting test on cnt e to 0 and test off cnt e to 0FFFFH will force the shunt arm to be off all the time.

TEST PATTERN E (0) (:30A)
TEST PATTERN E (1) (:30C)
.
.
.
TEST PATTERN E (15) (:328)

TEST PATTERN E (?) are used by the SEM self test code to perform memory read/write checks.

TORQ DROOP GAIN E (:17E)

TORQ DROOP GAIN E is the gain factor on the adjustment that can be made to the regenerative torque command during a source line voltage dip. The recommended setting is 0H.

TORQ VS SPD GAIN E (:180)

TORQ VS SPD GAIN E defines the function that limits torque command as a function of speed. This function is set up by specifying the desired torque limit at 1 pu speed. The torque limit at other speeds will be directly proportional. The relation will be:

TORQUE VS SPD GAIN E = 24 * pu torque limit @ 1 pu speed

The maximum allowable value for this variable is 1.8 pu counts (2DH). It is recommended that this function be set on the IMD to 1.25 pu (1EH) 0FFH.

TST CELL E (:060)

TST CELL LD E is used when trying to simulate commutation failure faults on either the load or source-side. In effect, it chooses on which phase the commutation failure will occur.

PH = 1 --> 02EH
2 --> 01EH
3 --> 01DH
4 --> 035H
5 --> 033H
6 --> 02BH

This function is active only when SYN FAULT SRC E is set (01H) to induce commutation faulures on the source-side or SYN FAULT LD E is set (01H) to induce commutation failures on the load-side. It is recommended to set this variable to 03FH when not in use.

TST INTRVL E (:062)

TST INTRVL LD E controls the number of firings beween successive induced commutation or diametric faulures on the source or load side. It is calculated:

1. Commutation Faulure - TST INTRVL LD E = (# firings between failures) / 6
2. Diametric Filure - TST INTRVL LD E = # firings between faulures

A recommended value is 32 firings (20H) between induced failures. This function is active only when SYN FAULT SRC E is set to induce commutation or diametric faulures on the source side or SYN FAULT LD E is set to induce commutation or diametric failures on the load-side.

UV TIME OUT LIM E (:040)

UV TIME OUT LIM E is the maximum time a source under voltage may exist before a source uv fault is declared. If the undervoltage exists for less than this time, then it is considered a bus transfer and the control will restart the motor when the source voltage returns. The scaling is:

1 count = 0.8msec

This variable dependent upon the application and the customer should be consulted to determine its setting. In the absense of other guidlines, set at 5 seconds (186AH).

XM GAP E (:0F6)

XM GAP E defines the per unit exciting reactance of the motor. The scaling is:

XM GAP E = 4 * pu motor exciting reactance * (Idci;/Idcm)

where

Ii = rated inverter current (rms)
Im = rated motor current (rms)

VAR DA1 PTR E (:022) (offset)
1 (:024) (base segment)
VAR DA2 PTR E (:026) (o)
2 (:028) (b)
VAR DA3 PTR E (:02A) (o)
3 (:02C) (b)

VAR DA? PTR E contains the address of a variable to be output to the D/A converter testpoint DA? on the HLCB where it can be monitored (by a brush recorder, etc). Since the output is limited to +/- 10.0vdc (+/- 127 counts), a scaling variable (DA? SCL E) is used in conjunction.

VDC GAIN E (:18A)
VDC GAIN E sets the gain of the dc link voltage feed forward path of the current regulator. The gain is calculated:

VDC GAIN E = 24 * (Vllm / Vlls)

where

Vlls = rated line to line source voltage
Vllm = rated line to line motor voltage

4-3.1. SECTION B: FIXED PUBLIC VARIABLES

4-3.2. PUBLIC MEMORY . Public Memory is a RAM memory space used to hold working variables that are constantly modified by the executing control software. A group of selected public variables have been segregated at the begining of the public memory ram in order to permanently fix there addresses. These fixed address variables are ones commonly used for monitoring and troubleshooting.

The addressing for the public memory is shown graphically:
base : off
20 : 40 = 20
+ 40
240 absolute

HMPG RAM MEMORY MAP

base:off		(absolute)
20:???		
	PUBLIC	
	MEMORY	
20:400		(600)
20:3FF	RESERVED	(5FF)
20:3F0		(5F0)
20:3EF		(5EF)
	SELF TEST	
	EEPROM SHADOW	
20:300		(500)
20:2FF		(4FF)
	EEPROM	
	SHADOW	
20:000		(200)

4-3.3. MEMORY MAP PUBLIC MEMORY DEFINITIONS. Note that the number in parenthesis following the variable name is the absolute address for that variable.

ALARM WORD (600)
ALARM WORD is a bit coded variable that contains the status of all alarms. A one indicates that alarm condition has occurred. The bit assignments are:

bit	function
1H	spare
2H	spare
4H	I/O ADDRESSING PROBLEM
8H	DIVISION BY ZERO
10H	IMPROPER INTERRUPT REQUEST
20H	IMPROPER RUNNING MODE FOR UPROCESSOR
40H	NUMERICAL OVERFLOW IN CALCULATION
80H	NON-MASKABLE INTERRUPT SERVICE
100H	IMPROPER HLCB INTERRUPT REQUEST
200H	SYSTEM TIME TIC INT RUNNING
400H	UNRESOLVED INTERRUPT REQUEST
800H	LOSS OF SPEED CONTROL
1000H	spare
2000H	spare
4000H	LOSS OF SPEED REFERENCE
8000H	spare

All alarm indications are cleared by the panel reset.

ALPHA (640)
PHA is the source firing angle. The scaling is:

fire ang (degrees) = 0.3516 * alpha

where the fire ang is referenced from alpha = 0 (full advance).

COM MODE (610)
COM MODE is a variable use to indicate which of several methods of inverter commutation is being used:

0H = self commutation
1H = source side forced commutation
2H = dc link forced commutation

CROSS TIE (642)
CROSS TIE is the spillover of the commanded source dc link voltage into the programed margin for source dc link voltage command. The scaling is:

256 counts = 1 pu source dc link voltage

CRSSTP (65A)
CRSSTP is the contribution to the flux setpoint from thr crosstie regulator. This signal works to lower the flux setpoint in order to sllow the source side to not fire at full advance. The scaling is:

8192 counts = 1 pu flux command

CUR COMP (63C)
CUR COMP is the commanded source dc link voltage which is the linear output of the current regulator. The scaling is:

16384 counts = 1 pu source dc link voltage

CUR REG STEP (630)
CUR REG STEP is a variable that is used to apply step changes to the dc link current command to the current regulator. This is useful for tuning up the current regulator. The scaling is:

1024 counts = 1 pu dc link current command

DCLC MODE (612)
DCLC MODE is a variable used to indicate which harmonic elimination chopping pattern is being implimented while using the dc link forced commutation scheme:

0H = chop to eliminate 5th and 7th harmonics (mode A)
1H = chop to eliminate 5th harmonic (mode B)
2H = no harmonic elimination (mode C)

DCLC STAT LATCH (604)
DCLC STAT LATCH is a bit coded variable which latches fault status indications from any of the dc link commutator gto gate drivers. The bit assignments are:

bit	dclc gto gate driver
1H	series "D"
2H	series "C"
4H	series "B"
8H	series "A"
10H	shunt "D"
20H	shunt "C"
40H	shunt "B"
80H	shunt "A"

A one in any bit position indicates a problem for that

gto/gate driver. All fault indications are cleared by the panel reset.

DL IERROR (63A)
DL IERROR is the clamped derivative of the dc link current error signal. The scaling is:

1024 counts = 1 pu dc link current

FAEAPP (656)
FAEAPP is the contribution to the flux setpoint from the "fire as early as possible" regulator. This signal works to raise the forx setpoint to make the source side to fire more advanced. The scaling is:

8192 counts = 1 pu flux command

FAULT WORD (602)
FAULT WORD is a bit coded variable that contains the status of some FAULTS. A one indicates that fault condition has occurred. The bit assignments are:

bit	function
1H	spare
2H	spare
4H	spare
8H	DC LINK OVERCURRENT SRC
10H	DC LINK OVERCURRENT LD
20H	IDC SOURCE SHOOT THRU
40H	IDC LOAD SHOOT THRU
80H	SRC UV FAULT
100H	SOURCE PLL FAULT
200H	LOAD PLL FAULT
400H	spare
800H	STACK FAULT
1000H	FAILURE TO START
2000H	EEPROM TRANSFER FAILURE
4000H	spare
8000H	spare

All fault indications are cleared by the panel reset.

FIRE ANG ABS (65E)
FIRE ANG ABS is the non quadrant compensated load side firing angle. It is the result of a single quadrant arc tangent function performed on the IX (torque) and IQ (flux) current commands. The scaling is

1 count = 0.3516 degrees

FIRE ANG CL (61C)
FIRE ANG CL is the quadrant adjusted, rate limited, clamped inverter firing angle command. The scaling is:

alpha (degrees) = 180 + (fire ang cl * 0.3516)

FLX ERR FIL (644)
FLX ERR FIL is the filtered flux error signal. The scaling is:

1024 counts = 1 pu flux

FLX FBK (628)
FLX FBK is the flux feedback signal for the flux regulator. The scaling is:

1024 counts = 1 pu motor flux

FLX PI SUM (646)
FLX PI SUM is the output of the proportional + integral flux regulator before the feed forward path gets added in.

FLX REG STEP (632)
FLX REG STEP is a variable that is used to apply step changes to the flux command to the flux regulator. This is useful for tuning up the flux regulator. The scaling is:

1024 counts = 1 pu flux command

FLX STAR1 (61E)
FLX STAR1 is the flux regulator setpoint. The scaling is:

1024 counts = 1 pu flux command

FLX STP (65C)
FLX STP is the flux regulator command signal. The scaling is

1024 counts = 1 pu flux

FLX STP SCL1 (654)
FLX STP SCL1 is the flux setpoint after the fire-as-early-as-possible compensation has been added in. The scaling is:

8192 counts = 1 pu flux command

ICAP FIL (648)
ICAP FIL is the filtered version of the calculated motor capacitor current. The scaling is:

1024 counts = 1 pu capaitor current (on inverter base)

IDC LOD (624)
IDC LOD is the load dc link current feedback signal. The scaling is:

1024 counts = 1 pu dc link current

IDC SRC (622)
IDC SRC is the source dc link current feedback signal. The scaling is

1024 counts = 1 pu dc link current

IDC STAR (618)
IDC STAR is the raw dc link current command formed as the vector sum of ix star and iq star. The scaling is:

1024 counts = 1 pu dc link current command

IERROR (638)
IERROR is the clamped dc link current error signal. The scaling is:

1024 counts = 1 pu dc link current

INVLIM (606)
INVLIM is a bit coded variable that is used to distribute information about conditions that require the dc link current to be extinguished (ie. fire the source converter at inversion limit). The bit assignments are:

bit	function
1H	ZERO CURRENT REQUEST
2H	spare
4H	spare
8H	spare
10H	LOAD IDLE REQUEST
20H	spare
40H	SOURCE OVER CURRENT
80H	spare
100H	DRIVE IDLE REQUEST
200H	LOSS OF HARDWARE SPEED
400H	LOAD OVER CURRENT
800H	SOURCE PLL ERROR
1000H	LOAD PLL ERROR
2000H	CROWBAR
4000H	spare
8000H	STACK OVERFLOW

IQ STAR (614)
IQ STAR is the command for the torque producing component of the inverter output current. The scaling is:

1024 counts = 1 pu torque command

ISTAR2 (61A)
ISTAR2 is the DCLC compensated and rate limited dc link current command. The scaling is:

1024 counts = 1 pu dc link current command

IX STAR (616)
IX STAR is the command for the flux producing component of the inverter output current. The scaling is:

1024 counts = 1 pu flux command

LAC MODE (60C)
LAC MODE indicates the mode that the load-side is in:

00H = idle mode
10H = fix fire mode
20H = run fire mode

LD AVG VDC (626)
LD AVG VDC is the calculated average dc link voltage on the load side. This signal is used in the feed-forward portion of the current regulator. The scaling is:

256 counts = 1 pu (on src base) dc link volts

LINE COM (63E)
LINE COM is the clamped and scaled version of the commanded source dc link voltage (output of the current regulator). The scaling is:

+256 counts = +1 pu source dc link voltage
- 256 counts = - 1 pu source dc link voltage

LOD STATUS (60E)
LOD STATUS is a variable that can be used to monitor the modes of inverter firing:

00H = idle mode
10H = fix fire mode
20H = run fire mode using src to force commutate
30H = run fire mode using DCLC to commutate and chopping to eliminate 5th and 7th
40H = run fire mode using DCLC to commutate and chopping to eliminate 5th
50H = run fire mode using DCLC to commutate and not chopping
60H = run fire mode self commutating

This variable is easy to monitor if placed on a d/a output.

MODE FLAG (608)
MODE FLAG is a variable used to communicate information between the foreground and background code. The bit assignments are:

bit	function
1H	spare
2H	RUN
4H	spare
8H	spare
10H	CROWBAR
20H	SWITCH GEAR CLOSED
40H	TWELVE PULSE
80H	MASTER CHANNEL
100H	GATE TEST MODE
200H	spare
400H	spare
800H	FACTORY DCLC TEST
1000H	FACTORY SOURCE A FIRING
2000H	FACTORY SOURCE B FIRING
4000H	FACTORY LOAD FIRING
8000H	FACTORY PUMP BACK

OPEN SPD REG (636)
OPEN SPD REG is a variable that will force the speed regulator to open and instead allow a constant torque to be commanded via TCOM DEFAULT E.

00H = normal operation of speed regulator
01H = open speed regulator

PI MIN CLMP (652)
PI MIN CLMP is the minimum clamp placed on the output of the proportional + integral portion of the flux regulator. This is normally fixed at -2000H (-2pu), however it may be greater if the drive is operating in a light load condition in order to clamp ixstar to some minimum positive value.

PLLC LOAD LD (62E)
PLLC LOAD LD is the output of the load-side phase-lock-loop regulator that is used to keep the load time counter synchronized with the motor flux. The scaling is:

motor electrical freq (Hz) = 7200 / pllc load ld

PLLC LOAD SRC (62C)
PLLC LOAD SRC is the output of the source-side phase-lock-loop regulator that is used to keep the source time counter synchronized with the source ac line. The scaling is:

ac line freq (Hz) = 7200 / pllc load src

SAC MODE (60A)
SAC MODE indicates the mode that the source side is in:

00H = idle mode

01H = run mode (running current reg and firing source converter scr's)

SPD (62A)
SPD is the speed feedback signal for the speed regulator. The scaling is:

683 counts = 1 pu speed (60 Hz)

SPD ERR FIL (64C)
SPD ERR FIL is the clamped and filtered speed error signal. The scaling is:

683 counts = 1 pu speed (60 Hz)

SPD REG STEP (634)
SPD REG STEP is a variable that is used to apply step changes to the speed command to the speed regulator. This is useful for tuning up the speed regulator. The scaling is:

683 counts = 1 pu speed command (60 Hz)

SPD STP (620)
SPD STP is the non rate-limited speed regulator setpoint. The scaling is:

683 counts = 1 pu speed command (60 Hz)

SPD STR SUM (64A)
SPD STR SUM is the rate limited speed regulator setpoint. The scaling is:

683 counts = 1 pu speed setpoint (60 Hz)

SRC VOLTS (658)
SRC VOLTS is a variable proportional to the source ac line voltage which is used for synchronizing. It is obtained via an a/d read of VABS signal on the source NLIB pwb. The scaling is dependant on the attenuation ratio:

1 pu SRC VOLTS = Vll * Kv * 2.2 * 204.8

TCOM (64E)
TCOM is the torque command output from the speed regulator. The scaling is:

1024 counts = 1 pu torque command

TSTR SGN (650)
TSTR SGN is the rate limited torque command. The scaling is:

1024 counts = 1 pu torque command

CHAPTER 5

SECTION A

5-1. SERIAL MONITOR AND HAND TERMINAL (OR OTHER DIAGNOSTIC TERMINAL)

Integrated into the firmware for the electronics is a monitor program which supports manipulation of system memory and I/O. The serial monitor drives an RS-232 port on the front of the microcomputer card. Communication with the microcomputer can be established by connecting the hand-held terminal supplied with the drive to the RS-232 port. This connection must be made using the HNMB null modem (supplied with the electronics module) as illustrated in Figure 5-1. Observe the plug orientation and Berg jumper positions marked on the null modem.

The mode switches in the terminal (beneath back access cover) should be set as follows:

SWITCH	1	2	3	4	5	6
POSITION	DN	UP	DN	DN	UP	UP

The hand terminal contains the full ASCII alphanumeric character set and operates at 300 baud.

The terminal can be used to communicate with RAM memory (HMPG), public memory and alterable memory (DMPB). The address for alterable memory is 3000.

The HMPG RS-232 ports are automatically compatible with the 300 baud hand terminal. When another diagnostic terminal is used, it should be set for a compatible baud rate (see table).

5-2. ON LINE MONITOR COMMANDS

The available commands are listed in the following table. The control-character commands may be used at any time during execution of other monitor commands.

On Line Monitor Commands:

COMMAND	ABBREVIATION	FUNCTION
CONTROL-C	CTRL C	Abort execution of pending moni
CONTROL-Q	CTRL Q	Resumes display after CNTRL S
CONTROL-S	CTRL S	Halts display
(CTRL key and letter command pressed simultaneously)		
ADJUST	A	Adjust variable in RAM memory
BAUD	B	Change terminal baud rate
CONVERT	C	Convert values between binary hexadecimal
DISPLAY	D	Display RAM, ROM, or EAROM v
INPUT	I	Read input port data
LIST	LL	Trigger diagnostic list
MOVE	M	Move block of memory
OUTPUT	0	Write data to output port
REDEFINE	RL	redefine diagnostic list control
SET	S	Display and change RAM value
TYPE	TL	Type diagnostic list
ZAP	Z	Display and write value to EARO command letter key abbreviation monitor responds by printing the command.

5-3. MONITOR COMMAND STRUCTURE

Symbols used in the monitor commands are as follows:

( ) A quantity enclosed in ( ) is considered to be an option. The user does not enter ( ), only the enclosed quantity.

< > A quantity enclosed in < > is a value which

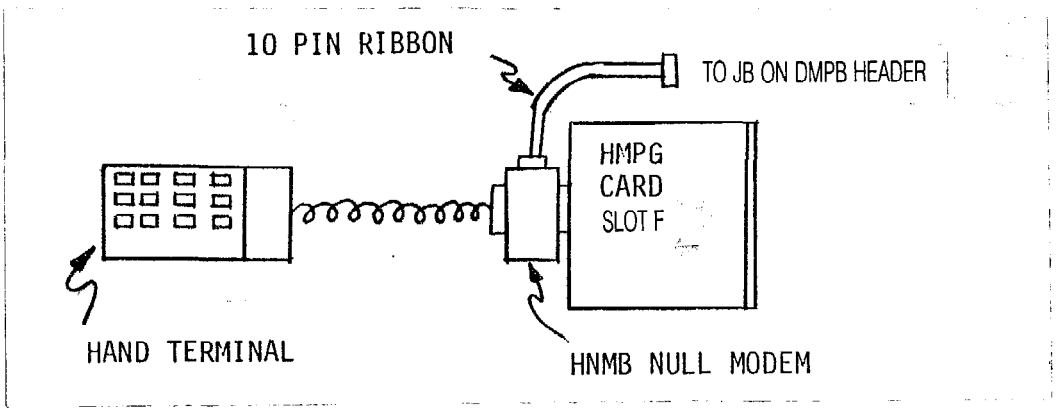


Figure 5-1.
Hand Terminal Connection.

must be entered by the user. Only the quan-"'tity is entered, not the < > .

W Denotes a data or I/O word having a value between 0 and FFFFH or 0 to 65535 decimal.

I Denotes a data or I/O integer having a value between -32768 and +32767 decimal. An integer is a 16 bit binary number in standard 2's complement form.

H Denotes HEX number.

CR Denotes a carriage return.

• Prompt

E Denotes EEPROM memory variable when added to end of word.

When using the monitor program, security is provided against unskilled access to the system by use of a password. Monitor commands which are capable of changing memory or I/O data will ask for this password before allowing the user to proceed further. Whenever such a command is entered, the response on the terminal will be KEY= and the user must respond with the password OK before the monitor will allow him to proceed. For example, the user enters O to output to a port. The terminal then responds KEY= and the user then keys in OK. The monitor responds by printing the command again and allows the user to continue. C complete sequence might be as follows:

<O> UTPUT
KEY = <OK> . . . . OK not printed
O <38 , FF CR>

Again, commands which merely read I/O or memory do not require a password.

The following is a list with detailed syntax of monitor commands:

Adjust and Display Memory:
<A> DJUST
KEY = <OK>
A <(W or I) ADDRESS TO BE ADJUSTED (,(W or I) ADDRESS TO BE DISPLAYED)>
<CR>
ADDRESS displayed
PRESENT CONTENTS displayed
STEP = <STEP VALUE = 0 to F CR>
<(+ or -), (+ or -),. . . . CR>

CAUTION: If W or I is not entered when required, the high byte of the affected address will be omitted or unchanged.

If one address is specified, the data at that address will be adjusted and displayed. If two addresses are specified, the data at the first address will be adjusted while the second address data is displayed. After entering the first <CR> , STEP = will be displayed. The step size is the amount by which the variable of interest is to be incremented or decremented. A <CR> or <1 CR> entered will set the step value to 1; 0 through F are the optional step decrement the byte, word, or integer at the address to be adjusted. If CR is entered, the adjust routine will be exited with the adjusted variable stored in RAM memory. Entering an S will allow changing to another step size. NOTE: This command, with a step of 0, is useful for observing the trend in a variable, without changing any data in memory.

Change Terminal Baud Rate:

<B>AUD <BAUD RATE VALUE CR>. . .<CR>
Available baud rate values are 75, 110, 300, 1200, 2400, 4800, 9600, and 19200. After typing the first carriage return, change the baud rate on the terminal, wait one second, and type the second carriage return. If the HMPG does not receive the second carriage return at the new baud rate within 60 seconds, the monitor changes back to the old baud rate and prints an error message.

Convert Numbers:

<C> ONVERT <B or D or H (+ or -) NUMBER TO BE CONVERTED CR>
This command converts between binary, hexadecimal, decimal word, and decimal integer numbers. The B, D, or H prefix specifies the type of number. This is followed by an optional + or - sign and the number to be converted. The monitor responds with the conversion in binary, hexidecimal, decimal integer, and decimal word representations. For example, if <C D 32 CR> is entered, the monitor responds:

H 0020
B 0000000000100000
DW 00032
DI +00032

Display Memory Location Value:

<D>ISPLAY <(W) START ADDRESS BASE and OFFSET (, END ADDRESS OFFSET only) CR>
If only START ADDRESS is specified, then only that memory location is displayed. If START ADDRESS are both specified, then all memory contents between the two locations are displayed.

Input From Port:
<I>NPUT <(W) PORT ADDRESS , (,). . .CR>
Displays data from port located at PORT ADDRESS. Each comma entered causes port data to be displayed again. CR terminates the command.

Trigger Diagnostic List(s):

<L>IST<L>
This command triggers the diagnostic list The list will not be printed unless it is in the auto-print mode.

Block Move of Memory:

<M>OVE
KEY = <OK>
M <(H)> START ADDRESS BASE and OFFSET , END ADDRESS OFFSET only, DESTINATION ADDRESS BASE and OFFSET CR>

Accepts three addresses and moves block of memory specified by START ADDRESS , END ADDRESS to a block of memory beginning at DESTINATION ADDRESS. If MH is entered, a block of memory (typically from ROM) is moved into EAROM. The destination must be a EAROM address and the block to be moved must not overflow the address range. In order to do the MH option, the programming supply on the HUMC card must be turned on.

This command does not work from the source microprocessor - it will disable the monitor, requiring a MEM reset with the correct value for MON SRC in EAROM.

Output to port:

<O >UTPUT
KEY = <OK>
O <(W) PORT ADDRESS , DATA (, DATA) . . . CR>
Outputs user supplied DATA to port having specified address. Data can be output repetitively to a port by entering new data followed by a comma. CR terminates the command.

Redefine Diagnostic List Controls:

<R>EDEFINE <L NEW LIST CONTROL VALUE CR>
This command redefines the DIAG LIST ENABLE word, initializes list parameters, and zeroes all diagnostic list memory locations. See section on diagnostic list and EEPROM dictionary for definition and usage.

Set memory:

<S>ET
KEY = <OK>
S <(W) ADDRESS , PRESENT CONTENTS displayed ((NEW CONTENTS),) CR>
Displays memory contents located at ADDRESS . IF NEW CONTENTS are specified, then NEW CONTENTS are substituted for present contents. When the second, and subsequent comma is entered, the MONITOR increments to the next byte or word/integer address, awaiting NEW CONTENTS to be entered there.

Type Diagnostic List:

<T>YPE <L CR>
This command causes the diagnostic list to print out. This command does not trigger the list and does nothing more than print out the list data, if any.

Write DMPB (EEPROM) -- Zap:

<Z>AP
KEY = <OK>
Z <ADDRESS , PRESENT CONTENTS displayed DATA CR>
If ADDRESS is a valid address (only the offset is necessary for Zap- the base is optional) DATA will be written into location ADDRESS . Before writing to EEPROM, the toggle switch on the DMPB card in slot 2F must be placed in the up position to enable memory writing power supply. A check is made to verify if the memory received data. the base addresse is 3000. A SEM reset must be performed after the Zap.

5-4. SYSTEM I/O ADDRESSING AND MANIPULATION

The system process inputs and outputs. These are contact signals to/from the signal electronics module inputs and outputs interface through the HISA (1B) cards and the HRDA (1A) cards, respectively, to the I/O module (DS3820IOM).

It is possible to read and manipulate system I/O with the terminal/monitor using the I (input) and 0 (output) commands. This is particularly useful during initial system checkout and may be used as a troubleshooting tool as well.

Input port data (including contact input cards) can be read at any time by using the Input command described earlier.

Data to output cards is normally controlled by software resident in the microcomputers (HMPG cards). Any attempt to change this data with the terminal/monitor will be overwritten unless the normal software is defeated. It is possible to lock out all the software except the serial monitor. In this case, the output ports (including the relay outputs) can be controlled exclusively by the terminal/monitor output command. To get in this mode, it is necessary to Zap EEPROM variable I/O test module from 0 to 1, then reset the SEM; or set RAM shadow of this variable from 0 to 1 (Addr. 20:offset).

Output port data can now be manipulated at will. Table 5-4 lists some of the I/O addresses, functions, and location of the I/O device in the electronics module that may be useful.

Restore IO TEST MODE E and/or its RAM shadow to 0 when I/O testing is completed.

5-5. DIAGNOSTIC LISTS

In addition to status monitoring lights for direct indication of system behavior, the electronics has the ability to take diagnostic lists. A diagnostic list is a list of samples of system variables (such as current) which

Table 5-4.
LCI System I/O Map

PORT ADDRESS	DEVICE/FUNCTION	LOCATION
8000	D/A converter # 1	HLCB; 2E
8002	D/A converter # 2	HLCG; 2E
8004	D/A converter # 3	HLCG; 2E
8006	D/A converter # 4(HI word)	HLCG; 2E
8008	D/A converter # 4(LO word)	HLCG; 2E
800A	Software testpoints 1-4 and IMOK	HLCG; 2E
8038	Inputs 0-7 from I/O module	HISA; 1B
803A	Inputs 8-F from I/O module	HISA; 1B
803C	Inputs 10-17 from I/O module	HISA; 1B
803E	Inputs 18-1F from I/O module	HISA; 1B
8040	Outputs 0-7 to I/O module	HISA; 1B
8044	outputs 8-1F to I/O module	HISA; 1A

is used to evaluate system behavior. This feature is particularly useful as a window into software-implemented functions which could not otherwise be monitored. The diagnostic list is stored in the memory of the system. The microcomputer can store 256 samples of 5 different variables. These samples may be read out via a printing terminal, and variables prior to a fault shutdown may be reconstructed and analyzed.

Diagnostic lists may be stored in linear, circular, or combination modes. In the linear mode, the lists fill with the next 256 samples after being triggered. In the circular mode, the lists repeatedly fill and refill until triggered to stop. In the combination mode, the lists fill and refill as in circular mode, but continue to store samples for 128 (or 192) passes after being triggered. Thus the combination lists contain 128 (or 64) samples before and 128 (or 192) samples after the triggering event.

It should be pointed out that the diagnostic list is a powerful diagnostic tool that requires detailed knowledge of the system to set up and use. It is not expected that the user would have this level of knowledge. However, factory or field service personnel can use this tool to more rapidly diagnose subtle or intermittent problems. It is to this end that the following rather detailed information is given.

5-6. SETTING UP THE LISTS

To set up the list it is necessary to select the desired mode options and to assign variable addresses to list pointers located in EEPROM.

The list type and options must be selected by properly setting the low byte of control word DIAG LIST ENABLE. This byte selects the various options as detailed in the following paragraphs.

Bit 0 of the list control word is used to enable/disable the diagnostic list. Setting this bit to 1 enables the list while setting it to 0 disables the list. When disabled, no list samples are taken.

Bit 2 is used to cause the list to sample data points at a slower than normal rate. "Slow" data points are sampled at 1/129 the "normal" rate. The "normal" rate is the repetition rate of the particular module of software from which the list sampling is done. For example, sampling done from the current regulator is done at a rate of 6 times source line frequency, 1/129 times this rate at the "slow" rate. The "slow" rate sampling is used to establish trend type information as would be done with a chart recorder.

When bit 2 is set, both "slow" and "normal" data are taken. If bit 2 is not set, only "normal" data is taken. "Slow" data may be taken with linear, circular, or combination lists as follows:

For a "slow" linear list the first 128 samples are at the "normal" rate and the last 128 samples are at the "slow" rate. For a "slow" circular list, the 128 "normal" samples begin with the triggering event. A "slow" combination list contains (in order of occurrence) 64 samples at the "slow" rate (which is 1/65 the normal rate, for the combination list), 128 samples at "normal" rate, and 64 samples at the "slow" rate. The first 64 samples occur prior to the triggering of the list, and the last 192 samples (128 "normal" and 64 "slow") occur after the list is triggered.

Bit 1 is used to select/deselect the combination list. Bit 3 and bit 1 together determine the list type, as shown in the following table:

BIT (3 2 1)	LIST MODE
0 0 0	Regular circular list
0 0 1	Regular combination list
0 1 0	Circular list with solw/fast samples
0 1 1	Combination list with slow/fast samples
1 0 0	Regular linear list
1 0 1	Combination list (same as 001)
1 1 0	Linear list with slow/fast samples
1 1 1	Combination list with slow/fast samples (same as 011)

Bit 4 of the list control words is used to control triggering of the list. This bit is set to 1 in HUMC only if it is desired for the list to trigger immediately after system reset. Ordinarily the manipulation of this bit is controlled (in its RAM shadow) by software in one of several ways. The monitor command "L" (list) sets this bit and will therefore trigger the list. This bit is also set and the list triggered when an INVLIM flag word bit (see section on INVLIM) matches up with any bit set in the EEPROM word DIAG LIST INVLIM MASK E. Since most faults result in the setting of one or more INVLIM bits, this feature is valuable for automatic data gathering on faults. Finally, the trigger bit is set when the list variable 2 (address stored in DIAG VAR PTR2 E) exceeds the value set in DIAG VAR 2 MAX LMT E. This feature is valuable as a means of triggering the list on events which are unpredictable and do not result in a fault. If this feature is not used, DIAG VAR2 MAX LMT E should be set to OFFFFH in EEPROM and/or RAM shadow of EEPROM.

Bit 5 is the previous trigger bit and should always be set to 0 in EEPROM. It is used by the list routine to save the status of the trigger (bit 4) from the previous pass.

Bit 6 is the auto-print bit. If this bit is set (1H), the diagnostic list will be automatically printed on the monitor terminal when the list is triggered and completed.

Bit 7 is the no-overwrite bit; it may be set to 1 in EEPROM to prevent the list from being triggered more than once (between system resets). If bit 7 is set to zero, then a subsequent list is enabled once the present list is triggered, completed, and printed (if the auto print option is selected).

5-7. ASSIGNING LIST DATA SAMPLING LOCATIONS

Inasmuch as the software consists of continuous looping background programs punctuated by various priority interrupts, it is desirable to be able to sample data in different modules of the software. For example, it might be desirable to take data samples after the execution of the current regulator calculation. This is made possible by choosing the appropriate value for the EEPROM variables DIAG LIST SITE E and writing this value into the EEPROM location. The bits of EEPROM variable are flags which select the point in the code execution where list data is taken for storage. The EEPROM directory give the available options for this variable.

It is also possible to sample data from more than one location by adding together the list site values for the desired locations and writing the resulting sum into the DIAG LIST SITE variable. In this case a special variable, MODE TRACKER, is useful for keeping track of which list site the list sample is taken from. The MODE TRACKER variable is located in local RAM and take on the value corresponding to the list site at the time of sampling. By placing the MODE TRACKER variable in the list, it is possible to correlate the list samples with the module of code from which the sample was taken.

5-8. ASSIGNING LIST VARIABLES

Any byte, word, or integer variable used in the software may be assigned to the list. To assign a variable to the list it is necessary only to write its address into a list variable pointer in the appropriate EEPROM memory location. There are 5 variable pointers. The first pointer must be used with a byte (8 bit) variable while the other 4 can be used for word or integer (16 bit) variables. (Byte variables may also be addressed by these pointers but the list will sample both the desired byte and an adjacent byte in memory.)

The variable pointer addresses are as follows:

POINTER	ADDRESS	
DIAG VAR PTR1E	:4	The base addresses are:
DIAG VAR PTR2E	:8	EEPROM - 3000
DIAG VAR PTR3 E	:C	RAM shadow - 20
DIAG VAR PTR4 E	:10	
DIAG VAR PTR5 E	:14	

To assign variable addresses, it is necessary to obtain the desired addresses from the variable maps. The addresses thus obtained will be in the form SEGMENT:OFFSET, and will generally be two 4-digit hexadecimal numbers (eg, SSSS:0000). Note: SEGMENT and BASE are used interchangeably. Using the Zap command, write the variable addresses into EEPROM memory, offset first and then base address. It will require writing 2 words into EEPROM to specify each variable pointer.

As an example, assume that it is desired to assign a source word variable with address FE76:18A2 to DIAG VAR PTR2 E. To do this it is necessary to write (using the Z command) the following into EEPROM:

18A2 into word address :14
FE76 into word address :16

5-9. LIST STORAGE LOCATIONS

The list data is stored in ten 256 entry arrays in common memory starting at location 4090:0. The list data may be accessed from microcomputer RS-232 port. The list data arrays are located in memory as shown in the following table:

DATA VARIABLE ARRAY	ADDRESS RANGE	
Variable #1	:0,FF	The base address:
Variable #2	:100,2FF	1090
Variable #3	:500,6FF	3090
Variable #4	:500,6FF	
Variable #5	:700,8FF	

It is possible to display the list data on a terminal by using the monitor command D (display memory bytes) for the first variable, or by using the command DW (display memory words) for all other variables. However it is more convenient and preferable to print out the data in list format by using the monitor commands TL (type list).

5-10. TRIGGERING THE DIAGNOSTIC LIST

The "trigger" is a software signal which requests the monitor to complete a diagnostic list. There are three ways to trigger the diagnostic list:

1. Using the monitor command "L"
2. Matching an INVILIM bit against the EEPROM variable DIAG LIST INVLIM MASK E
3. Triggering when list variable 2 exceeds the value of EEPROM variables DIAG VAR2 MAX LMT E.

The use of the monitor command "LL" (see previous section in this chapter on Serial Monitor) allows manual triggering of the list. The "LL" command may be used to repetitively trigger the list unless the "no overwrite bit" of the DIAG LIST ENABLE E word(s) is set. In this case, the list must be reset by executing a

MEM reset or by use of the monitor command "RL". The "RL" command is used to dynamically redefine the DIAG LIST CONTROL E word in RAM memory. RL NN sets DIAG LIST CONTROL E to NN where NN is the byte code for the desired list mode and options. Use of the "RL" command automatically resets the list parameters and zeros the old list samples.

Triggering with the INVLIM word occurs whenever any bit set in INVLIM matches with a bit set in DIAG LIST INVLIM MASK E. The INVLIM words are variables for which any bit being set will initiate a call of the converters to inversion limit. Each bit of the INVLIM words represents a particular condition for which it is desired to bring current to zero. See the EEPROM Dictionary for bit coding of DIAG LIST INVLIM MASK E. To disable this mode of triggering, set DIAG LIST INVLIM MASK E to zero. After triggering the list on a particular INVLIM bit, the SEM must be reset or the corresponding bit of DIAG LIST INVLIM MASK E in local ram must be reset with the monitor in order to re-enable INVLIM triggering on that bit.

The third method of triggering the list makes use of DIAG VAR2 MAX LMT E. If the value of list variable 2 exceeds the value of DIAG VAR2 MAX LMT E respectively, then the list will be triggered. When the list is triggered using this method, the local RAM value of DIAG VAR2 MAX LMT E is set to OFFFH. To re-enable the list DIAG VAR2 MAX LMT E may be set to the desired value in RAM or a SEM reset may be executed. To disable this method of triggering, set the DIAG VAR 2 MAX LMT E variable to OFFFFH.

5-11. USE OF THE DIAGNOSTIC LIST

The diagnostic list facility may be thought of as a five-channel digital storage oscilloscope. The signal inputs to each channel are connected by entering the desired software variable address into the selected channel variable pointer (DIAG VAR PTR1, etc.).

The horizontal base of the scope is set by DIAG LIST SITE E. Instead of setting the time period to a value (5 microseconds for instance, which would store a sample after each 5 microsecond interval), set DIAG LIST SITE E to a site value (80H or PLL for instance, which stores a sample after each pass through PLL). This is not, strictly speaking, a time base; however, it is handled similarly and represents the horizontal axis of the scope picture. The horizontal base of the scope may be modified by DIAG LIST ENABLE E. The normal screen picture is given by a regular linear list, which writes a "screen trace" immediately after being triggered. However, on the digital scope, the "trigger cursor" may be set to a position at the center or at the right-hand border of the screen. Triggering this list will result in a picture of data immediately before the trigger.

Setting the slow bit of DIAG LIST ENABLE E to 1 modifies the scope picture by taking half the data at a much slower rate. For a linear slow list, these slow points are put on the right-hand half of the "screen". For a circular slow list, the points take up the left-hand half of the "screen". A combination slow list has slow points at the left and right extremities, each taking up one-fourth of the screen.

The "scope" may be triggered in one of three ways. A trigger pulse will be acknowledged when the user invokes the "LL" command, when channel 2 (DIAG VAR PTR2 E) exceeds its "trigger sensitivity" (DIAG VAR2 MAX LIMIT E), and when an unmasked INV LIM occurs. The latter two types of triggering must be reset before being reused. The "Channel 2 Max" trigger can be reset by entering the appropriate maximum value in the local RAM variable DIAG VAR2 MAX LIMIT E, using the "S" command. The "INV LIM" trigger can be reset by setting the appropriate bit in the local RAM value of DIAG LIST INVLIM MASK E to 1 using the "S" command.

The single-sweep storage mode is entered by setting the no-overwrite and auto-print bits of LIST CONTROL each to 1. Now when the trigger signal is pulsed in one of the above ways, a "screenful" of data is printed on the monitor terminal. The "screen" may be refreshed by using the "T" command (useful if auto-print was not selected, terminal was not connected, or print-out was lost for some other reason).

The free-running mode is entered by resetting the no-overwrite bit and setting the auto-print bit of DIAG LIST ENABLE. A "screenful" of data will be printed after each trigger of the type outlined above.

5-12. EXAMPLES OF DIAGNOSTIC LIST SET UP

Before setting EEPROM for a particular list, enter the terminal width (in number of characters per line) in the EEPROM variable DIAG LINE WIDTH E for 80 characters per line (50H):

Z 1E, 50

I. Linear list, no overwrite, auto print:
Given: Sample Variable - PLLC LOAD (FE67:09A2)
Sample Site - PLL REG (LOAD SIDE)

1. PLLC LOAD is a word, so use variable 2 on the loadside. Write the address of PLLC LOAD address in the pointer variable by zapping EEPROM.

   ZW 8, 09A2
   ZW A, FE67

2. Set DIAG LIST ENABLE E = 0C9H for a linear list with no overwrite and auto print, and set DIAG LIST SITE E = 100H for site PLL

(load).

ZW 54, 0C9
ZW 56, 100

3. Reset microprocessor to transfer EEPROM data into RAM.
4. When user wants the linear list to begin, type LL. The list will be taken and will stop automatically when full.
5. As the auto print bit is set, the list will print out automatically when full.

II. Circular List, no overwrite, auto print

Given: SAMPLE VARIABLE = LINE COM (FFBE:0020)
SAMPLE SITE = CROSSOVER (Source Side)

1. LINE COM is an integer. Use Variable 2 on source side. Load the address of LINE COM in DIAG VAR PTR2 E.

ZW 14, 0020
ZW 16, FFBE

2. Set DIAG LIST CONTROL E = 0C1 for a circular list with no overwrite and auto print, and set DIAG LIST SITE E = 1H for CROSSOVER.

ZW 24, C1
ZW 26, 1

3. Reset microprocessor
4. The list will begin immediately. To stop the list, type LS.
5. As the auto print bit is set, the list will print out automatically when triggered to stop sampling.

III. Slow combo list, no over-write.

Given: Sample Variable - FAST SLOT (FC2E:40E1)
Sample Sites - CROSSOVER, RUN FIRE (load side)

1. FAST SLOT is a byte. Use Variable 1 on the load side. Put the address of FAST SLOT into DIAG VAR PTR1 E.

ZW 40, 40E1
ZW 42, FC2E

2. Set DIAG LIST ENABLE E = 87H for a slow combination list with no overwrite, and set DIAG LIST SITE E = 900 H.

ZW 54, 87
ZW 56, 900

3. Reset microprocessor.
4. Trigger the list by entering LL. This will stop the half-slow, half-fast circular list and start the half-fast, half-slow linear list.
5. As the auto print bit of DIAG LIST ENABLE E was not set, print out the list by typing TL.

CHAPTER 6

INSTALLATION AND STARTUP GUIDE (OUTLINE)

I. Keep detailed installation log for each drive.

II. Install Equipment
(Align motor, setup coupling to run uncoupled from driven device. Setup motor protectives - lubrication, thermocouples, RTDs, vibration detectors. Check-out transformers and reactors - fault devices, alarm devices; source and load breakers - close and trip circuits, overcurrent devices.).

III. Interconnect equipment

IV. Wire check

V. Check-out source and load breakers

VI. Megger

VII. Run motor in "Bypass".

VIII. Check Berg jumpers, daughter board components.

IX. Apply control power
Check starters and blowers
Check power supplies
Check relays, lights

X. Operate in gate-test-mode
Check source and load firing
Check process inputs and outputs
Set current feedback

XI. Apply source voltage
Check attenuator signals
Set scaling and trip levels

XII. Simulate load voltage
Check attenuator signals
Set scaling and trip levels

XIII. Perform crowbar test

XIV. Perform "locked" rotor test

XV. Run motor uncoupled
Check bearings, vibration
Check acceleration, deceleration, regeneration

XVI. Check transfer schemes

XVII. Run motor coupled
Tune control
Check loaded transfer schemes
Evaluate control

XVIII. Clean-Up
Send back to factory corrected prints
Make 3 lists of EEPROM Default Values
(Customer, A&ES, Factory)

6-1. DETAILED INSTALLATION INSTRUCTIONS

It is assumed that all interconnecting wire and cables are in place and are connected. The erection and alignment of the motors is not described here, nor is the routine checkout of the circuit interrupters. It is advised that a detailed installation log be maintained for each drive.

6-2. MEGGER TEST

a. Short all power buses together in the IMD panel. (L1A, L2A, L3A, L1B, L2B, L3B, P2, SB1, N1, N2, PC, M1, M2, M3)
b. Pull the ribbon cables that connect to the front edge of the HPTK cell firing cards.
c. Disconnect ribbon cable to each HPTK card in the Leg module of the bridge compartment. Also remove the ground (GND) stab on this card and on the XPTN.
d. Disconnect "JA" and "GND" wires connected on the XPTNs in the capacitor panel. These wires are the attenuator grounds from the capacitor PTN card.
e. Perform the 1000 VOLT megger test and record the test values.

MASTER

_____ Megaohms at_____ Volts

SLAVE

_____ Megaohms at_____ Volts

6-3. RUN MOTOR UNCOUPLED - BYPASS

CONSIDERATIONS:
Bearing monitoring protection should be functioning at this point to protect the motor. The motor bearing temperature rise, vibration, and oil rings (slingers) should be checked.

6-4. CHECKOUT MOTOR ON BYPASS CONTACTOR

a. Stop the drive and rack-out and lock-out the drives load circuit interrupter. Rack-in

the bypass interrupter. Source interrupter should be closed. (Drive's isolation switch should be open).

b. Place INVERTER/BYPASS switch in BYPASS. Start motor with start switch. Check motor direction of rotation - matches drive. Check motor vibration, magnetic center, etc. Stop motor.

6-5. PRELIMINARY CONSIDERATIONS

a. Verify the variable component values in the SEM per sheet DA of the drive elementary.
b. Verify Berg jumper locations as shown on drive elementary sheet DA.
c. Inspect all compartments for loose hardware/debris removing any such material prior to applying power.
d. Inspect all stab connections in both SCR and GTO for shorts.
e. Open all circuit breakers; refer to drive elementary.
f. Open or turn off all switches and power supplies.
g. Partially extract all PWB's from the SEM.
h. Check each SRC + 670 for shorts.

6-6. AUXILIARY POWER

a. Open the power supplies circuit breakers.
b. Apply the 480VAC and/or 120VDC control power.
c. Check bridge cooling fans for proper direction of rotation, air should enter the front of each of the bridge compartments and exit the top of the blowers when the doors are closed.

6-7. POWER SUPPLIES

a. Turn on control power supply, PSFF.
b. Verify presence of P5, P15, N15 and P28 at the electronics module backplane.
c. Insert the PWBs and verify that the power supply remains constant.
d. With all PWBs in, adjust the e power supply to obtain the following voltages at the MEM.
   1. Adjust R1 on NPPB until P5 = + 5.0 +/- 0.1 volts.
   2. Adjust R6 on NPSK until P15 = +15 +/- 0.2 volts.
   3. Adjust R7 on NPSK until N15 = - 15.0 +/- 0.2 volts.

6-8. ELECTRONICS MODULE INITIALIZATION

Null all meters.

6-9. RUN IMD SELFTEST:

Turn-on (up) NLCB test switch. Reset HMPG by jumpering TP1(RST) to TP2(DCOM). Follow the indicated instruction and investigate any fault messages. Repeat till unit passes all tests.

6-10. OPERATING SET-UP ADJUSTMENTS AND EEPROM MEMORY INITIALIZATION:

Measure and set the voltages for all the OV, SUPP, RTD, UVR, OSP test rings as tabulated in the system elementary sheet 1DC.

Reset SEM, using the HLCB RESET switch; the HMPG IMOK LED should come on. Jumper testpoint TINO on HLCB-2E to DCOM, turn-on(up) WREN switch on DMPB(HMPG), WREN LED on the DMPB card should come on. Reset the HMPG. Remove the jumper and turn off WREN switch. Dump EEPROM memory on a printing terminal and compare to values shown on system elementary (sheet 1DD, 1DE).

6-11. BRIDGE CURRENT FEEDBACK CALIBRATION (MVB):

Verify resistors and Berg jumpers on the XTFH against system elementary (sheet DA). Remove CT lead plugs JR and JT from the XTFH. Measure the resistance across each of the burden resistors on the XTFH; it should equal the "R BURDEN SETTING" value per the elementary. Adjust the corresponding pot to fine tune. Re-connect plugs JR and JT.

6-12. GATE TEST MODE OPERATION

Refer to the paragraph entitled GATING TESTS in Chapter 7 for a description of procedure.

Verify that the source-side circuit interrupt and load-side circuit interrupter are open.

Follow the procedure described in GATING TESTS.

6-13. CHECK PROCESS INPUTS

Verify that the source-side circuit interrupter and load side circuit interrupter are open and locked out.

The purpose of the tests in this section is to operate each external input shown on system elementary sheets BD, BE into the SEM. Operate each of these inputs by the most convenient means and confirm that the correct LED lights with a closed circuit in each case.

6-14. CHECK PROCESS OUTPUTS

Verify that the source contactor and load isolator are open and locked out.

Connect the hand terminal (or equivalent) to the HMPG and Z(ap) EEPROMs variable (IO TEST MODE) to "FF" to temporarily disable the interrupts. (See Chapter 5, section MONITOR - section Z(ap) for procedure.)

Using the OUTPUT command, operate the output relays shown on system elementary sheets BG, BJ in accordance with the table 6-1. Confirm that the relays operate the proper device(s) per the system elementary.

Relays may be operated in any desired combination by adding the appropriate DATA values from the table.

Re-zap IOTESTMODE to original value.

Table 6-1

SLOT 1A

ADDRESS	:	DATA	:	LED ON
8040	:	01	:	0
8040	:	02	:	1
8040	:	04	:	2
8040	:	08	:	3
8040	:	10	:	4
8040	:	20	:	5
8040	:	40	:	6
8040	:	80	:	7

ADDRESS	:	DATA	:	LED ON
8044	:	01	:	8
8044	:	02	:	9
8044	:	04	:	A
8044	:	08	:	B
8044	:	10	:	C
8044	:	20	:	D
8044	:	40	:	E
8044	:	80	:	F

6-15. NO-LOAD FULL-VOLTAGE TESTS

The tests in this section require that the source power be connected and available. Open and lock-out the load side circuit interrupter. (If no interrupter is available disconnect leads to motor.)

Apply the control power, reset the SEM and check that all the IMOK LEDs are on. Check for any fault indications. Investigate and clear any indicated faults. Null all meters.

6-16. DC OFFSET CHECK

This procedure should be performed whenever the NLIB card is replaced by a niw one.

a. With source power off and the control not in GATE-TEST-MODE, measure TP4(FACR), TP5 (FCBR), and TP6 (FACE) on NLIB's in slots 1C and 1D with a DVM in the millivolts dc range. Adjust dc offset pots R1, R2 and R3, respectively on the NLIB MOTHER BOARD so that TP4, TP5, and TP6 read 0.0 millivolts dc (±5).

b. Close source side circuit interrupter. Observe that all source side bridge neon lamps are illuminated.

6-17. VOLTAGE PHASING AND BALANCING

A. Connect an oscilloscope to simultaneously observe the waveforms on testpoints TP7 (VBA), TP8 (VCB), and TP9 (VAC) on the NSFC-2C. Confirm that the phase sequence is 1-2-3 and that all these are approximately equal.

The actual system voltage = (ATTN RATIO) X VAC

where ATTN RATIO is given on the system elementary sheet 1EA.

Using a DVM, measure the DC voltage offset at TP7, TP8 and TP9. This value should be less than 5 millivolts.

NOTE: Must measure between TP and ACOM testpoint on the card.

B. Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1C. Adjust R1(FLXAB) to obtain voltage as follows:

$$\text{FAB} = \frac{\text{ACTUAL SYSTEM VOLTAGE}}{\text{NOMINAL SYSTEM VOLTAGE}} \times 4.75 \text{ VAC RMS}$$

Nominal system voltage is given on the system elem. Sh. 1EA.

Now connect the voltmeter to TP2 (FBC) and adjust R2(FLXBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3(FLXCA). Balance signals as accurately as possible (+5 millivolts). Check for DC offset at TP4, TP5 and TP6 of NLIB (<10 millivolts).

C. Check for presence of zero crossing pulses at TP8 (SFT1) on the HLCB-2E. These are narrow 5v pulses evenly spaced at 2.78 millisecond intervals.

D. Connect the voltmeter to measure TP13 (VABS) on the NLIB-1C and check that the DC voltage level is approximately 2.229 X VAC at TP9 of NSFC-2C.

6-18. SET OVERVOLTAGE:

Connect the voltmeter to TP18 (OVR) on the NLIB-1C. Turn R8 (OV) slowly counterclockwise until the "OV" LED comes on. The source side circuit interrupter should trip.

Observe the DC voltage on TP-18 and re-adjust R8 so that the voltage is [per unit source overvoltage] times the trip value. (If incoming line is high or low, compensate by the ratio of RATED LINE VOLTS/ACTUAL LINE VOLTS.) See elementary sheet DC for per unit source overvoltage.

6-19. SET UNDERVOLTAGE.

Clear any faults and close the line contactor. Connect voltmeter to TP19 (UVR) of NLIB-1C.

Connect an oscilloscope to backplane pin 55-1C; this should be a logic zero. Turn R9 (UV) slowly CW until pin 55 just transitions high (may alternate high and low).

Observe TP19-1C and adjust UV(R9)-1C so that the measured value is per unit source undervoltage X trip level. (If incoming line is high or low, compensate by the ratio of RATED LINE VOLTS/ACTUAL LINE VOLTS.) See system elementary sheet 1DC for per unit source undervoltage.

6-20. SLAVE BRIDGE:

Using an oscilloscope, observe the waveforms on NSFC (2A) VBA (TP7), VCB (TP8), and VAC (TP9). Confirm that the phase sequence is TP7, TP8, TP9.

Use a DVM to measure the voltage - record each voltage.

TP7 = _____ TP8 = _____ TP9 =_____

The actual L-L system voltage = (ATTN RATIO) X V1-n avg.

where ATTN RATIO is given in system elem. (sh. EA)

6-21. CHECK PHASING BETWEEN MASTER AND SLAVE:

Using an oscilloscope, observe TP7 (VBA) - NSFC(2C) and TP7 (VBA) - NSFC(2A). Verify master(2C) leads slave(2A) by 30 degrees.

6-22. LOAD SIDE BRIDGE VOLTAGE TESTS WITH SIGNALS FROM SOURCE SIDE BRIDGE

Trip source circuit interrupter. Disconnect ribbon cable JA from NSFC-2B and move the ribbon cable JA from NSFC-2C to NSFC-2B. Connect a test jumper from 1C-55 to 1C-1 (disables source undervoltage fault).

6-23. VOLTAGE SIGNAL PHASING AND BALANCING

Close source circuit interrupter.

A. Connect oscilloscope to simultaneously observe the waveforms on testpoints TP7 (VBA), TP8 (VCB), and TP9 (VAC) on the NSFC-2B. Confirm that the phase sequence is 1-2-3 and all balanced. The testpoints should measure the same as for the source-side. Using a DVM, measure the DC voltage offset at TP2, TP3 and TP4. This value should be less than 5 millivolts.

B. Connect an RMS-measuring voltmeter to TP1 (FAB) on the NLIB-1D. Adjust R1 (FLXAB) to obtain voltage as follows:

$$FAB = \frac{\text{ACTUAL VOLTAGE}}{\text{NOMINAL MOTOR VOLTAGE}} \times 4.75\text{VAC RMS}$$

$$\times \frac{\text{Rated Freq.}}{60} \times \frac{\text{LOAD ATTN RATIO}}{\text{SRC ATTN RATIO}}$$

Rated frequency and nominal motor volts from elementary sheet 1EA.
Now connect the voltmeter to TP2 (FBC) and adjust R2 (FLXBC) for the same rms voltage as above. Repeat for TP3 (FCA) and R3 (FLXCA). Balance signals as accurately as possible (+5 millivolts). Check for DC offset at TP4, TP5 and TP6 of NLIB (< 10 millivolts).

C. Check for the presence of zero-crossing pulses at TP9 (SFT2) on the HLCB-2E. These are narrow 5vdc pulses evenly spaced at 2.78 millisecond intervals.

D. Connect the voltmeter to measure TP13 (VABS) on the NLIB-1D and check that the DC voltage level is the same as for the source-side 2.229 X VAC at TP9 of NSFC-2B

6-24. SET OVERVOLTAGE

Connect voltmeter to TP18 (OVR) on the NLIB-1D. Turn R8 (OV) slowly CCW until the "OVA" LED comes on. The motor contactor should trip. Observe the DC voltage on TP18 and re-adjust R8 so that the voltage is as follows:

$$\text{"OVA" volts} = 1.2 \times \text{trip value} \times \frac{\text{RATED MOTOR VOLTS}}{\text{ACTUAL LINE VOLTS}}$$

6-25. SET SPEED AND VOLTAGE INDICATORS

Since speed is derived from the motor voltage frequency, the source 60 Hz can be used to set the overspeed.

Set OSP -1E to +11VDC. Close-in line contactor. Connect a voltmeter to TP1-1E (SPD). Adjust R1 (SPD) until TP1 = 10.0 VDC.

Speed Indication:

Adjust R4 (SPM)-1E so that the speed indicator on the control panel reads rated motor speed for 60 Hz.

Motor Voltage Indication:

Adjust VM (R7) -1D on DLIB so that the motor voltmeter reads actual line voltage.

Set Overspeed:

Connect voltmeter to testpoint OSP (TP6)-1E and slowly turn OSP (R2)-1E CCW until the OSP LED comes on. The voltage on TP6 should be +10 VDC at the trip point. See elementary sheet DC for per unit overspeed.

Adjust OSP until TP6 = [per unit overspeed] X trip level.

Trip the source circuit interrupter.

Restore NSFC ribbon cables to correct configuration.

Remove jumpers.

6-26. CROWBAR TEST

The crowbar test permits controllable dc link current to be circulated without involving the motor windings. On the load-side, the dc current circulates though one or more diametric cell pairs bypassing the motor. The signals to be observed on the oscilloscope will indicate whether the source-side CTs are phased correctly.

a. Zap the following EEPROM variables:

CURLLSRC	to	60
CROWBARTESTMODE	to	1F
MANUALTCOM	to	85

b. Connect channel one of the scope to NSFC-2C (1F) SIG1 (TP1). Connect channel two of the oscilloscope to NLIB (1C) "OIA" (TP11). Trigger on channel one.
c. Reset SEM and apply primary power. Clear all existing faults.
d. Connect hand terminal to source-side HMPG.
e. Initiate a run sequence; i.e., sequence the drive to pick-up the software logic ladder's "RUN". The current regulator should now be active but regulating min current. Check that word MODEFLAG = 32H.(See elem. Sh. 1FF for sofrware logic "RUN").
f. Using the adjust command, increase (+) MANUALTCOM by steps of 8 to 100H.
g. Verify that test point "IFB" on NLIB (1C) = 1.25 VDC (+/- 30%).
h. Connect channel one to NLIB (1C), 0IA (TP11). Channel 2 to NSFC (2C), SIG1 (TP1). Compare the display to Figure 6-8.
i. Connect channel one to NLIB (1C), 0IB (TP10). Channel 2 to NSFC (2C), SIG3 (TP3). Compare the display to Figure 6-8.
j. Connect channel one to NLIB (1C), 0IC (TP9). Channel 2 to NSFC (2C), SIG5 (TP5). Compare the display to Figure 6-8.
k. Check NSFC (2C) - Vdc - (TP11) for saw-tooth (360 Hz) wave form.

6-27. CHECK SLAVE BRIDGE

a. Verify that testpoint IFB2(TP4) - NCCC(1F) = 1.25 VDC (+/- 30%).
b. Connect channel one to NCCC (1F), 0IA2 (TP1). Channel 2 to NSFC (2C), SIG1 (TP1). Compare the display to Figure 6-8.
c. Connect channel one to NCCC (1F), 0IB2 (TP2). Channel 2 to NSFC (2B), SIG3 (TP3). Compare the display to Figure 6-8.
d. Connect channel one to NCCC (1F), 0IC2 (TP3). Channel 2 to NSFC (2B), SIG5 (TP5). Compare the display to Figure 6-8.
e. Check NSFC(2A) - Vdc - (TP11) for saw-tooth (360 Hz) wave form.

Return CURLLSRC, CROWBARTESTMODE, and MANUALTCOM to original values.

6-28. "LOCK ROTOR"

The following test is designed to check the phasing of the load-side CTs. The motor may start and turn slowly, therefore, the lubrication system should be operating at this time.

a. Zap the following EEPROM variables:

MANUELFIXFIRE	to	1H
MANUELFREQ	to	CH
MANUALTCOM	to	85H

b. Reset the SEM and apply primary power.
c. Connect a brush recorder to the following signals; channel A to 0IA-1D (TP11) and channel B to SIG1-2B (TP1) Start brush recorder and then drive. Move brush to 0IB-1D (TP10) and SIG3-2B (TP3). Move brush to 0IC-1D (TP9) and SIG5-2B (TP5). Verify traces against Figure 6-8.

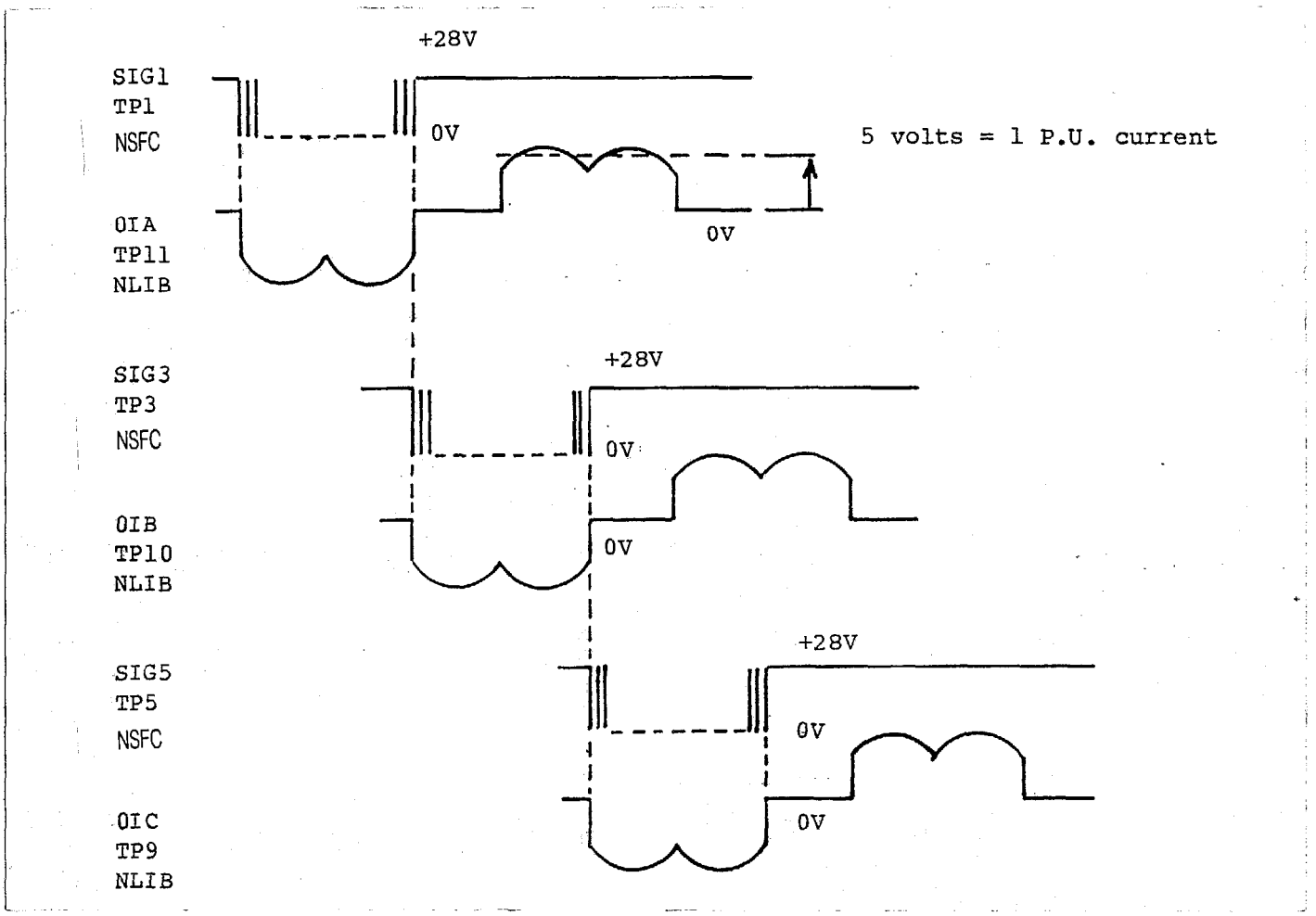


Figure 6-8. Firing Signals and Current Signals

d. After a few seconds move channel B of the brush to OIB-1D (TP10). After a few seconds move channel B to OIA-1D (TP11). Compare current brush traces against Figure 6-10.
e. Verify IFB-1C = 0.65 VDC (+/- 30%). Stop drive.
f. Return MANUELFIXFIRE, _FREQ, and _TCOM to original values.

6-29. RUN MOTOR UNCOUPLED - BYPASS

CONSIDERATIONS:
Bearing monitoring protection should be functioning at this point to protect the motor. The motor should be run to 50% speed once started and bearing temperature rise, vibration, and oil rings(slingers) checked.

NOTE: Because of no-load (or light load) condition, motor will run up to approximately [70-90% (shunt only)][30-35% (series-shunt)] speed - rather than set speed . Speed can be adjusted between this point and top speed only. Rated speed range will be available when normal load conditions exist.

A. Test Connections

1. Make the following connections to a six channel chart recorder:

CH1 - DA1(TP1) - 2E
CH2 - OIA(TP11) - 1D
CH3 - FCBR(TP5) - 1D
CH4 - OIC(TP9) - 1D
CH5 - IFB(TP7) - 1C
CH6 - SPD(TP10) - 1E2.

2. Temporarily Zap the following EEPROM variables:

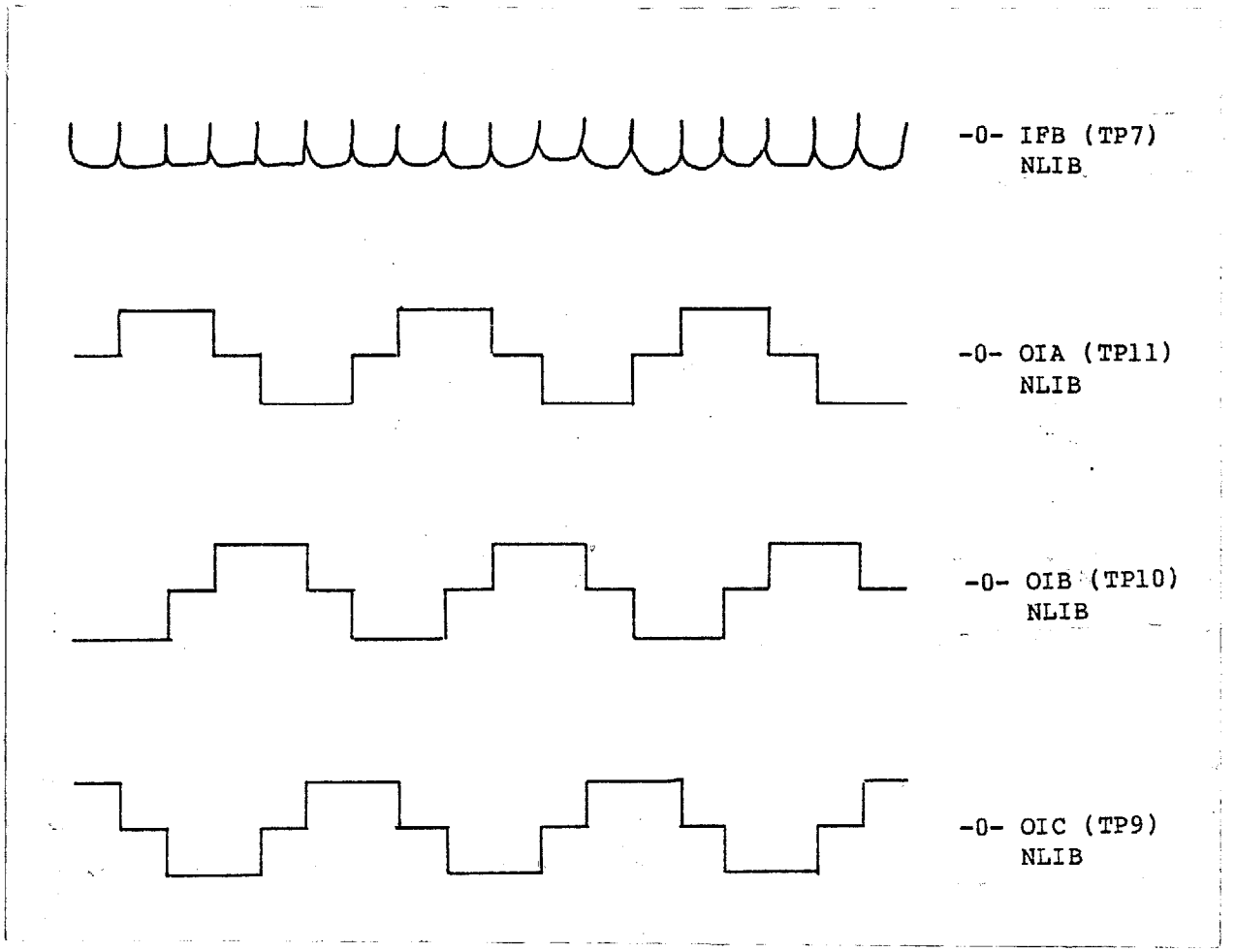


Figure 6-10.

SPDG1 to 8H
SPDG2 to 2H
STARTTCOM to FOH
SPDREGMAX to 156H

B. Start and Run Motor

1. Check that auxillary and control power is applied.
2. Clear all alarms and faults.
3. Close source circuit interrupter and rack-in load circuit interrupter (bypass remains racked-out).
4. Select local speed reference and set it to minimum.
5. Initiate run.

Monitoring MODE changes on DA1-2E:
starts from mode LOOK START(idle) 0 VDC
when run initiated
switches to mode FIX FIRE(open loop) 1.25 VDC
at approx. 5HZ
switches to mode DCLC(square-wave) 6.25 VDC
at approx. 13-15Hz DCLC starts in
switches to mode CHOPA(reduces 5th & 7th harmonics) 3.75 VDC
at approx. 20HZ
switches to mode CHOPB(reduces 5th harmonic) 5.0 VDC
at approx. 28HZ
switches to mode DCLC(no chopping-normal) 6.25 VDC
depending on load (firing angle)
switches to mode SELF(no DCLC) 7.5 VDC

If motor fails to start, then decrease STARTFREQ by steps of 1H (minimum of 6H) until motor starts.

6. Check motor rotation. If motor rotates backward, check IMD to motor wiring; swap two motor leads if necessary.
7. Initiate stop and motor should coast stop.

C. Speed Range Test

Start drive and gradually increase the drive speed to 50% speed . Check magnetic center, vibration, and bearing temperature rise. Verify panel speed meter reads rated rpm.

D. Speed Ramp Test

While monitoring speed with the brush recorder, decelerate the drive as fast as possible by changing the speed reference to minimum as quickly as possible. Then check accleration by increasing ref. as fast as possible.

E. Remote Operation

Place drive control in remote and verify drive follows remote reference and remote stop/start commands.

6-30. OPERATE WITH MOTOR COUPLED

a. Test Connections
Make the following connections to a six channel chart recorder.

CH1 - DA1	TP1-2E
CH2 - 0IA	TP11-1D
CH3 - FCBR	TP5-1D
CH4 - 0IC	TP9-1D
CH5 - IFB	TP7-1C
CH6 - SPD	TP10-1E

b. Starting Tune-up
The start-up of the IMD should be optimized by varying EEPROM variables STARTFREQ and STARTTCOM. The procedure is as follows:
1. Attempt to start drive and monitor chart recorder.
2. If you fail to get load flux, start decreasing STARTFREQ until system starts(starts getting flux) (to maximum 6H).
3. If IMD still fails to start, return STARTFREQ to original value, increase STARTTCOM, and repeat step 2.

c. Motor Flux Check
1. Start the drive and run up to some speed between 50% and 75%. Connect oscilloscope to testpoint "VAC" (TP7-2B). The waveform should appear as shown in Figure 6-11:

$$f = \frac{1}{T}$$

Calculate motor volts/Hz (flux).

$$\overline{V/Hz} = \frac{(\text{attn ratio})(Vpk/1.414)/f}{(\text{rated (rated motor volts})/60Hz}$$

The flux integrators on 1D should then have peak voltages of:

$Vf = (\overline{V/Hz}) * 6.718 =$ _____Vpk

d. Capacitor Current

1. Measure NSFC(2B)- V_{AC}(TP9) and drive frequency(freq)
2. Calculate capacitor current I_{cap}

$$I_{cap} = \frac{V_{LN}}{X_C} = V_{AC} * \text{ATINRATIO} * 2 * PI * FREQ * C$$

where ATINRATIO and C are given on system elem. sh. 1EA.

3. Adjust EEPROM variable LOADCAPE so that RAM variable

$$\text{ICAPFIL} = \frac{I_{CAP} * 1024}{\text{Rated Drive Current}} (\pm 5\%)$$

4. Zap LOADCAPE if changed.

e. Checking XMGAP value

1. Run drive at the middle of the operating (between the speed limits) range. The current should be greater than 15%.
2. Put the output of the sum of the flux regulator's proportional and intrgral gain paths - FLXPISUM (646) - out on D/A #1 (divide by 16).
3. Adjust EEPROM variable XMGAPE until FLXPISUM (132) approximates 0 Vdc. Then set +1 count so that FLXPISUM is slightly positive (+).

f. Flux Regulator

1. Check stability of the flux regulator and response (approx. 7 ±2 radian) .
2. Connect TP1 (DA1) -2E and TP7 (IFB)-1C to a chart recorder. Output FLXERR on DA1
3. Start the drive and run it up to approximately 60% speed. Apply a 5-10% step to the flux

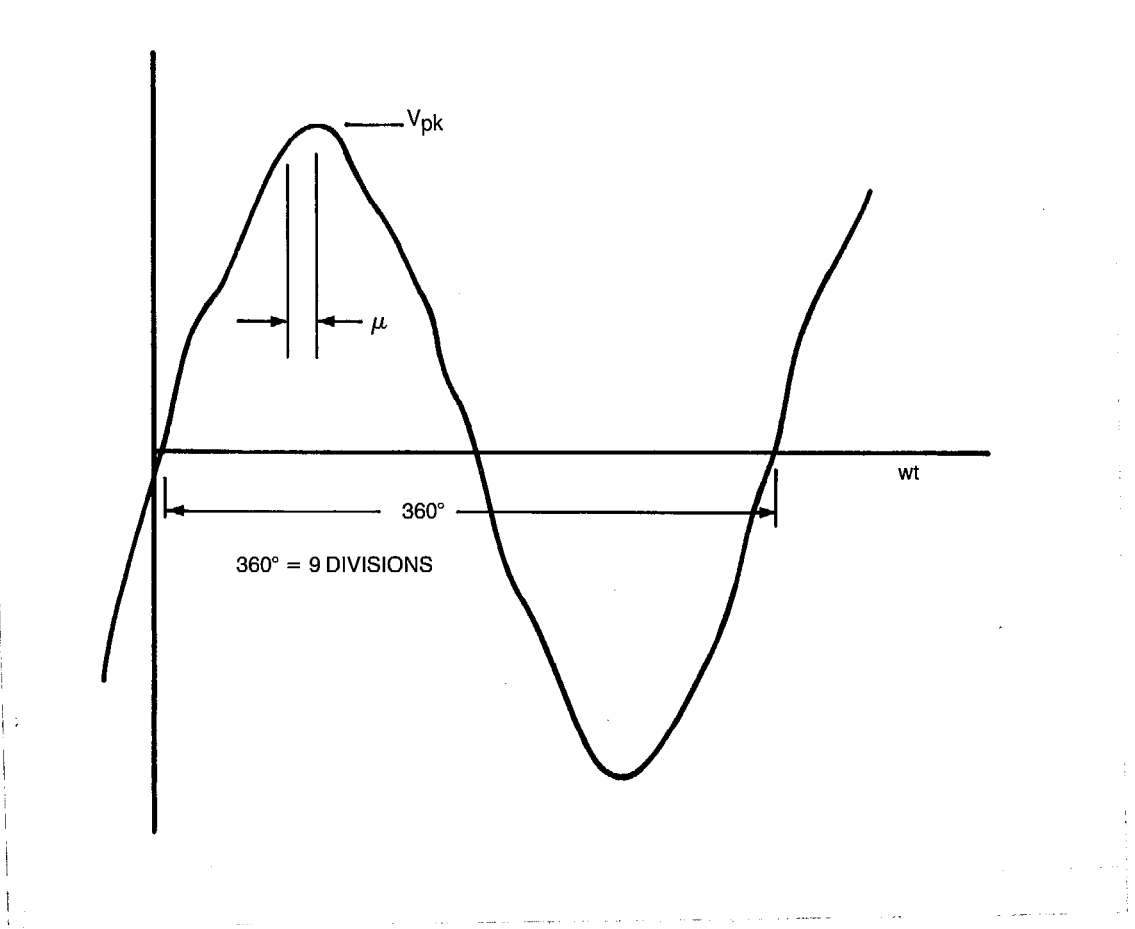
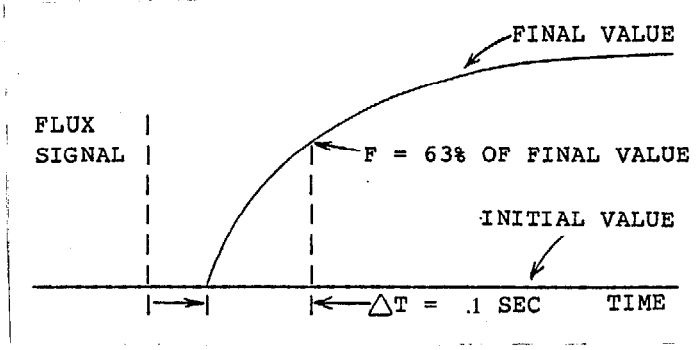


Figure 6-11.

regulator using the monitor SET command to place a value into the common memory variable FLUX STEP. A 5% step is equal to 33AH.

4. Observe the response on the chart recorder. Response should be critically damped and non-oscillating. Measure and record response.



5. If the response is oscillating, decrease EEPROM variables FLX INT GAIN and FLX PRP GAIN by the same ratio until the nearest desired response is obtained.
6. If the response is overdamped or underdamped, increase or decrease the integral gain FLX INT GAIN to obtain critical damping. If damping is now correct but response is not, return to step 3.
7. Zap EEPROM variables if changed.
8. Verify the NSFC (2B) - VDC(-) (TP11) for sawtooth (6X load freq) waveform.

g. Checking VDCGAIN value

1. Run drive at the middle of the operating range. The current should be greater than 15%.
2. Put CURCOMP out on D/A #1 (divide by 64).
3. Adjust EEPROM variable VDCGAINE until CURCOMP (/64) approxinates 0 Vdc. Then zap SCT +1 count so that CURCOMP is slightly positive (+).

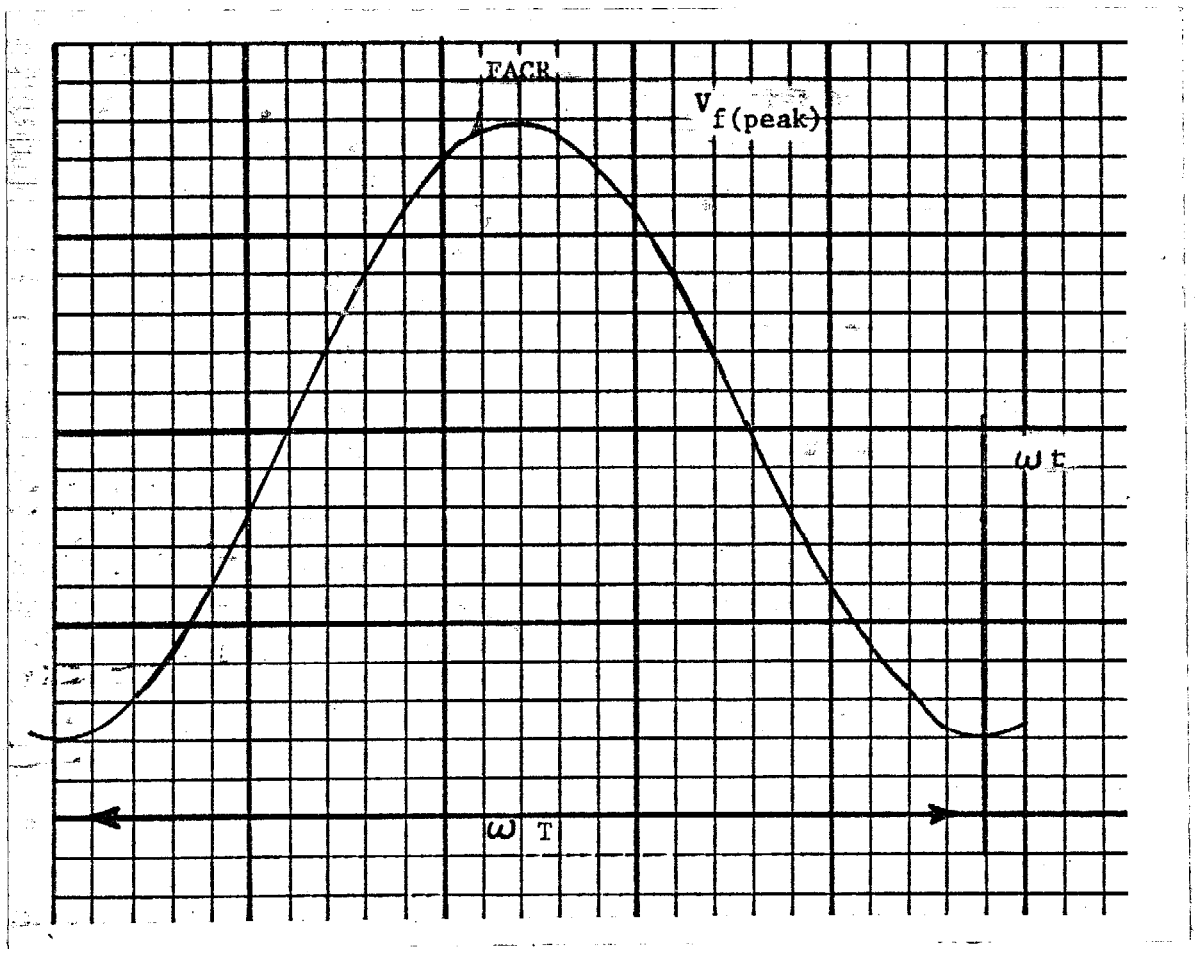
h. Current Regulator

The current regulator for the LCI drive is calculated for a 150 (±25) radian response. The parameters involved in this calculation

are fairly well known and measurement of the response is impractical.
The stability of the regulator can be checked by the following procedure:

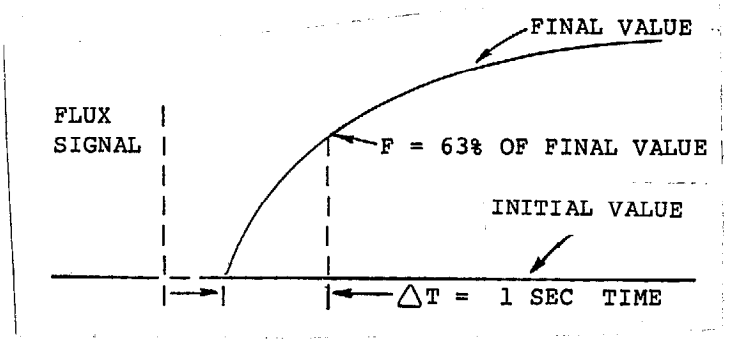
Run drive to middle of operating range and read several samples of the current command TCOM. Connect the brush recorder to IFB-1C 2;(TP7).
Step the current regulator by changing IREGSTEP - i.e., set to (+15% 66H) then immediately set it back to 0. The brush recorder should be running at its fastest speed.
Verify IFB-1C does not overshoot or oscillate on both the up-step to a higher current and speed and the return step to the lower current and speed.

i. Speed Regulator
Run the drive speed up to middle of operating speed range. Verify that CURDB = 00H. Use memory variable SPDREGSTEP to step the speed regulator. Select a step size (about 20H) to obtain as large a step as possible without hitting current limit (observe IFB on chart recorder).
NOTE: It may be necessary to buck-out the speed signal with a battery voltage to obtain enough amplification. If step down response is different from step up, use step down response.
Observe the step and adjust SPDG1 and SPDG2 for a 1 radian critically damped response as shown below. Maintain the same ratio between G1/G2 at all times:



The voltage at testpoints FBAR, FCBR, and FACR should appear as follows: It is important that these be balanced as closely as possible.

Zap these values for SPDG1 and SPDG2 into EEPROM. Record.



j. Source-side Line Impedance
Normally the source-side operates in the rectifying quadrant. However, when the drive is forced to slow down, the source converter may have to regenerate (invert). To avoid commutation failures, the angle of retard must be programmed as a function of current. Since a regenerating condition is difficult to hold steady-state, a measurement of system impedance will be made and this value will be programmed into EEPROM.
Run the drive to some convenient speed (the higher the better). Connect channel 1 of the oscilloscope to testpoint "VAC"-1G. Connect channel 2 to testpoint "IA"-1G. The signal should be similar to those shown in figure:

$\overline{V1\text{-}1}$ = (attn ratio)*(Vpk/1.414)/Vnom where Vnom = transformer nominal voltage

$\overline{Id} = I/5$

$\overline{Xc} = (\overline{V1\text{-}1}\ \overline{Id})\ [\cos\gamma - \cos(\gamma + \mu)]$

LINEIMP = 90 $(\overline{Xc})$
=_____D = _____H

Round-up to nearest integer and Zap into EEPROM.
Take drive to top spd, then decelerate as fast as possible. If drive trips on overcurrent then increase RTARDMARGSRC.

k. 1. Run drive to middle of operaing range and measure the drive horsepower (using chapter 6 - Section 6-31 -A.2.II and B.3)

2. Adjust NLCB(1E) -HPM so that horsepower meter reads measured horsepower.

l. 1. Measure motor current (IMM) if possdible and check that MIFB(TP) -NCCC(IF) =

$$\frac{IMM * 5Vdc}{\text{Rated Motor Current}}$$

where rated motor current is listed on sys. elem.sh, 1EA

2. Adjust GAIN-NLIB(1D) so that ammeter reads correct motor amps.

m. 1. Run motor to full speed and measure motor rpm and speed reference input variable SPD_STR.

2. Adjust EEPROM variable SLIPGAINE so that motor rpm = requested rpm

$$RPM = \frac{SPD_STR}{683} * (\text{Rated RPM @ 60Hz})$$

6-31. POWER READINGS

a. PCM input power factor and power readings (per channel) use Figure 6-13.

1. Power factor (PF)
   a. Measure μ
   b. Measure γ

$$\text{Power factor} = \frac{\cos(\gamma - \mu) + \cos\gamma}{2}$$

2.I. Kilowatts (KW)
   a. Measure power factor (PF).
   b. Measure OIA (peak average) (OIA).
   c. Measure peak voltage of VAC (Vacpk).

$$\text{Kilowatts} = \frac{\text{Vacpk} \bullet \text{attn ratio}}{1000 \bullet 2} * \frac{\text{OIA}}{5\text{V/unit}}$$

$$* \text{ rated bridge ac current} \bullet \text{PF}.$$

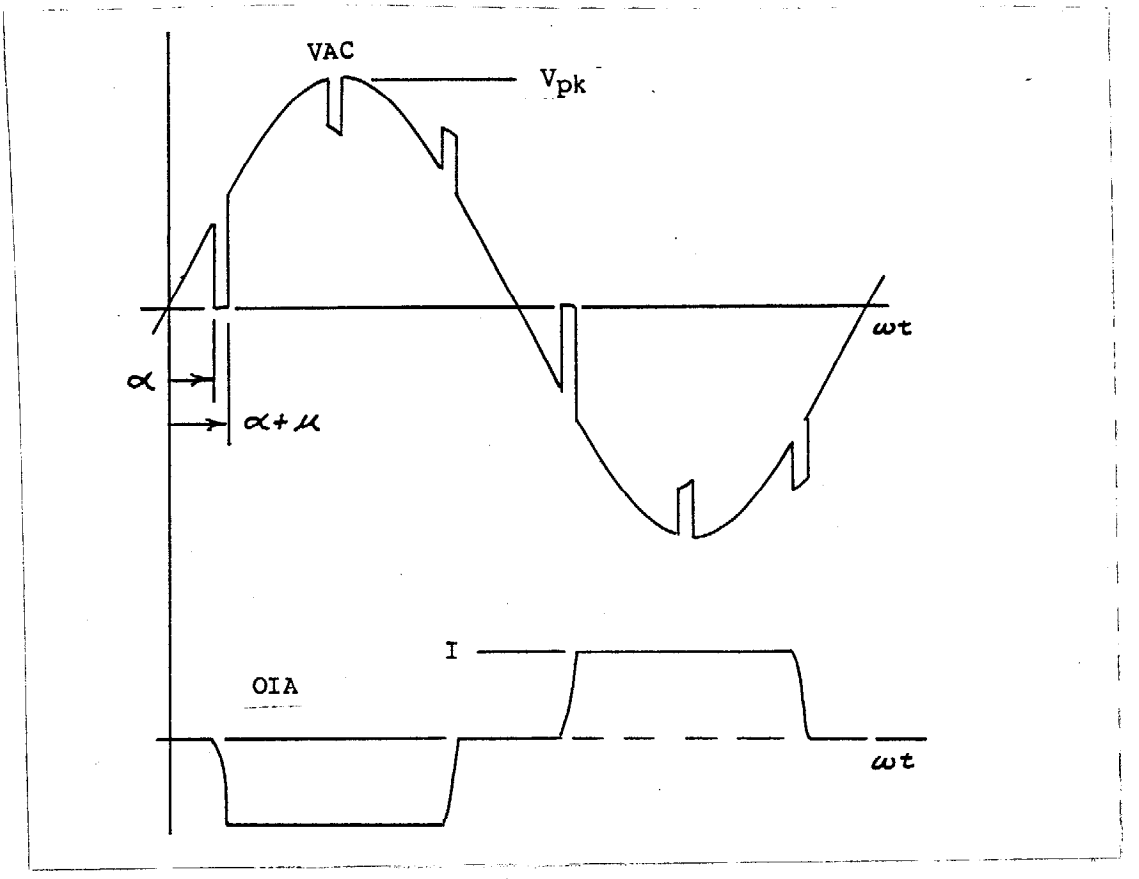
2.II Alternative to 2 (eliminates P.F reading)
   a. If VDC not being used, turn to min (full CCW).
   b. Measure VDC
   c. Measure IFB

$$\text{Kilowatts} = \frac{\text{IFB}}{5\text{V/volt}} \bullet \text{rated bridge ac current} \bullet 1.2825 * \frac{\text{VDC} \bullet \text{ATTN RATIO}}{1000}$$

b. Motor power factor and power readings - use Figure 6-12.
   1. Power factor (PF)
   With an oscilloscope, observe the potential transformer (PT) signal -VA and the current transformer (CT) signal - IA (with a current probe).
   Measure the power factor
   PF = cos (PF°)

   2. Kilowatts (KW)
      a. Measure power factor (PF)
      b. Measure voltage peak (V_{pk})
      c. Measure current peak - (I_{pk})

$$\text{Kilowatts} = \left[\frac{I_{pk}}{\sqrt{2}} \bullet \text{CT Ratio}\right] \bullet \left[\frac{V_{pk}}{\sqrt{2}} \bullet \text{PT ratio}\right] \bullet \left[\frac{1}{1000}\right] \bullet \text{PF}$$



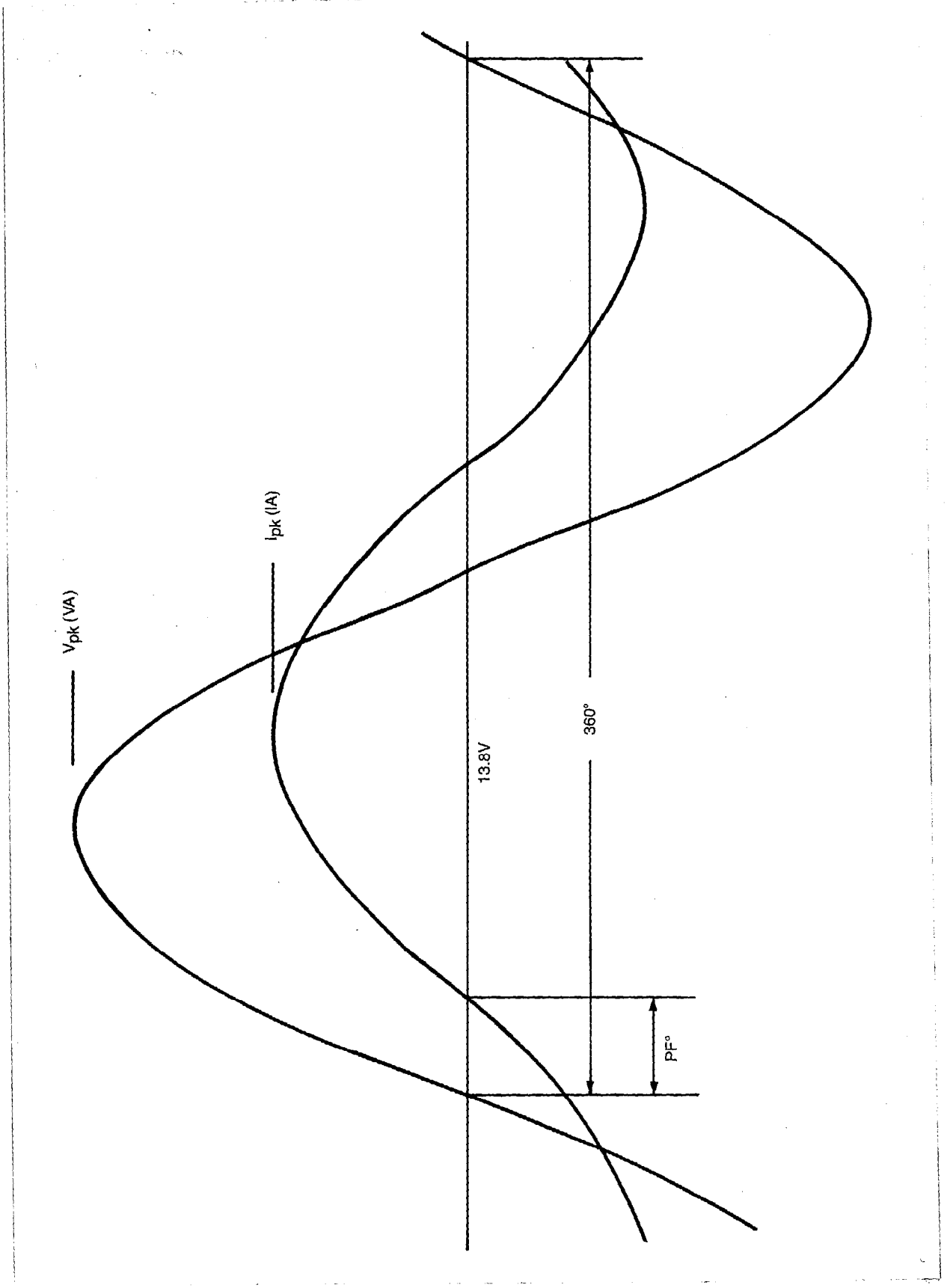


Figure 6-12.

3. Horsepower (HP)

a. Measure Kilowatts (KW)
Horsepower = • KW • Eff./7457
[Eff - efficiency can be estimated at 0.94 for large motors (see motor data sheets for accurate data)].

4. Torque (τ)
a. Measure horsepower (HP)

b. Measure SPD (RPM)
Put oscilloscope's time base on calibration for this entire measurement. Use as much as possible at the screen to display a half-cycle. Measure the time of this half cycle (T).
SPD = [(rated frequency) . rated RPM at rated frequency)/(2-t)]

$$\text{Torque} = \frac{5450 \bullet \text{HP}}{\text{SPD}}$$

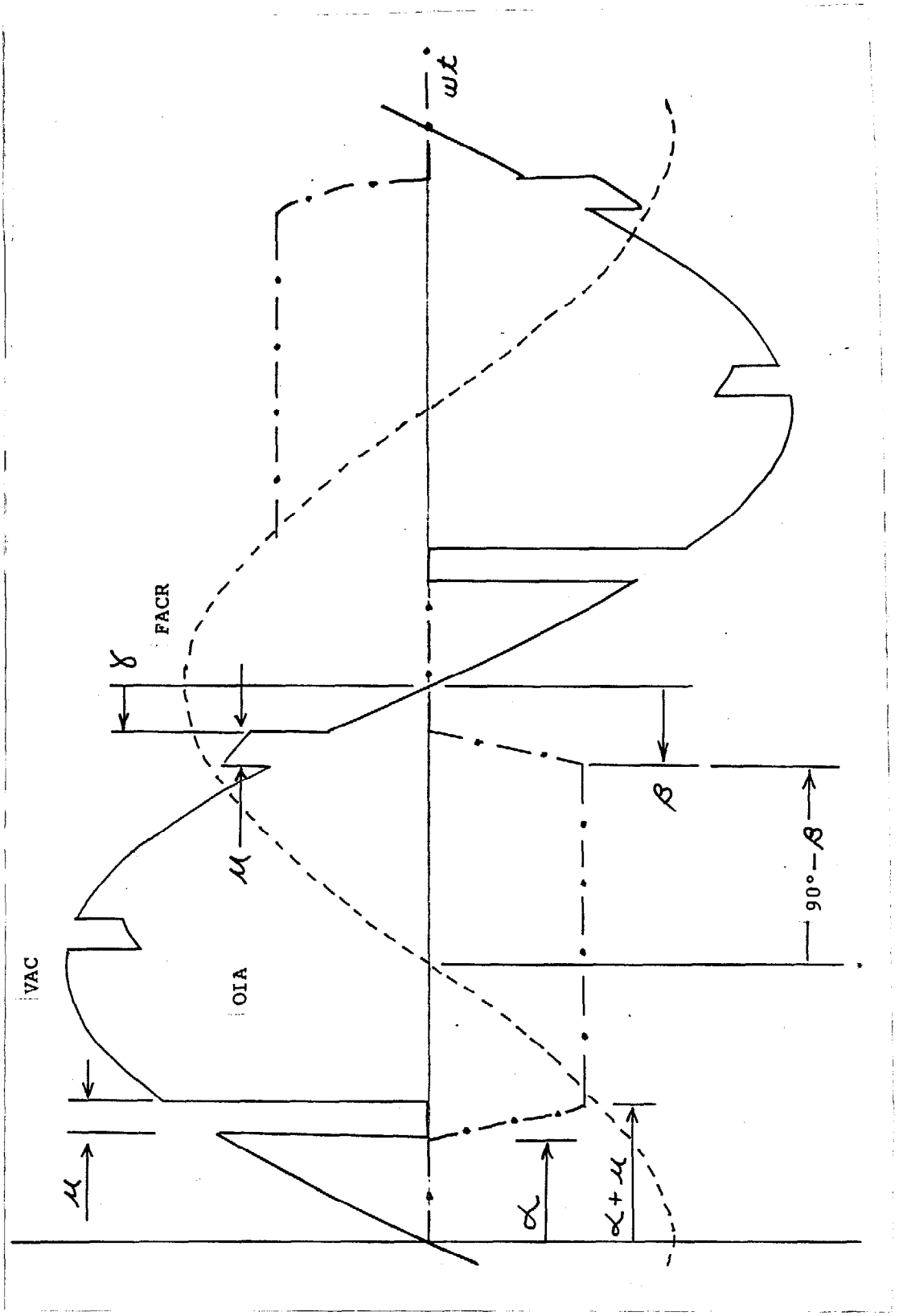


Figure 6-13.

CHAPTER 7

DIAGNOSTICS AND TROUBLESHOOTING

7-1. INTRODUCTION

Included in the drive power and electronic circuitry is a multi-layered diagnostic and troubleshooting capability. The main elements of this capability are the following:

1. Self testing of critical microcomputer system elements every time the control is reset or power applied.
2. A memory type fault monitor which displays the cause of shutdown in a printed message at the printer.
3. Lights and test points on front of cards provide the following information:
   a. Attenuated system bus voltages and currents on test points allow a check of magnitude and phasing.
   b. Gate pulse signals to converter bridge available at test points.
   c. Neon cell voltage indicating lights across each thyristor in power converters for indication of shorted cells.
   d. Overcurrent, overvoltage, and overspeed indicating lights in electronics module.
   e. Card front lights indicate status of all switching inputs and outputs to/from electronics module.
4. Gate-test-mode allows checkout of analog signal conditioning circuitry, microprocessor firing control, and cell gate signal integrity while off-line.
5. Serial monitor software which allows examination of system memory, inputs, and outputs-.(Chapter 5)
6. Capability to store diagnostic lists of system variables preceding and following fault conditions.(Chapter 4)

7-2. DIAGNOSTICS INITIALIZATION SELF TEST

SECTION A - SELFTEST

7-3. BASIC STRUCTURE

The Selftest routines are PLM86 procedures which may be executed by the HMPG resident in the Structured control Small Electronics Module (SEM) to test individual PWB's and functions. These routines are executed upon power-up if the TEST switch on the NLCB is in the up position. These routines may also be run when the drive is in a non-RUN condition by operator commands received via a terminal connected to the HMPG's RS-232 port or by depressing a test switch. The function of the Selftest is to test most of the control panel for the Structured DCLC-IMD, with great emphasis placed on the testing of the PWB's in the SEM.

Testing of the panel begins by performing tests of the hardware elements within a neighborhood of the HMPG's microprocessor and continues to tests of increasingly remote hardware. All of the tests incorporated within the selftest may be classed into 3 groups:
1. PWB-Specific tests
2. Paired PWB tests
3. Panel-level tests

7-4. PWB-SPECIFIC TESTS:

A great amount of effort has been placed on performing tests that will pinpoint problems on specific PWB's first. In this manner, if any easily diagnosed problem exists, it will be clearly spelled out to the operator, instead of vaguely defined by a list of possible solutions. Tests such as these comprise the PWB-Specific test group.

7-5. PAIRED PWB TESTS:

Tests which depend on the operation of several PWB's are classed within the Paired-PWB group. Diagnostic messages for failures of these tests will indicate a functional problem and also a list of possible solutions which may be general in nature.

CAUTION

Selftest excercises the output relays; therefore, the output terminal boards should be disconnected from the I/O module, and/or source and load breakers should be racked-out and drive stopped (if 12-pulse).

7-6. PANEL-LEVEL TESTS:

Panel-level tests also require correct operation of several PWB's or modules, however, these tests are segregated to allow the user to perform either a module-level test (involving only the control SEM) or a complete panel-level test which would include this last catagory of procedures. From a practical point of view, the sequence of testing within the control SEM progresses in the following manner:

1. HMPG
2. HLCB
3. HAIA
4. NSFC (source)
5. NSFC (load)
6. NSFC(DC LINK)
7. NLCB
8. NLIB (source)
9. NLIB (load)
10. HISA
11. HRDA

Further tests which include PWB's outside the SEM, progress in the following sequence:

1. Cables leading from the SEM.
2. the I/O Module.

7-7. EXECUTING THE TESTS

The self test routine may be initiated by resetting (toggle TP1 (RESET) to TP2 (DCOM) the HMPG with the TEST switch (on the NLCB) in the UP position. The entire test is completed in approx 20 seconds. At its completion, status information will be printed out at the door-mounted printer. In the present product, the entire self test routine is executed; however, provisions for executing small sections of the code are in place.

Upon completion of the self-test, the drive will operate in the gate-test-mode. To return to normal operation, the NLCB's test switch must be returned to the normal position and the HMPG must be reset (Toggle TP1 (RESET) to TP2 (DCOM)).

CAUTION

Selftest excercises the output relays; therefore, the output terminal boards should be disconnected from the I/O module, and/or source and load breakers should be racked-out and drive stopped (if 12-pulse).

7-8. DIAGNOSING FAILURES

As mentioned above, failure of any of the selftest procedures will produce a diagnostic message on the door-mounted printer. The format of the diagnostic message is:

DATE TIME	
FAILED	
(aabb) DURING TEST OF:	aa=TEST NUMBER
description of test	bb=SUB-TEST NUMBER
PROBLEM:	
problem statement	
POSSIBLE SOLUTIONS	
solution 1	
solution 2	
solution 3	

Usually, a failure can easily be diagnosed by the problem statement. At least one solution will be suggested in the print out. It is suggested that troubleshooting be performed at the PWB level (Replace the PWB not its component). Although the diagnostic messages may indicate more specific failures, it is not recommended that PWB components be replaced or repaired (always write or attach the diagnosis to the card)

The TEST NUMBER and SUB-TEST NUMBER uniquely define a failure within the Selftest procedure. The TEST NUMBER is incremented from 1 to 31 (hex) as the tests progress. For different test numbers, the SUB-TEST parameter will vary from 1 up to the number of sub-tests within that major test. The Description of Tests section that follows details the complete Selftest procedure according to the TEST NUMBER and SUB-TEST NUMBER parameters.

7-8.1. 6 DESCRIPTION OF TESTS. The following lists all tests available with sub-test information:

TEST	DESCRIPTION
1	HMPGROMTST: Reads all of ROM memory (F000:0 to F000:FFFF). Possible failures are:
.0	BusHog error
.1	- Read error
2	HMPGRAMTST: Reads all of RAM memory (0:0 to 0:3FFF).Possible failures are:
.0	- BusHog error
.1	- Read error
.2	- Write error
3	HMPGINTRTST
.1	- Verifies control and data lines to HMPG's 8259
4	HMPG_TMR
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
.6	- FOR TMR0: Checks for null count = 1
.7	- FOR TMR0: Loads TMR; checks for null count = 0
.8	- FOR TMR0: Verify TMR time out
.9	- FOR TMR0: Verify TMR reloads
.A	- FOR TMR1: Checks for null count = 1
.B	- FOR TMR1: Loads TMR; checks for null count = 0
.C	- FOR TMR1: Verify TMR time out
.D	- FOR TMR1: Verify TMR reloads
.E	- FOR TMR2: Checks for null count = 1
.F	- FOR TMR2: Loads TMR; checks for null count = 0
.10	- FOR TMR2: Verify TMR time out
.11	- FOR TMR2: Verify TMR reloads
4	HMPG_SIO_PRES
.1	- Check that trans and rec buffers are empty
5	HMPG_STALL
.1	- Verifies software reset of Stall circuit.
.2	- Initiate stall timer; checks for no STALL after 100
.3	- Verifies STALL asserted after 1.5 seconds.
7	HMPG_INTR
.1	- Verify operation of BusHog interrupt.
.2	- Verify that BusHog doesn't occur for legitimate read.
.3	- Verify operation of Time Tic interrupt.
8	DMPB
.1	- Checks that Write Enable Switch is not asserted.
.2	- Checks for no BusHog during EEPROM reads.
.3	- Checks for expected pattern in EEPROM.
9	HLCB_PRES
.1	- Enables IMOK; checks for 0IMOK=0.
.2	- Disables IMOK; checks for 0IMOK=1.
9	HLCB_DAK
.1	- Checks for no BusHog during reads of HLCB.
A	HLCB_TMR_U22
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
.6	- FOR TMR0: Checks for null count = 1
.7	- FOR TMR0: Loads TMR; checks for null count = 0
.8	- FOR TMR0: Verify TMR time out
.9	- FOR TMR0: Verify TMR reloads
.A	- FOR TMR1: Checks for null count = 1
.B	- FOR TMR1: Loads TMR; checks for null count = 0
.C	- FOR TMR1: Verify TMR time out
.D	- FOR TMR1: Verify TMR reloads
.E	- FOR TMR2: Checks for null count = 1
.F	- FOR TMR2: Loads TMR; checks for null count = 0
.10	- FOR TMR2: Verify TMR time out
.11	- FOR TMR2: Verify TMR reloads

TEST	DESCRIPTION
B	HLCB_TMRU23
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
.6	- FOR TMR0: Checks for null count = 1
.7	- FOR TMR0: Loads TMR; checks for null count = 0
.8	- FOR TMR0: Verify TMR time out
.9	- FOR TMR0: Verify TMR reloads
.A	- FOR TMR1: Checks for null count = 1
.B	- FOR TMR1: Loads TMR; checks for null count = 0
.C	- FOR TMR1: Verify TMR time out
.D	- FOR TMR1: Verify TMR reloads
.E	- FOR TMR2: Checks for null count = 1
.F	- FOR TMR2: Loads TMR; checks for null count = 0
.10	- FOR TMR2: Verify TMR time out
.11	- FOR TMR2: Verify TMR reloads
.12	- FOR TMR2: Initialize to Mode 0; verify status.
.13	- FOR TMR2: Load with 10 ms delay; verify status.
.14	- FOR TMR2: After 4 ms, verify status.
.15	- FOR TMR2: After 15 ms, verify time out.
.16	- FOR TMR2: Check that count = 0.
C	HLCB_INTR_PRE
.1	- Verify addr and data lines to HLCB's 8259.
D	HLCB_SIO
.1	- Initialize and check status of HLCB's 8251.
.2	- Verify that single byte of data is send, by sensing status.
.3	- Verify transfer of data pattern.
E	HAIA_PRES
.1	- Checks that HAIA is present by reading single bit.
F	HAIA_DAK
.1	- Checks for no BusHog during reads of HAIA.
.2	- Checks for no BusHog during writes to HAIA.
10	HAIA_TMR
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
.6	- FOR TMR0: Checks for null count = 1
.7	- FOR TMR0: Loads TMR; checks for null count = 0
.8	- FOR TMR0: Verify TMR time out
.9	- FOR TMR0: Verify TMR reloads
.A	- FOR TMR1: Checks for null count = 1
.B	- FOR TMR1: Loads TMR; checks for null count = 0
.C	- FOR TMR1: Verify TMR time out
.D	- FOR TMR1: Verify TMR reloads
.E	- FOR TMR2: Checks for null count = 1
.F	- FOR TMR2: Loads TMR; checks for null count = 0
.10	- FOR TMR2: Verify TMR time out
.11	- FOR TMR2: Verify TMR reloads
11	HAIA_CONVERSION
.1	- Start A/D conversion; check that BUSY=1.
.2	- Check that 1DAVAL = 0.
.3	- Wait for conversion to complete; check that BUSY=0.
.4	- Check that 1DAVAL = 1.
12	HAIA_PSUPPLY
.1	- Check that N15 supply is read correctly.
.2	- Check that P15 supply is read correctly.
13	NSFC_S_PRES
.1	- Set IMOK true; read back IMOK signal.
.2	- Set IMOK false; read back IMOK signal.
14	NSFC_S_TMR_PRES
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7

TEST	DESCRIPTION
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
15	NSFC_S_TMODE
.1	- FOR TMR0: Checks for null count = 1
.2	- FOR TMR0: Loads TMR; checks for null count = 0
.3	- FOR TMR0: Verify TMR time out
.4	- FOR TMR0: Verify TMR reloads
.5	- FOR TMR1: Checks for null count = 1
.6	- FOR TMR1: Loads TMR; checks for null count = 0
.7	- FOR TMR1: Verify TMR time out
.8	- FOR TMR1: Verify TMR reloads
.9	- Perform operations to set pulse train to 0; verify.
.A	- In TEST Mode, write all on pattern, expect echo of same
.B	- In TEST Mode, walk 1 thru primary & firing latch.
.C	- In TEST Mode, see if pulse train is stuck low.
.D	- In TEST Mode, see if pulse train is stuck high.
.E	- In TEST Mode, check that pulse train is toggling.
16	NSFC_L_PRES
.1	- Set IMOK true; read back IMOK signal.
.2	- Set IMOK false; read back IMOK signal.
17	NSFC_L_TMR_PRES
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
18	NSFC_L_TMODE
.1	- FOR TMR0: Checks for null count = 1
.2	- FOR TMR0: Loads TMR; checks for null count = 0
.3	- FOR TMR0: Verify TMR time out
.4	- FOR TMR0: Verify TMR reloads
.5	- FOR TMR1: Checks for null count = 1
.6	- FOR TMR1: Loads TMR; checks for null count = 0
.7	- FOR TMR1: Verify TMR time out
.8	- FOR TMR1: Verify TMR reloads
.9	- Perform operations to set pulse train to 0; verify.
.A	- In TEST Mode, write all on pattern, expect echo of same.
.B	- In TEST Mode, walk 1 thru primary & firing latch.
.C	- In TEST Mode, see if pulse train is stuck low.
.D	- In TEST Mode, see if pulse train is stuck high.
.E	- In TEST Mode, check that pulse train is toggling.
19	NSFE_PRES
.1	- Set IMOK true; read back IMOK signal.
.2	- Set IMOK false; read back IMOK signal.
1A	NSFE_TMR_PRES
.1	- Verifies addr lines & data bits 6 & 7
.2	- Verifies addr lines & data bits 6 & 7
.3	- Verifies addr lines & data bits 6 & 7
.4	- Verifies data bits 5-0 with walking 1 pattern
.5	- Verifies that all outputs are free to toggle.
1B	NSFE_TMODE
.1	- FOR TMR0: Checks for null count = 1
.2	- FOR TMR0: Loads TMR; checks for null count = 0
.3	- FOR TMR0: Verify TMR time out
.4	- FOR TMR0: Verify TMR reloads
.5	- FOR TMR1: Checks for null count = 1
.6	- FOR TMR1: Loads TMR; checks for null count = 0
.7	- FOR TMR1: Verify TMR time out
.8	- FOR TMR1: Verify TMR reloads
.9	- Perform operations to set pulse train to 0; verify.
.A	- In TEST Mode, write all on pattern, expect echo of same.
.B	- In TEST Mode, walk 1 thru primary & firing latch.

TEST	DESCRIPTION
.C	- In TEST Mode, see if pulse train is stuck low.
.D	- In TEST Mode, see if pulse train is stuck high.
.E	- In TEST Mode, check that pulse train is toggling.
1C	NSFE_DIAG
.1	- Tests for dormant field status information.
1D	NSFC_S_PLL
.1	- FOR TMR2: Enables PLL; Checks for null count = 1
.2	- FOR TMR2: Loads TMR; checks for null count = 0
.3	- FOR TMR2: Verify TMR time out
.4	- FOR TMR2: Verify TMR reloads
1E	NSFC_L_PLL
.1	- FOR TMR2: Enables PLL; Checks for null count = 1
.2	- FOR TMR2: Loads TMR; checks for null count = 0
.3	- FOR TMR2: Verify TMR time out
.4	- FOR TMR2: Verify TMR reloads
1F	NSFE_PLL
.1	- FOR TMR2: Enables PLL; Checks for null count = 1
.2	- FOR TMR2: Loads TMR; checks for null count = 0
.3	- FOR TMR2: Verify TMR time out
.4	- FOR TMR2: Verify TMR reloads
20	ROLL_CALL
.1	- Test for ribbon cable connecting HAIA(JA) to SEM(JG).
.2	- Enable NLCB IMOK; check for IMOK true.
.3	- Disable NLCB IMOK; check for IMOK false.
.4	- Enable NLIBS IMOK; check for IMOK true.
.5	- Disable NLIBS IMOK; check for IMOK false.
.6	- Enable NLIBL IMOK; check for IMOK true.
.7	- Disable NLIBL IMOK; check for IMOK false.

TEST	DESCRIPTION
21	NLCB_ANALOG_DORMANT
.1	- Test for 0 volts at REFO.
.2	- Test for 0 volts at TREFO.
.3	- Test for 0 volts at 12PREFO.
.4	- Inject current in REFO circuit; test for change in output.
.5	- Inject current in TREFO circuit; test for change in output.
.6	- Inject current in 12PREFO circuit; test for change in output.
.7	- Test for 0 volts at SP (speed).
.8	- Reset exixting faults; test for 1OSP (overspeed) = 0.
22	NSFE_ANALOG
.1	- Test for EI (exciter current) = 0 volts.
.2	- Test for no current imbalance
.3	- Inject signal in current circuit; test for output change.
.4	- Again, test for no current imbalance.
.5	- Excite current circuit with 55 Hz inputs; verify that currentimbalance is detected.
.6	- Wait 100 ms; verify no current imbalance condition.
23	NLIB_S_ANALOG
.1	- Reset existing faults; verify UV & not OV condition.
.2	- Verify not SUPPression conditon.
.3	- Verify 1UV70=0.
.4	- Test C17 for time delay. (Overvoltage circuit)
.5	- Test C17 for time delay. (Overvoltage circuit)
.6	- Test for 0 volts at IFBS.
.7	- Test for 0 volts at FLUX BAS.
.8	- Inject current in source current condition circuit. Detect change in IFBS.
.9	- Detect change in FLUX BAS.
.A	- For FLUX BAS: Check for 0 volts dc offset.
.B	- For FLUX BAS: Check level of positive peak.
.C	- For FLUX BAS: Check level of negative peak.
24	NLIB_L_ANALOG
.1	- Reset existing faults; verify UV & not OV condition.

TEST	DESCRIPTION
.2	- Verify not SUPPression conditon.
.3	- Verify 1UV70=0.
.4	- Test C17 for time delay. (Overvoltage circuit)
.5	- Test C17 for time delay. (Overvoltage circuit)
.6	- Test for 0 volts at IFBL.
.7	- Test for 0 volts at FLUX BAL.
.8	- Inject current in source current condition circuit. Detect change in IFBL.
.9	- Detect change in FLUX BAL.
.A	- For FLUX BAS: Check for 0 volts dc offset.
.B	- For FLUX BAS: Check level of positive peak.
.C	- For FLUX BAS: Check level of negative peak.
.D	- For FLUX CBS: Check for 0 volts dc offset.
.E	- For FLUX CBS: Check level of positive peak.
.F	- For FLUX CBS: Check level of negative peak.
.10	- For FLUX ACS: Check for 0 volts dc offset.
.11	For FLUX ACS: Check level of positive peak.
.12	For FLUX ACS: Check level of negative peak.
25	HLCB_INTR
.1	- Enable test oscillator; verify occurance of source zero crossing interrupt (slave intr 0).
.2	- Verify frequency of source zero crossing intr.
.3	- Enable test oscillator; verify occurance of load zero crossing interrupt (slave intr 1).
.4	- Verify frequency of load zero crossing intr.
.5	- Verify occurance of source firing interrupt.
.6	- Verify occurance of load firing interrupt.
.7	- Verify occurance of field firing interrupt.
26	NLCB_ANALOG_ACTIVE
.1	- Enable test oscillator; verify speed circuits reads 55 Hz.
.2	- Verify no overspeed condition.
27	HISA_PRES
.1	- Disable IMOK; verify by reading back bit.
.2	- Enable IMOK; verify by reading back bit.
28	CABLE_CONT
.1	- Tests for continuity in ribbon cable from SEM,JA to I/O Module.
.2	- Tests for continuity in ribbon cable from SEM, JKto I/O Module.
.3	- Tests for continuity in ribbon cable from SEM, JHto I/O Module.
.4	- Tests for continuity in ribbon cable from SEM, JB to Door Module.
29	HISA_IO
.1	- Set all inputs from I/O Module high; verify by reading back inputs to HISA.
.2	- Set all inputs from I/O Module low; verify by reading back inputs to HISA.
2A	P105_SUPPLY
.1	- Check that 105 volt dc power supply is OK.
2B	HROA_1A_IO: In sequence, pick-up single relays on the HROA in slot 1A of the I/O module. Expect feedback signal to indicate change.
.1	-Force all HROA(1A) relays to drop-out; verify by feedback.
.2	-Force 105 Vdc relays to pick-up; verify by feedback.
.3	-Drop out 105 Vdc relays, wait 40 ms, verify by feedback.
.4	-Pick-up single 28 V relay; verify by feedback.
.5	-Drop-out all relays; verify by feedback.
.6	-Pick-up single 28 V relay; verify by feedback.
.7	-Drop-out all relays; verify by feedback.
.8	-Pick-up single 28 V relay; verify by feedback.
.9	--Drop-out all relays; verify by feedback.
.A	--Pick-up single 28 V relay; verify by feedback.

TEST	DESCRIPTION
.B	--Drop-out all relays; verify by feedback.
2C	HROA_1B_IO: In sequence, pick-up single relays on the HROA in slot 1B of the I/O module. Expect feedback signal to indicate change.
.1	-Force all HROA(1A) relays to drop-out; verify by feedback.
.2	-Force 105 Vdc relays to pick-up; verify by feedback.
.3	-Drop out 105 Vdc relays, wait 40 ms, verify by feedback.
.4	-Pick-up single 28 V relay; verify by feedback.
.5	-Drop-out all relays; verify by feedback.
.6	-Pick-up single 28 V relay; verify by feedback.
.7	-Drop-out all relays; verify by feedback.
.8	-Pick-up single 28 V relay; verify by feedback.
.9	-Drop-out all relays; verify by feedback.
.A	-Pick-up single 28 V relay; verify by feedback.
.B	-Drop-out all relays; verify by feedback.
.C	-Pick-up single 28 V relay; verify by feedback.
.D	-Drop-out all relays; verify by feedback.
.E	-Pick-up single 28 V relay; verify by feedback.
.F	-Drop-out all relays; verify by feedback.
2D	NLCB_R_DVR
.1	- Verify presence of HROA in slot 1B of I/O Module.
.2	- Verify presence of HROA in slot 1A of I/O Module.
.3	- Force relay drivers on NLCB on; verify response from I/O Module via feedback.
.3	- Force relay drivers on NLCB on; verify response from I/O Module via feedback.
2E	DOOR_MOD_PRES
.1	- Test for CTS asserted.
.2	- Test for DSR asserted.

After the microcomputers have successfuly powered up and/or self-test operation have been finished successfuly, the printer can then display faults/alarms as described below.

7-9. FAULT/ALARM MONITOR

Built into the electronics software module are fault/alarm monitors.

7-10. FAULT DETECTION

In the event of a fault condition, the IMD will shut down and a fault indication will appear on the SEM. The cause of the shutdown will be displayed on the printer terminal located onthe control panel door. If more than one fault has occurred, the fault finder will store and print the subsequent faults (up to 10 total).

To reset the control after a fault condition, it is necessary to press the reset switch on the HLCB card in the SEM on the reset pushbutton on the control panel door. But first record the fault/alarm/relay status, and I/O states. If the fault condition is clear, the fault light will go out after a door reset, and stay out when the reset is completed.

Consult the system elementary ladder diagram for a complete list of faults. Most of the faults are standard and are defined in the following paragraphs, along with suggested corrective action.

LOSS OF COOLING FAN- This fault indicates a loss of bridge cooling fan(s). In units having redundant fans, this fault occurs only after both the normal and backup fans have failed.

Possible action to be taken:

1. Replace fan motor fuses
2. Replace defective fan motor (and/or centrifugal switch)
3. Replace HRDA boards
4. Replace I/O module

SOURCE, LOAD, MASTER, SLAVE HARDWARE OVERCURRENT - This fault indicates that the source or load side current has exceeded an acceptable level for more than 26 milliseconds. Under normal circumstances, the electronic control can reduce current to zero in less than 26 milliseconds. Actuation of the hardware overcurrent generally indicates a severe problem from which the control cannot recover.

Possible action to be taken:

1. Check converter bridges for shorted cells
2. Replace NLIB card

MICROPROCESSOR STALL - This microprocessor has failed to execute its program in a normal amount of time.

Possible action to be taken:

1. Replace HMPG card
2. Replace HITA card
3. Replace NLCB and NLIB cards

LOAD, SOURCE, MASTER, SLAVE SOFTWARE OVERCURRENT - Source and load current are compared in software to an adjustable overcurrent thresh-

old. If the current exceeds this threshold, both source and load converters will act to reduce the current to zero. If it is not possible to reduce current to zero and regain control, a fault will be declared and the drive will be tripped.

Possible action to be taken:

1. Go into gate-test-mode and look for missing pulse
2. Replace in order NLCB, NLIB, HAIA cards
3. Reset and try again

SOURCE, LOAD OVERVOLTAGE - Indicates an excessive level of voltage on the source or load ac buses.

Possible action to be taken:

1. If source overvoltage, check incoming voltage from switchgear.
2. Replace HLCB, NLIB cards

OVERSPEED - Indicates excessive motor speed has occurred.

Possible action to be taken:

1. Replace NLCB PWB
2. Check attenuators

GROUND FAULT - Indicates a ground exists in the converter bridges, transformer, reactor, or motor. The ground detector will also actuate if the load converter ac line filter circuits are unbalanced due to an open fuse or defective capacitor.

Possible action to be taken:

1. Check AC and DC filter fuses and capacitors in bridges
2. Check for grounds in motor, transformer, reactor (megger)
3. Replace NLCB PWB

For other possible faults and solution patterns see Appendix C.

7-11. ALARM DETECTION

Alarms are conditions which are cause for concern but which do not warrant immediate shutdown of the drive but shoud be investigated as soon as possible. A composite alarm signal is generally supplied to operate a light on the control panel door, a remote annunciator supplied by the user, or both.

Some of the alarms will vary from one installation to the next, so the software ladder diagram section of the system elementary must be consulted for a complete list of. The following alarms are standard (if applicable) and so will be defined here.

LOSS OF SPEED REFERENCE - When the drive is controlled from a 4 - 20 milliamp speed reference, a check is made to determine if this signal has gone to zero (Less than 4 mA). If so, an alarm is declared and the drive speed is locked at its present value.

The remaining alarms to be defined deal with unused or unexpected interrupts. In case one of these interrupts occurs, software is present to record data, declare an alarm, and then return to normal processing. The data saved upon execution of one of these interrupts is stored in a stack in RAM memory. There is also a counter for each of these interrupts so that the number of occurrences can be logged. The stack variable and event counter names are given in Table 5-2. The definition of the information contained in the stack is given in Table 5-3. It is not expected that the user would be able to fully utilize this information, but it is given here in the interest of completeness.

Rapid repetition of these interrupts can result in shutdown of the system but it has been shown that the system may recover from these if they are infrequent. The cause of these interrupts is generally due to noise or other unusual conditions stated in the following definitions.

I/O ADDRESSING - Whenever any memory or I/O device is addressed by the microprocessor, it is expected that that device will respond with an acknowledge signal. If an acknowledge is not received, the microprocessor will wait for up to 80 microseconds, generate a bus hog interrupt, and then proceed to the next task. Failure to receive this acknowledge indicates that some memory or I/O device is "hogging" the data bus, and an alarm is declared. Actually, addressing an undefined address with the terminal will invoke a bus hog alarm. The bus hog alarm may be tested in this manner by executing the following command:

DW500:500 CR

Continued receipt of the bus hog alarm usually indicates that a digital card is defective or not plugged in properly, or a wrong address was requested.

The following are combined to form one composite alarm - MICROPROCESSOR ALARM - DUMMY INTERRUPT (DUM INT) - If any of the masked (disabled) interrupts should occur, a dummy interrupt alarm will occur.

DIVISION BY ZERO INTERRUPT (DIV ZERO) - This is a non-maskable interrupt that is invoked whenever the microprocessor attempts a division by zero.

POWER DOWN INTERRUPT (POW DN) - This is a non-maskable interrupt that places the microprocessor in a special mode where all instructions and data must be one byte in length. This interrupt is generated whenever the result of a calculation exceeds the available register size.

SINGLE STEP INTERRUPT (SING STP) - A single step alarm is an alarm declared whenever the non-maskable interrupt used to single step the microprocessor execution is invoked. Single step mode is not used in the LCI control.

SYSTEM TIMING INTERRUPT (SYS TIM) - This is a non-maskable interrupt from an external timer that is ordinarily used to give a "heartbeat" interrupt for overall system timing functions. This interrupt is not used on the LCI.

USART RECEIVE, TRANSMIT (USART R, USART T) - USART receive, transmit is an interrupt that is generated whenever the USART fails to indicate that it is ready to receive, transmit data. These interrupts are not used in the LCI control.

Table 5-2.
INTERRUPT ALARM COUNTER AND STACK VARIABLES

ALARM	: STACK ARRAY VARIABLE	: TRAP CNTR BH
DUM INT	: SAVE STACK DUM	: TRAP CNTR DUM
DIV ZERO	: SAVE STACK 0	: TRAP CNTR 0
POW DN	: SAVE STACK 2	: TRAP CNTR 2
ONE B	: SAVE STACK 3	: TRAP CNTR 3
OV FL	: SAVE STACK 4	: TRAP CNTR 4
SING STP	: SAVE STACK 1	: TRAP CNTR 1
SYS TIM	: SAVE STACK SYS TIM	: TRAP CNTR SYS TIM
USART R	: SAVE STACK US R	: TRAP CNTR US R
USART T	: SAVE STACK US T	: TRAP CNTR US T

NOTE: In order to access information in the stacks, it is necessary to obtain the address of the desired SAVE STACK variable from a set of memory maps. Then the data from SAVE STACK ADDRESS to SAVE STACK ADDRESS + 1AH must be printed. To do this, use the following command structure:

<DW SAVE STACK ADDRESS , SAVE STACK ADDRESS + 1AH CR>

NOTE: See paragraph entitled "Monitor Command Structure" for an explanation of < >.

To obtain the trap counter value, the following must be done:

<DW TRAP CNTR ADDRESS CR>

7-12. SECTION B - GATE-TEST-MODE

7-12.1. Gate-Test-Mode. The gate test mode allows off-line testing of much of the analog signal conditioning, microprocessor firing control, and gate pulse generation circuitry. In the gate-test-mode, a local test oscillator supplies 3 phase, approximately 60 Hz signals in place of the attenuated bus signals from the source and load bridges. The electronic control can then lock onto these signals and generate signals

Table 5-3.
MAP FOR INTERRUPT SAVE STACKS

STACK OFFSET	REGISTER	DEFINITION
0	BP	
2	ISR	CALL RETURN (OFFSET)
4	ISR	CALL RETURN (SEGMENT)
6	DI	
8	DI	
A	BX	
C	DX	
E	CX	COUNT REGISTER
10	AX	16BIT ACCUMULATOR
12	DS	DATA SEGMENT
14	ES	EXTRA SEGMENT
16	IR	INTERRUPTED ROUTING (OFFSET)
18	IR	INTERRUPTED ROUTING (SEGMENT)
1A	FLAG	FLAG REGISTER

to "dry fire" the thyristors in both bridges. Thyristor gate pulses may then be examined for proper magnitude and phasing. Firing may be adjusted between advance and retard limit for each bridge. In the gate-test-mode, the control produces pulse train firing, thus it is necessary to run the converter ventilating fans for this test.

To perform the following tests, a dual channel oscilloscope with voltage probes and a current probe is required.

7-12.2. Getting Into Gate-Test-Mode

a. The drive should be stopped, primary switchgear and/or transformer isolation switch (if any), and load isolation device (if any) should be open. Check the software ladder diagram in the system elementary for any other conditions that must be met.
b. Put the NLCB (1E) "TEST" switch (SW1) in the test position.
c. Reset the electronics module by pressing the "RESET" (SW1) switch on the HLCB (2E or 2J).

Note: Always use DC coupling on the oscilloscope channels.

7-12.3. Gate-Test-Mode Checkout

a. Connect channel one of the oscilloscope to NLIB (1C) "FCBR" (TP5) and channel two of HLCB (2E) "VA1" (TP12). Trigger the scope on channel one and verify the waveforms look like figure 7B-1.
b. Connect channel one of the oscilloscope to NLIB (1D) "FCBR" (TP5) and channel two to HLCB (2E) (TP13). Trigger the scope on channel one and verify the waveforms look like figure 7B-1.
c. Connect channel one of the oscilloscope to HLCB (2E) "VA1" (TP12) and trigger on

this signal. Adjust the variable time/division scale so that one period equals six major divisions. (i.e, 60 degrees per major division or 12 degrees per minor division).

d. One at a time, connect channel two to SIG1 thru SIG6 on the source side "A" NSFC (2C) (TP1 thru TP6) and verify that the gate pulses are positioned as in figure 7B-2A. Note: The firing angle can be adjusted by varying the speed demand reference (place in "LOCAL" and adjust "LOCAL REF").

e. One at a time, connect channel two to SIG1 thru SIG5 on the source side NSFC (2A) (TP1 thru TP6) and verify that the gate pulses are positioned as in figure 5-2B. Note: The slave firing lags the master firing by 30°.

f. Connect channel one of the oscilloscope to HLCB (2E) "VA2" (TP13) and trigger on this signal. Adjust the variable time/division scale so that one period equals six major divisions. (i.e., 60 degrees per major division or 12 degrees per minor division).

g. One at a time, connect channel two to SIG1 thru SIG6 on the load side NSFC (2B) (TP1 thru TP6) and verify that the gate pulses are positioned as in Figure 7B-2A.

h. Verify that each of the red gate pulse LEDS on the HPTN cards in the leg cell modules is illuminated indicating that the cell is getting gate pulses.

i. With a current probe, look at the gate pulse of each thyristor while simultaneously monitoring the corresponding gate pulses on the NSFC card. The two signals should line up with each other as shown in Figure 7B-4A.

j. Connect a current probe to the scope and trigger on this channel. Hook the current probe to each of the gate leads (white wire) of the SCRs with the arrow on the current probe pointing away from the gate lead termination point. Verify that the waveform has the same characteristics as Figure 7B-4B. If the pulse fails the criterion, the HPTN card in that leg cell module is most likely the cause, or the NTDA connectio0ns may be reversed.

k. Connect channel one of oscilloscope to NLCB (2E) "VA2" (2E) and channel two to NCCC (1F) "GSH" (TP13). Verify against figure 7B-5A. Note that the top GTO pulse train is for the source angle at approximately 120°. As the angle increases the time is reduced until the bottom GTO train is reached. The GTO doesn't fire when the angle is less than 120°. Also verify that "GSR" (TP14) NCCC (1F) is as shown.

l. Connect channel one of oscilloscope to NCCC (1F) "GSH" (TP13) and with a current probe look at the gate pulse at each of the shunt GTOs (A-D). Verify Figure 7B-5B. The off pulse varys in width and amplitude, increasing as the actual drive current increases. Also check the series GTOs (A-D) (if used).

m. Put the NLCB (2E) "TEST" switch (SW1) in the OFF position and reset the electronics module by pressing the "RESET" (SW1) switch on the HLCB (2E).

Table 2.
Master Bridge

CORRESPONDING TP ON NSFC	GATE LEAD OF EACH SCR
SIG1(TP1, 2B)	LTM1
SIG1(TP2, 2B)	LTM2
SIG1(TP3, 2B)	LTM3
SIG1(TP4, 2B)	LTM4
SIG1(TP5, 2B)	LTM5
SIG1(TP6, 2B)	LTM6

Table 2A.
Master Bridge

CORRESPONDING TP ON NSFC	GATE LEAD OF EACH SCR
SIG1(TP1, 2C)	STM1A
SIG1(TP2, 2C)	STM2A
SIG1(TP3, 2C)	STM3A
SIG1(TP4, 2C)	STM4A
SIG1(TP5, 2C)	STM5A
SIG1(TP6, 2C)	STM6A

Table 2B.
Master Bridge

CORRESPONDING TP ON NSFC	GATE LEAD OF EACH SCR
SIG1(TP1, 2A)	STM1B
SIG1(TP2, 2A)	STM2B
SIG1(TP3, 2A)	STM3B
SIG1(TP4, 2A)	STM4B
SIG1(TP5, 2A)	STM5B
SIG1(TP6, 2A)	STM6B

7-13. SECTION D - SERIAL MONITOR AND HAND TERMINAL (OR OTHER DIAGNOSITIC TERMINAL)

Integrated into the firmware for both source and load electronics is a monitor program which supports manipulation of system memory and I/O. The serial monitor drives an RS-232 port on the front of each microcomputer card. Communicaion with the microcomputer can be established by connecting the handheld terminal suppllied with the drive to RS-232 port. This connection must be made using the HNMB null modem (supplied) with the electronics module) as illustrated in Chapter 5. Observe the plug orientation and Berg jumper positions marked on the null modem.

Figures 7D-1 and 7D-2 are sample cable connections to other terminals.

7-13.1. SECTION C - SOFTWARE REGULATOR BLOCK DIAGRAMS. Following are symbolic block diagrams of various regulators and functions found in the restructured LCI software. Note - most of the regulator diagrams are presented in both "S" domain (analog) and Z Domain (digital). Also, most of the variables listed (in "S" domain) are available by name through the monitor (Review Chapter 5) either as EEPROPM variable (Chapter 4, Section A) or public variables (Chapter 4, Section B).

These diagrams should be useful for understanding the overall control of the drive and for troubleshooting by identifying variable that could be checked or used on the diagnostic list or D/A function.

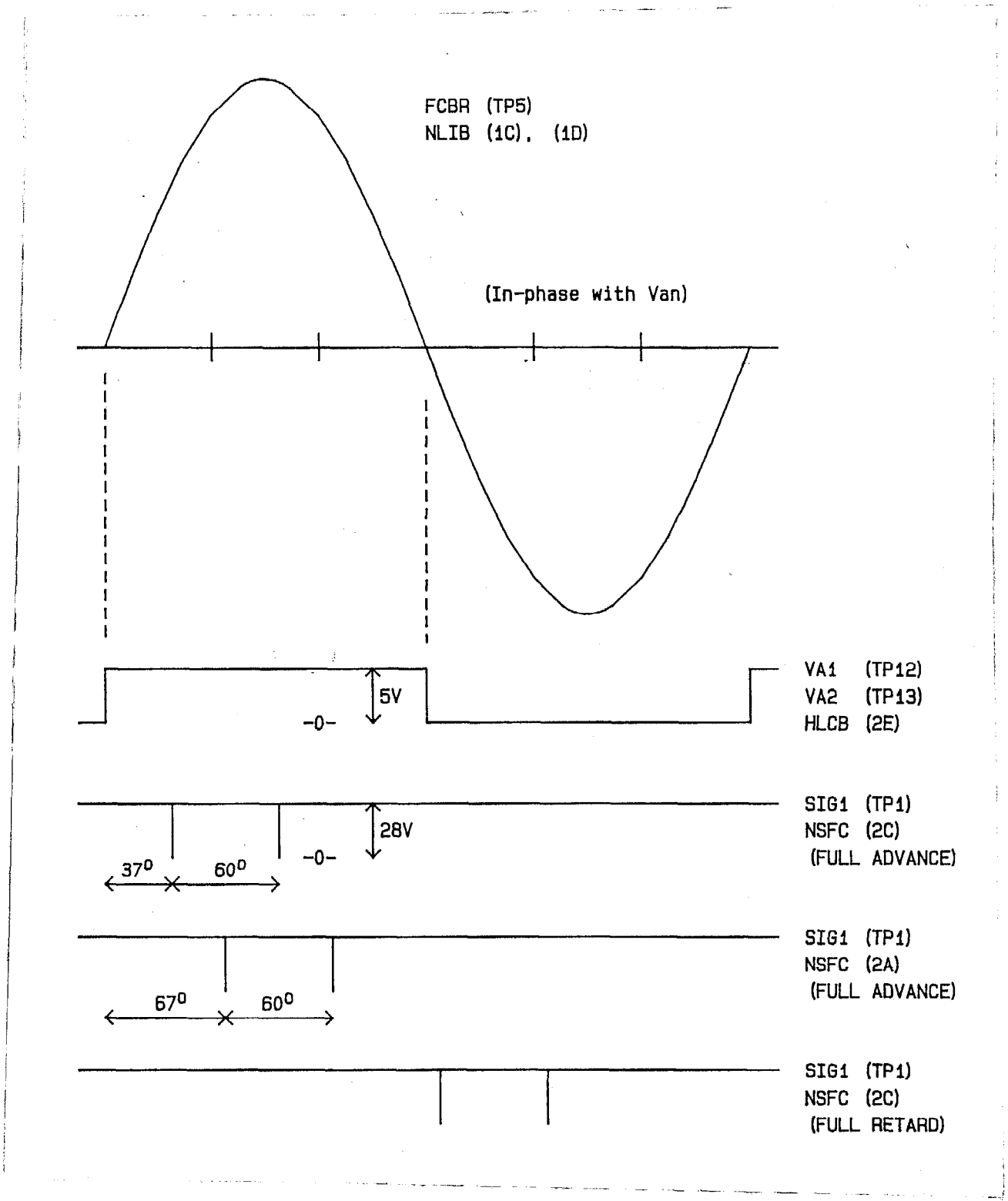
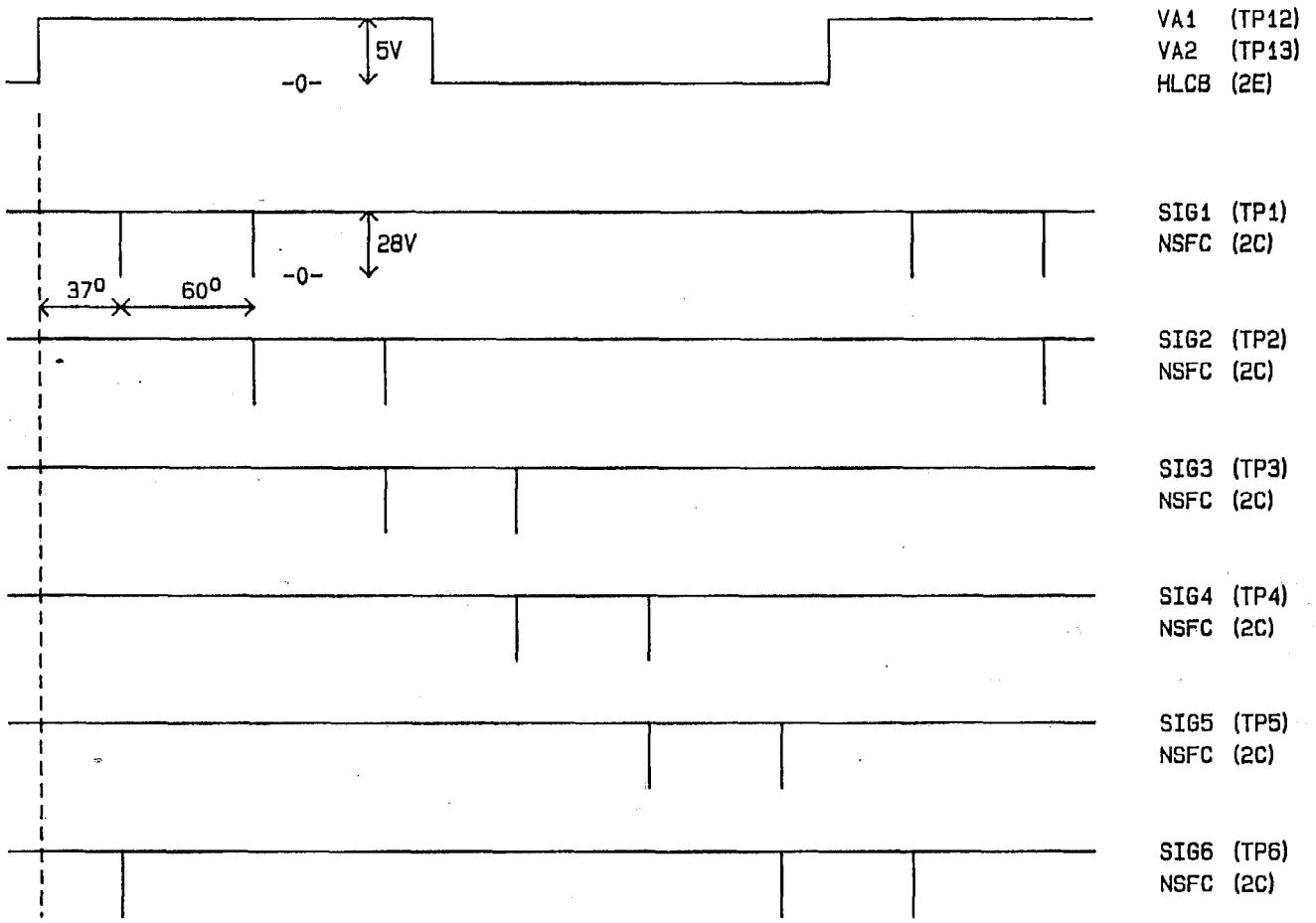
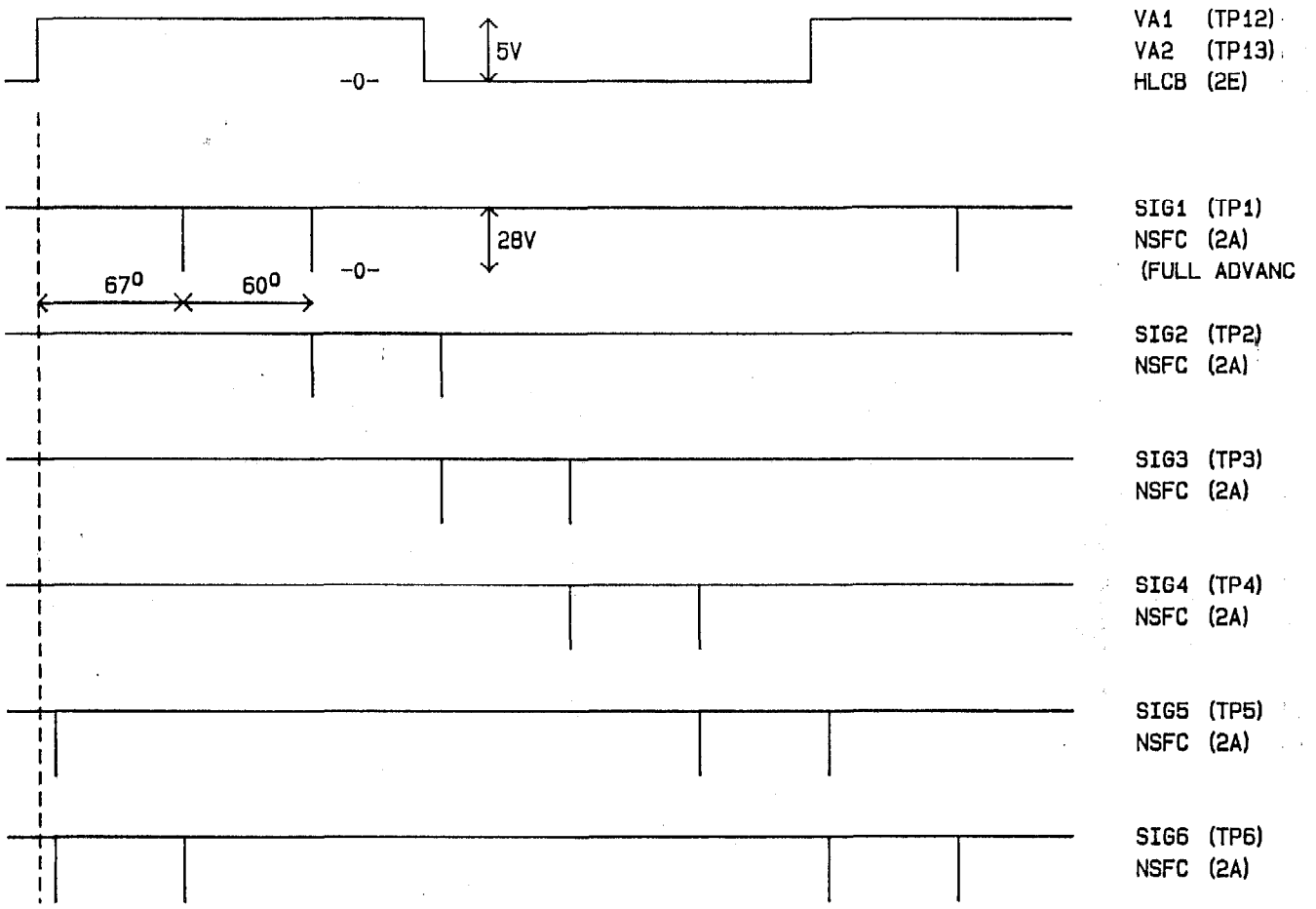

FCBR (TP5)
NLIB (1C), (1D)
(In-phase with Van)
5V
-0-
VA1 (TP12)
VA2 (TP13)
HLCB (2E)
28V
-0-
37°
60°
SIG1 (TP1)
NSFC (2C)
(FULL ADVANCE)
67°
60°
SIG1 (TP1)
NSFC (2A)
(FULL ADVANCE)
SIG1 (TP1)
NSFC (2C)
(FULL RETARD)

Figure 7B-1.



SOURCE FIRING (MASTER) - FULL ADVANCE

Figure 7B-2A.



Source firing (slave) - full advance

Figure 7B-2B.

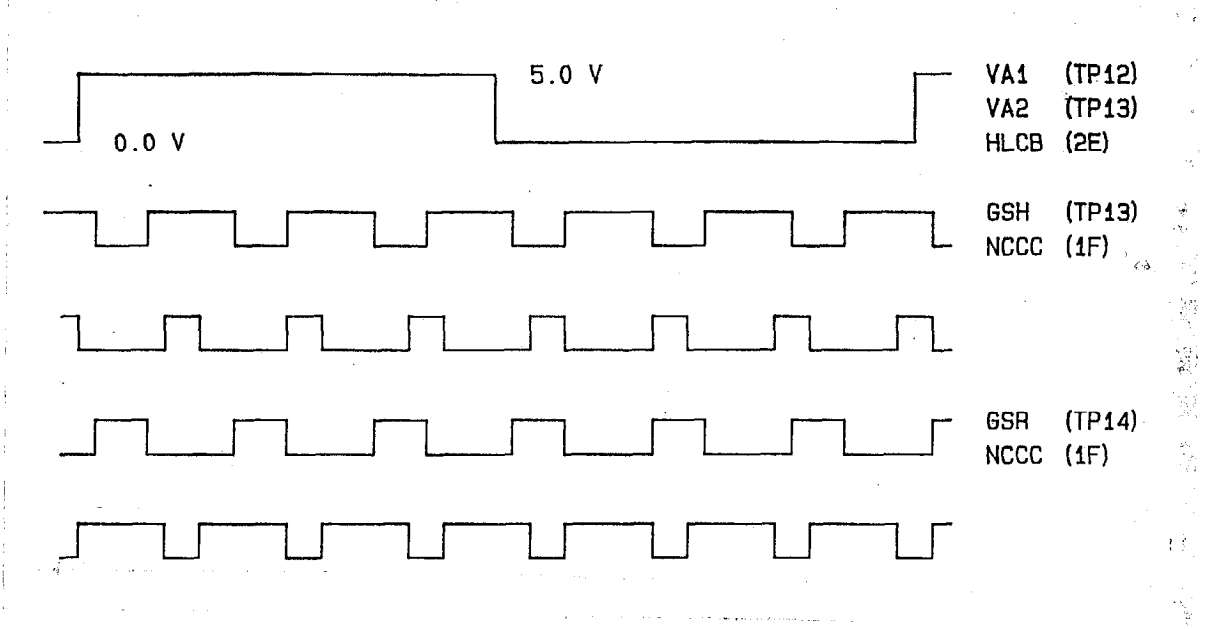

5.0 V
0.0 V
VA1 (TP12)
VA2 (TP13)
HLCB (2E)
GSH (TP13)
NCCC (1F)
GSR (TP14)
NCCC (1F)

Figure 7B-5A.

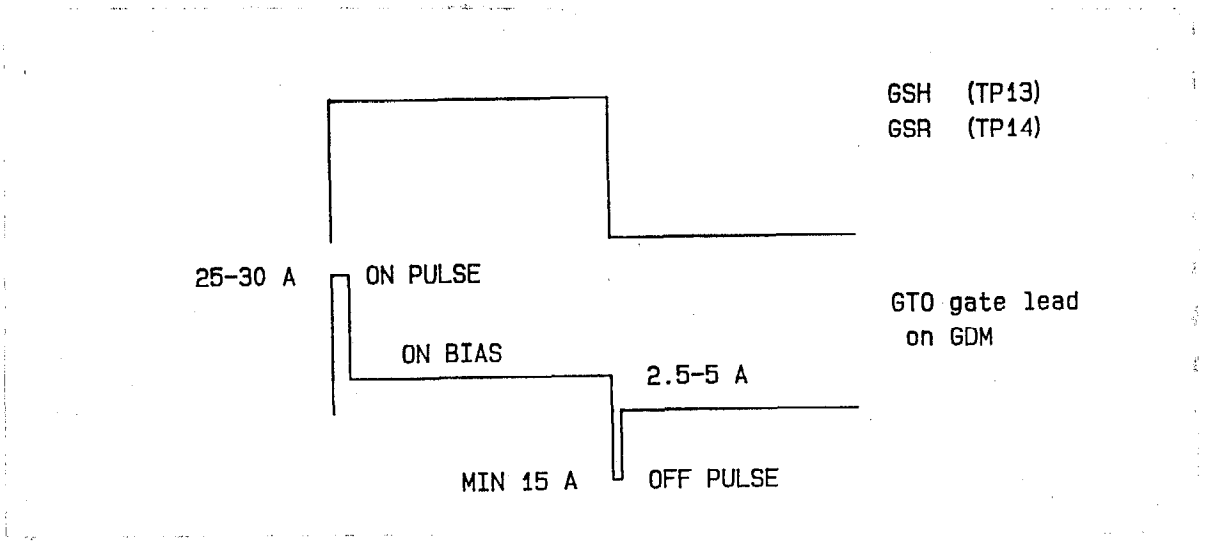
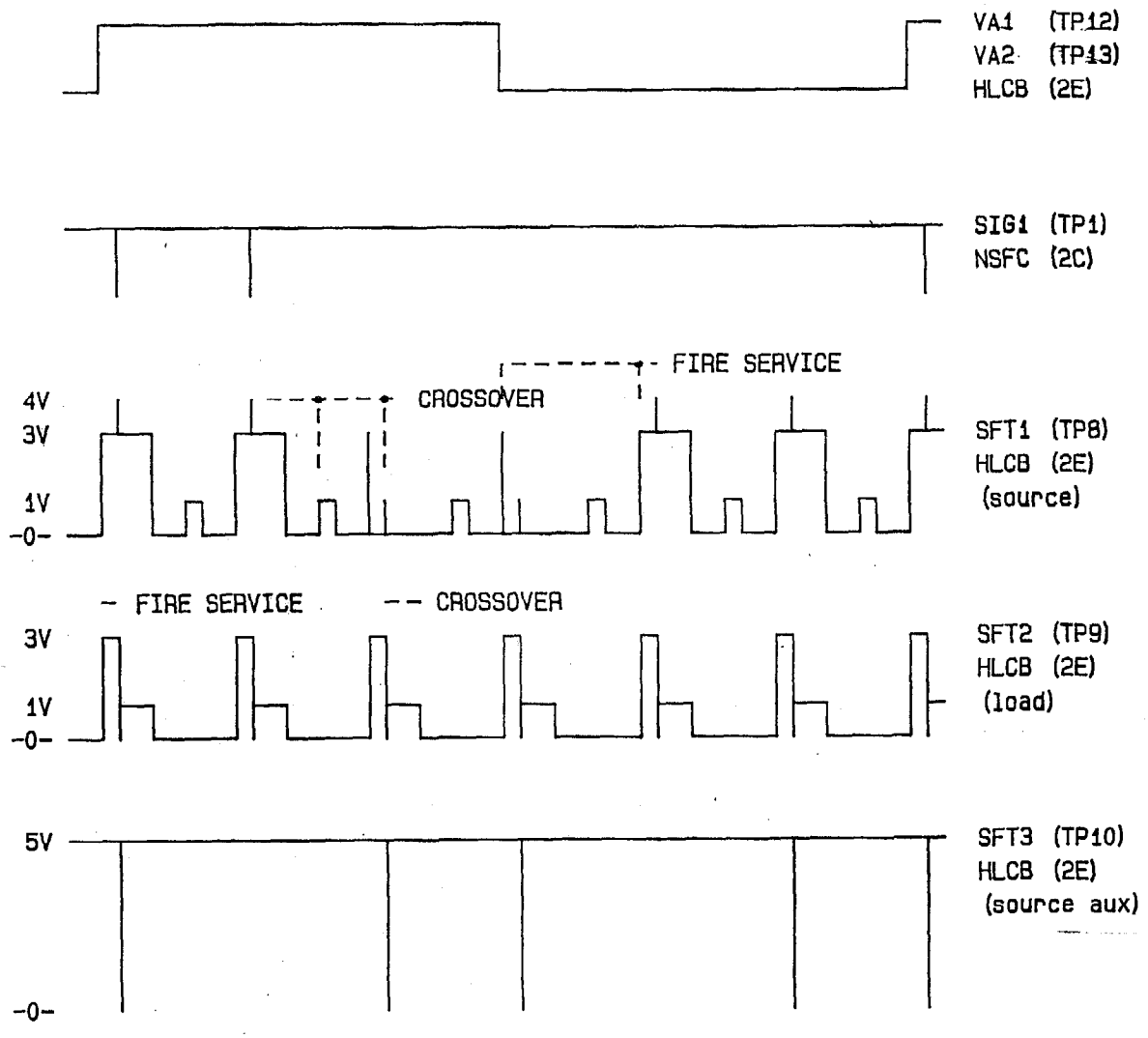

GSH (TP13)
GSR (TP14)
25-30 A
ON PULSE
GTO gate lead
on GDM
ON BIAS
2.5-5 A
MIN 15 A
OFF PULSE

Figure 7B-5B.



Software interrupts - full advance (source)

Figure 7B-6.

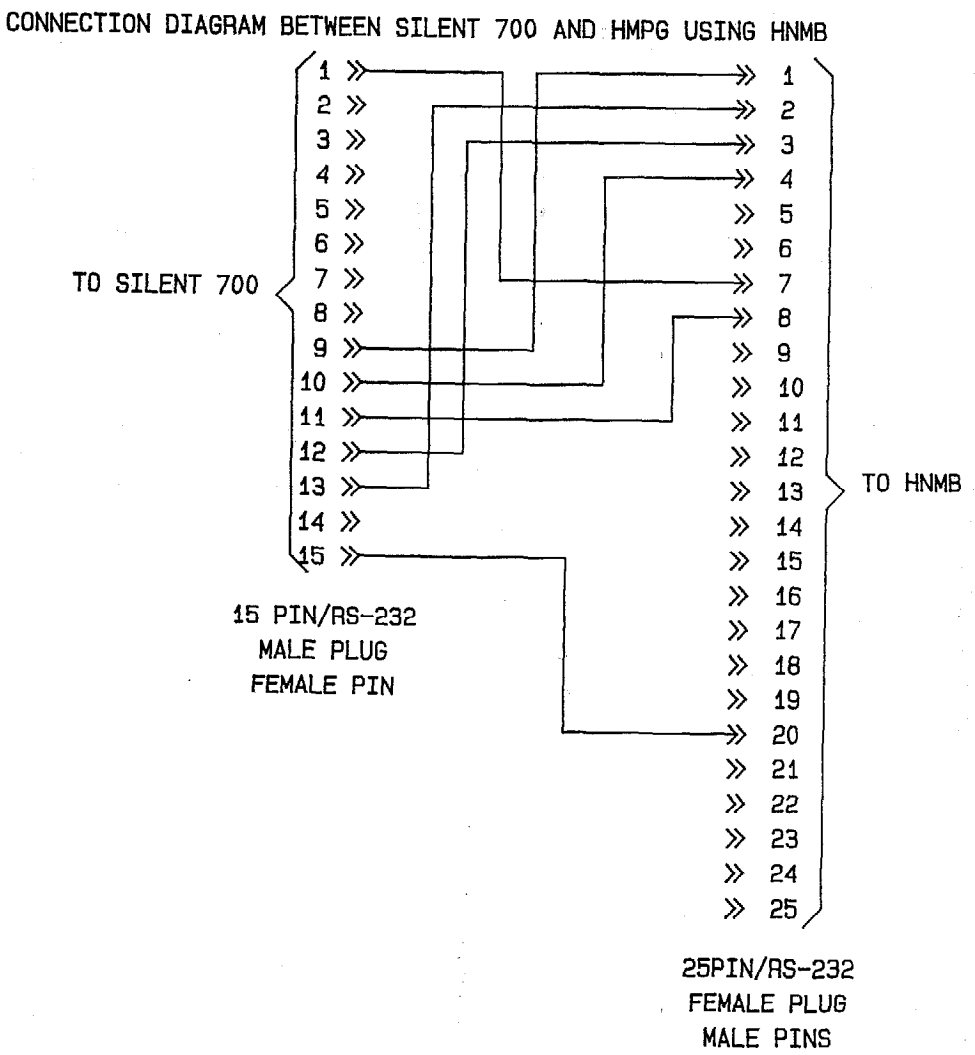

CONNECTION DIAGRAM BETWEEN SILENT 700 AND HMPG USING HNMB
TO SILENT 700
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
15 PIN/RS-232
MALE PLUG
FEMALE PIN
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
TO HNMB
25PIN/RS-232
FEMALE PLUG
MALE PINS

Figure 7D-1

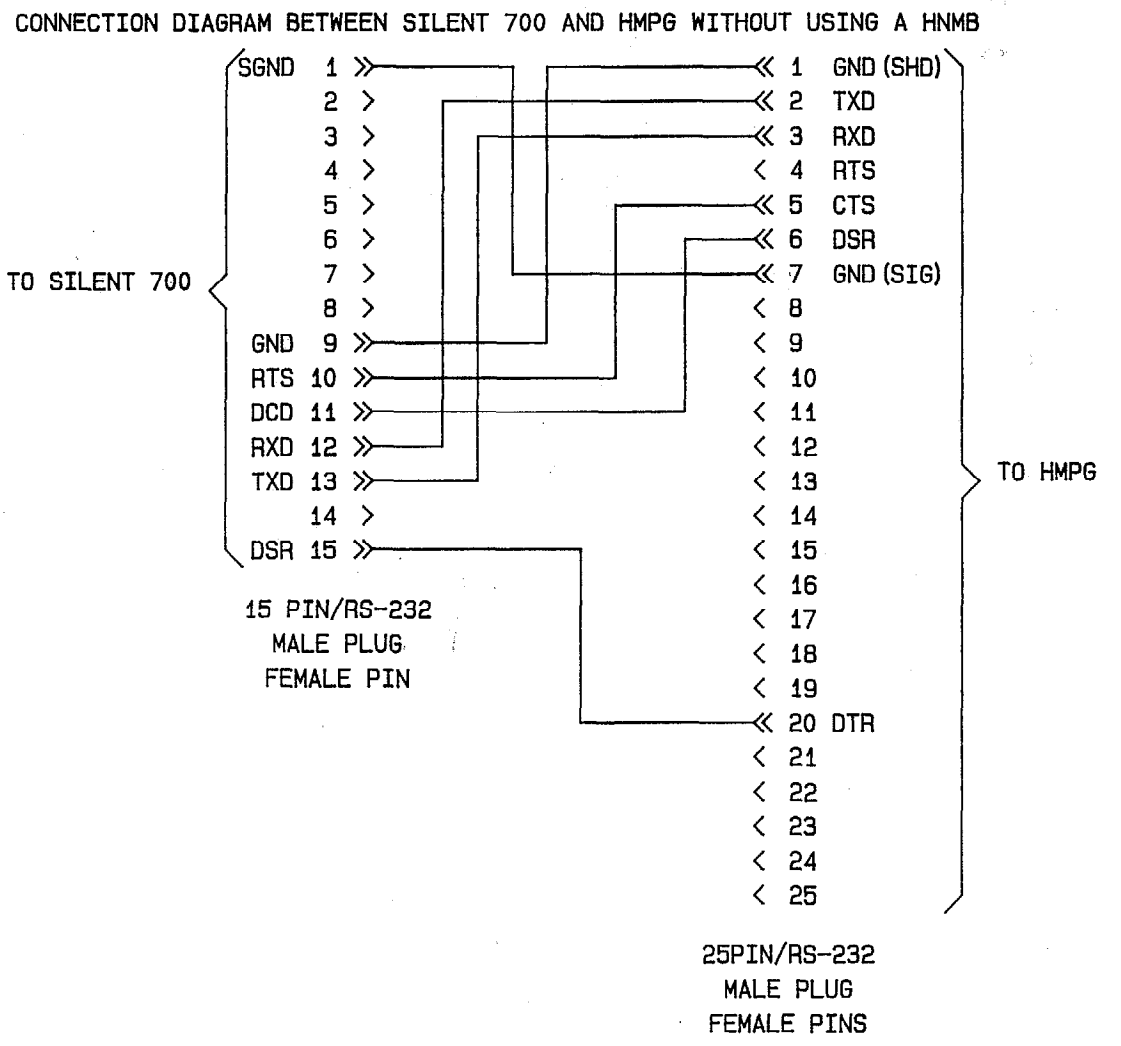
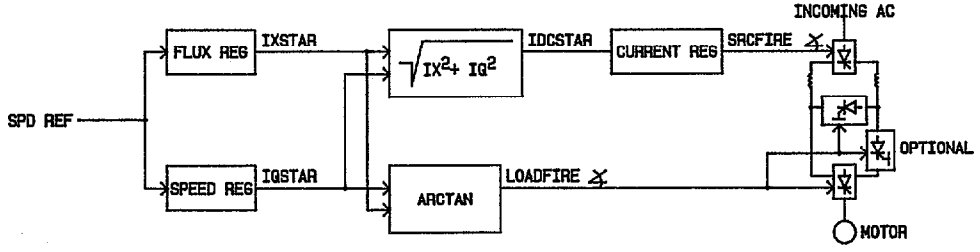

CONNECTION DIAGRAM BETWEEN SILENT 700 AND HMPG WITHOUT USING A HNMB
TO SILENT 700
SGND 1
2
3
4
5
6
7
8
GND 9
RTS 10
DCD 11
RXD 12
TXD 13
14
DSR 15
15 PIN/RS-232
MALE PLUG
FEMALE PIN
1 GND (SHD)
2 TXD
3 RXD
4 RTS
5 CTS
6 DSR
7 GND (SIG)
8
9
10
11
12
13
14
15
16
17
18
19
20 DTR
21
22
23
24
25
TO HMPG
25PIN/RS-232
MALE PLUG
FEMALE PINS

Figure 7D-2



ONELINE

DCLC-IMD 8AA

SH. NO. 0AF

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REV. 3

REV. 4

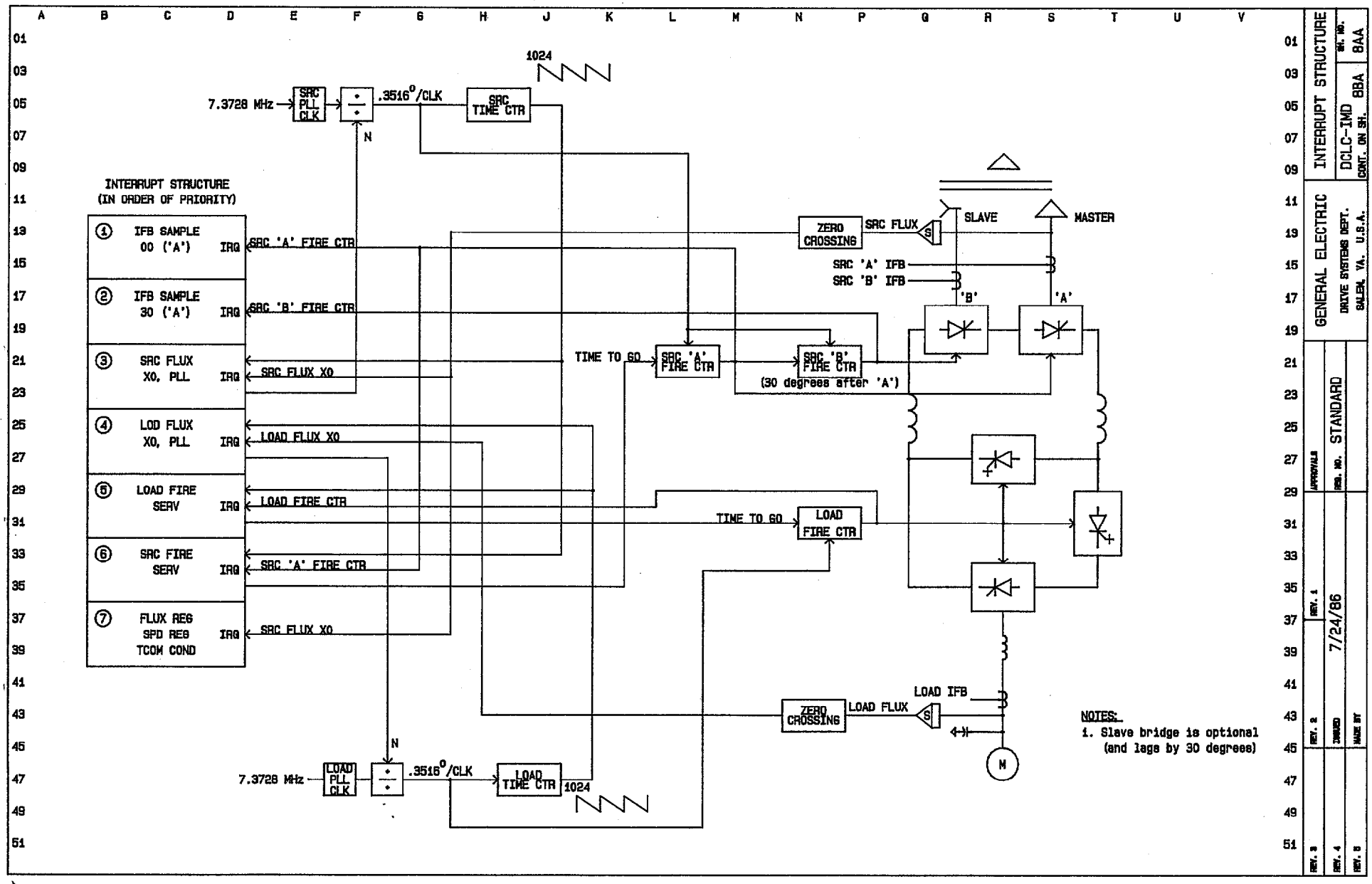
REV. 5

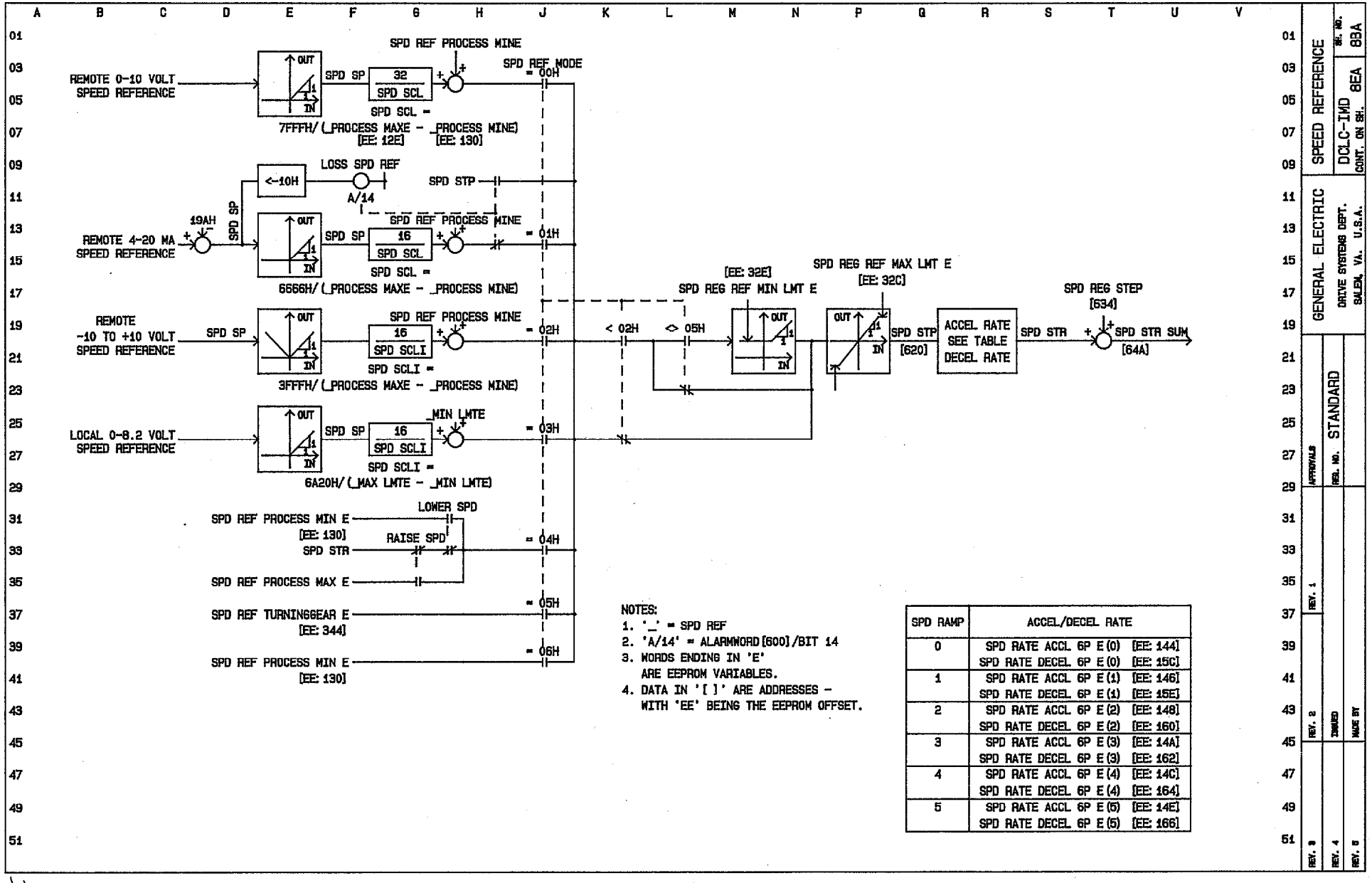
REL. NO. STANDARD

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PCAD\DRAWINGS\0AF.DAF

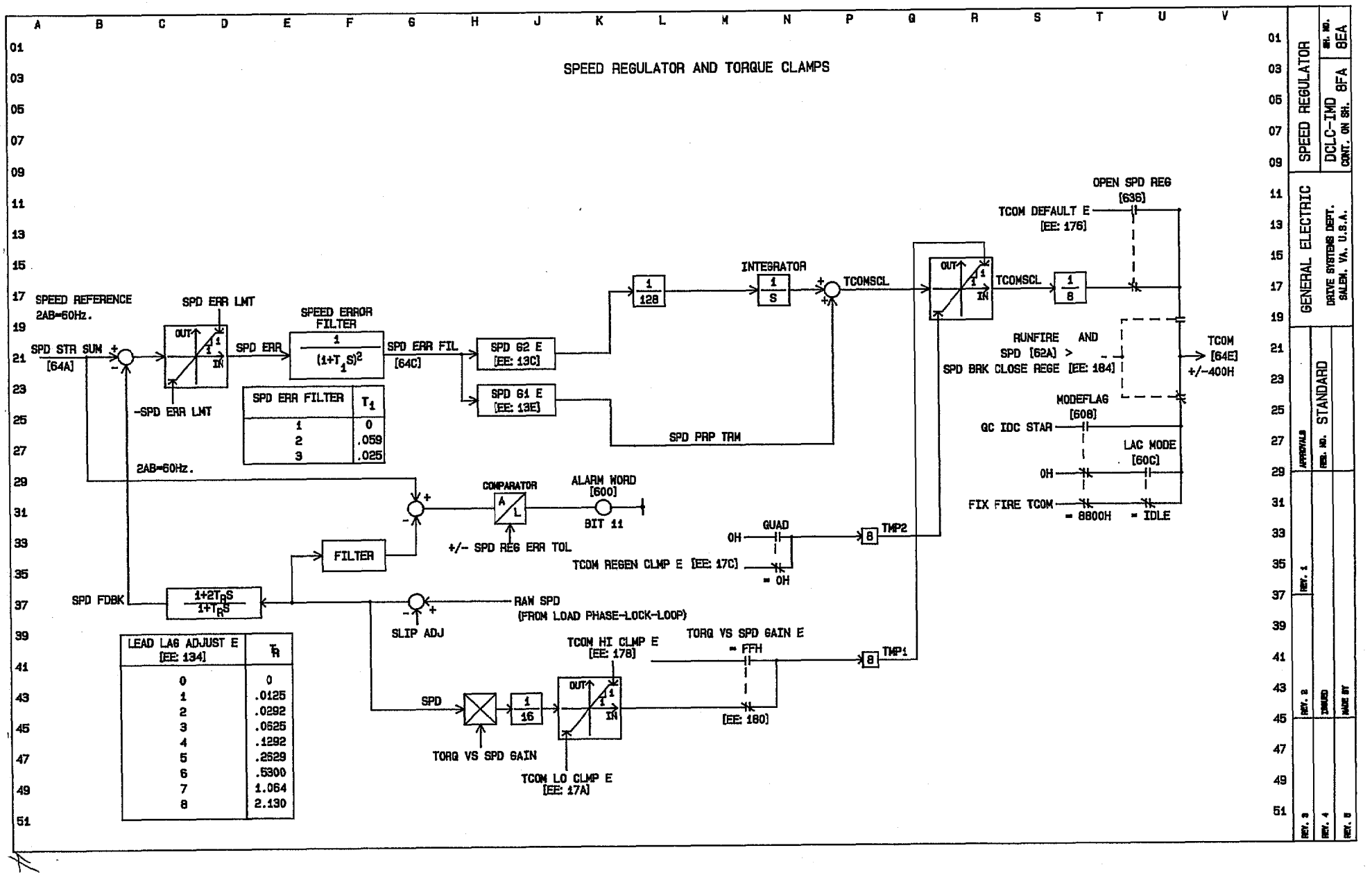

INTERRUPT STRUCTURE
(IN ORDER OF PRIORITY)
① IFB SAMPLE 00 ('A') IRQ
② IFB SAMPLE 30 ('A') IRQ
③ SRC FLUX X0, PLL IRQ
④ LOD FLUX X0, PLL IRQ
⑤ LOAD FIRE SERV IRQ
⑥ SRC FIRE SERV IRQ
⑦ FLUX REG SPD REG TCOM COND IRQ
7.3728 MHz
SRC PLL CLK
.3516°/CLK
SRC TIME CTR
1024
N
SRC 'A' FIRE CTR
SRC 'B' FIRE CTR
SRC FLUX X0
LOAD FLUX X0
LOAD FIRE CTR
TIME TO GO
ZERO CROSSING
SRC FLUX
SLAVE
MASTER
SRC 'A' IFB
SRC 'B' IFB
'B'
'A'
(30 degrees after 'A')
LOAD FIRE CTR
LOAD IFB
LOAD FLUX
LOAD PLL CLK
LOAD TIME CTR
M
NOTES:
1. Slave bridge is optional (and lags by 30 degrees)
INTERRUPT STRUCTURE
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DRIVE SYSTEMS DEPT.
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STANDARD
7/24/86

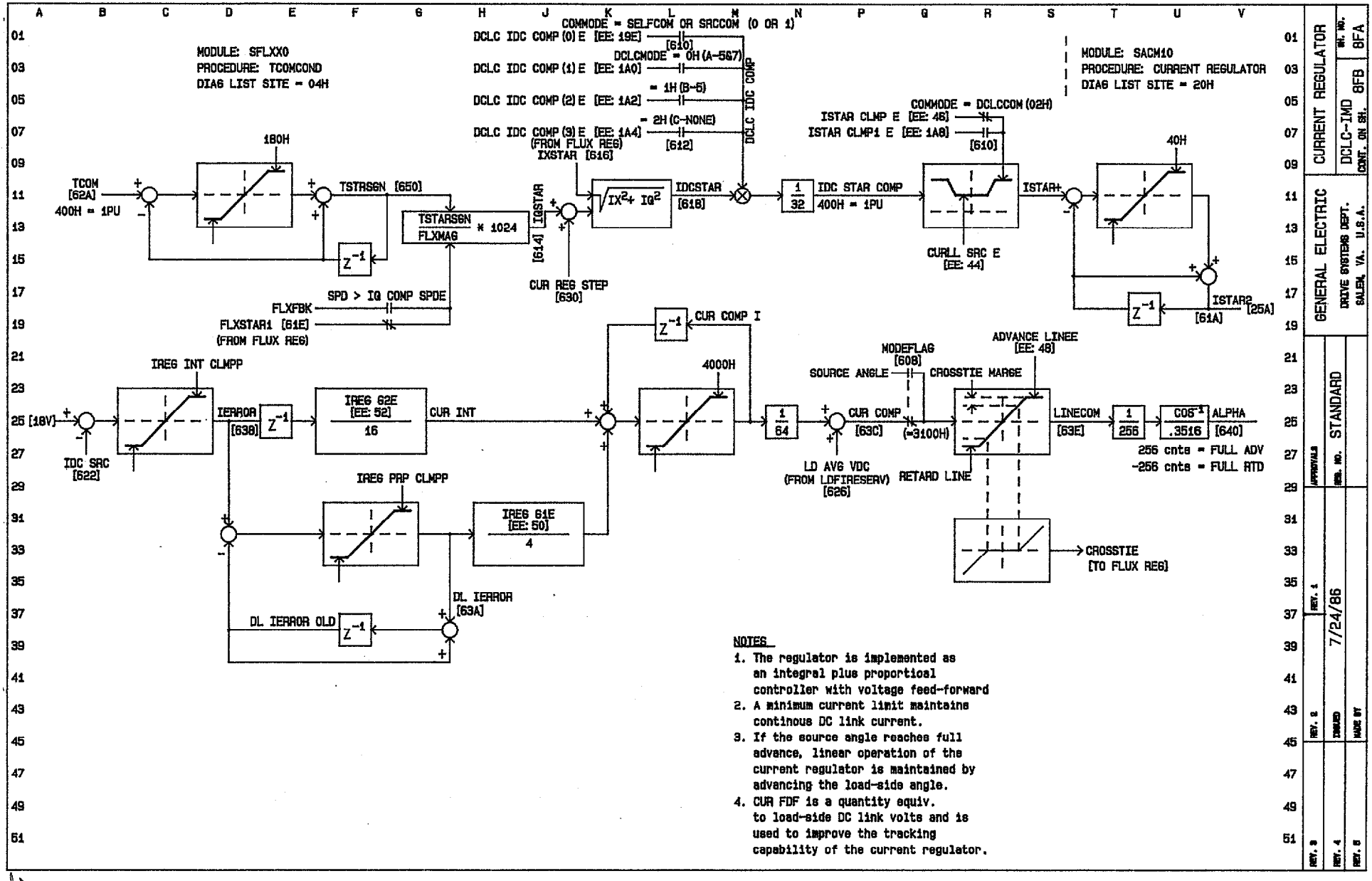


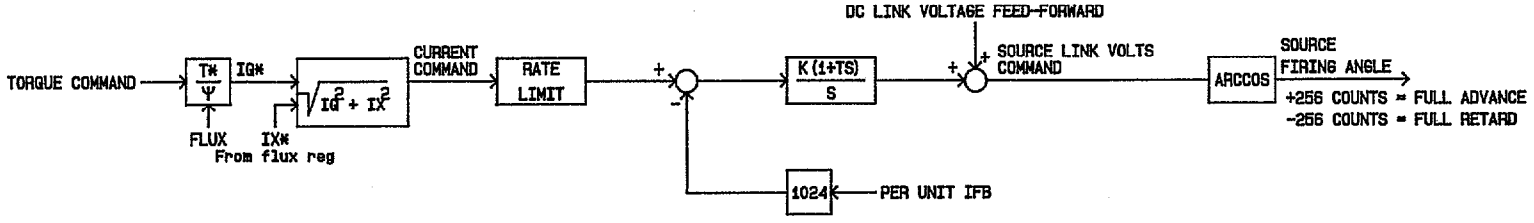
NOTES:
1. '_' = SPD REF
2. 'A/14' = ALARMWORD[600]/BIT 14
3. WORDS ENDING IN 'E' ARE EEPROM VARIABLES.
4. DATA IN '[ ]' ARE ADDRESSES - WITH 'EE' BEING THE EEPROM OFFSET.

SPD RAMP	ACCEL/DECEL RATE
0	SPD RATE ACCL 6P E(0) [EE: 144] SPD RATE DECEL 6P E(0) [EE: 15C]
1	SPD RATE ACCL 6P E(1) [EE: 146] SPD RATE DECEL 6P E(1) [EE: 15E]
2	SPD RATE ACCL 6P E(2) [EE: 148] SPD RATE DECEL 6P E(2) [EE: 160]
3	SPD RATE ACCL 6P E(3) [EE: 14A] SPD RATE DECEL 6P E(3) [EE: 162]
4	SPD RATE ACCL 6P E(4) [EE: 14C] SPD RATE DECEL 6P E(4) [EE: 164]
5	SPD RATE ACCL 6P E(5) [EE: 14E] SPD RATE DECEL 6P E(5) [EE: 166]

SPEED REFERENCE
DCLC-IMD CONT. ON SH. 8EA
SH. NO. 8BA
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT. SALEM, VA., U.S.A.
STANDARD
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REV. 1
REV. 2
REV. 3
REV. 4
REV. 5
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SPEED REGULATOR AND TORQUE CLAMPS
SPEED REFERENCE
2AB=60Hz.
SPD STR SUM
[64A]
SPD ERR LMT
-SPD ERR LMT
SPD ERR
SPEED ERROR FILTER
1/(1+T1S)2
SPD ERR FIL
[64C]
SPD ERR FILTER
T1
1 0
2 .059
3 .025
SPD G2 E
[EE: 13C]
SPD G1 E
[EE: 13E]
1/128
INTEGRATOR
1/S
TCOMSCL
SPD PRP TRM
OUT
IN
TCOMSCL
1/8
OPEN SPD REG
[636]
TCOM DEFAULT E
[EE: 176]
RUNFIRE AND
SPD [62A] >
SPD BRK CLOSE REGE [EE: 184]
TCOM
[64E]
+/-400H
MODEFLAG
[608]
QC IDC STAR
LAC MODE
[60C]
0H
FIX FIRE TCOM
= 8800H
= IDLE
2AB=60Hz.
COMPARATOR
A/L
ALARM WORD
[600]
BIT 11
+/- SPD REG ERR TOL
FILTER
GUAD
0H
TCOM REGEN CLMP E [EE: 17C]
= 0H
8
TMP2
SPD FDBK
1+2TRS / 1+TRS
LEAD LAG ADJUST E
[EE: 134]
TR
0 0
1 .0125
2 .0292
3 .0525
4 .1292
5 .2529
6 .5300
7 1.064
8 2.130
SLIP ADJ
RAW SPD
(FROM LOAD PHASE-LOCK-LOOP)
TCOM HI CLMP E
[EE: 178]
TORQ VS SPD GAIN E
= FFH
[EE: 180]
TMP1
SPD
1/16
TORQ VS SPD GAIN
TCOM LO CLMP E
[EE: 17A]
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
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SPEED REGULATOR
DCLC-IMD
CONT. ON SH. 8FA
SH. NO. 8EA
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REV. 2
REV. 3
REV. 4
REV. 5
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MODULE: SFLXX0
PROCEDURE: TCOMCOND
DIAG LIST SITE = 04H
COMMODE = SELFCOM OR SRCCOM (0 OR 1)
DCLC IDC COMP (0) E [EE: 19E]
[610]
DCLCMODE = 0H (A-567)
DCLC IDC COMP (1) E [EE: 1A0]
= 1H (B-5)
DCLC IDC COMP (2) E [EE: 1A2]
= 2H (C-NONE)
DCLC IDC COMP (3) E [EE: 1A4]
[612]
DCLC IDC COMP
(FROM FLUX REG)
IXSTAR [616]
MODULE: SACM10
PROCEDURE: CURRENT REGULATOR
DIAG LIST SITE = 20H
COMMODE = DCLCCOM (02H)
ISTAR CLMP E [EE: 46]
ISTAR CLMP1 E [EE: 1A8]
[610]
180H
TCOM
[62A]
400H = 1PU
TSTRSGN [650]
TSTARSGN / FLXMAG * 1024
IQSTAR [614]
IDCSTAR [618]
1/32
IDC STAR COMP
400H = 1PU
CURLL SRC E
[EE: 44]
ISTAR+
40H
ISTAR2 [25A]
[61A]
SPD > IQ COMP SPDE
FLXFBK
FLXSTAR1 [61E]
(FROM FLUX REG)
CUR REG STEP
[630]
CUR COMP I
IREG INT CLMPP
4000H
MODEFLAG
[608]
SOURCE ANGLE
CROSSTIE MARGE
ADVANCE LINEE
[EE: 48]
[18V]
IDC SRC
[622]
IERROR [638]
IREG 62E
[EE: 52]
16
CUR INT
1/64
CUR COMP
[63C]
(-3100H)
LD AVG VDC
(FROM LDFIRESERV)
[626]
RETARD LINE
LINECOM
[63E]
1/256
COS-1 .3516
ALPHA
[640]
256 cnts = FULL ADV
-256 cnts = FULL RTD
IREG PRP CLMPP
IREG 61E
[EE: 50]
4
DL IERROR
[63A]
DL IERROR OLD
CROSSTIE
[TO FLUX REG]
NOTES
1. The regulator is implemented as an integral plus proportional controller with voltage feed-forward
2. A minimum current limit maintains continous DC link current.
3. If the source angle reaches full advance, linear operation of the current regulator is maintained by advancing the load-side angle.
4. CUR FDF is a quantity equiv. to load-side DC link volts and is used to improve the tracking capability of the current regulator.
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
CURRENT REGULATOR
DCLC-IMD
DWG. NO. 8FB
CONT. ON SH. 8FA
APPROVALS
REV. 1
REV. 2
REV. 3
REV. 4
REV. 5
DES. NO. STANDARD
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CURRENT REGULATOR

DCLC-IMD 8GA

SH. NO. 8FB

CONT. ON SH.

GENERAL ELECTRIC

DRIVE SYSTEMS DEPT.

SALEM, VA. U.S.A.

STANDARD

APPROVALS

REV. 1

REV. 2

REV. 3

REV. 4

REV. 5

ISSUED

MADE BY

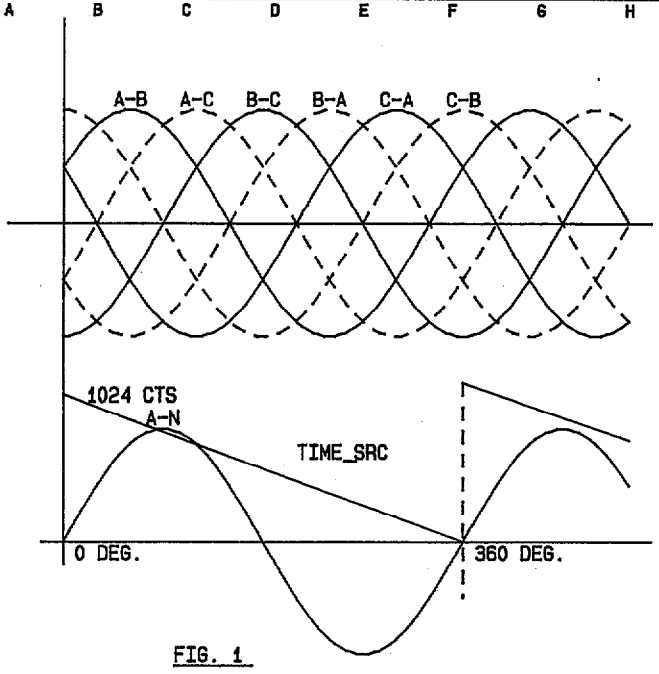


FIG. 1

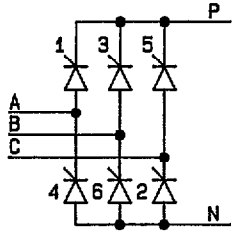


FIG. 2

PH TABLE

L-L	PH	OFFSET (CTS)
A-B	1	3ABH
A-C	2	300H
B-C	3	255H
B-A	4	1ABH
C-A	5	100H
C-B	6	055H

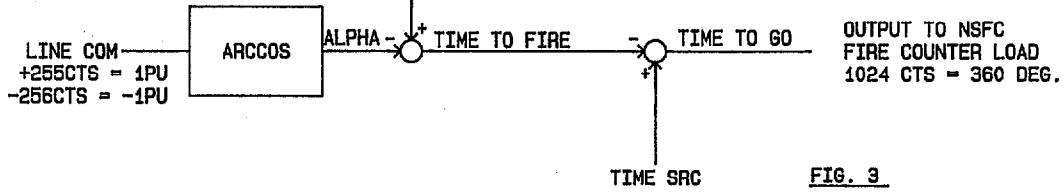
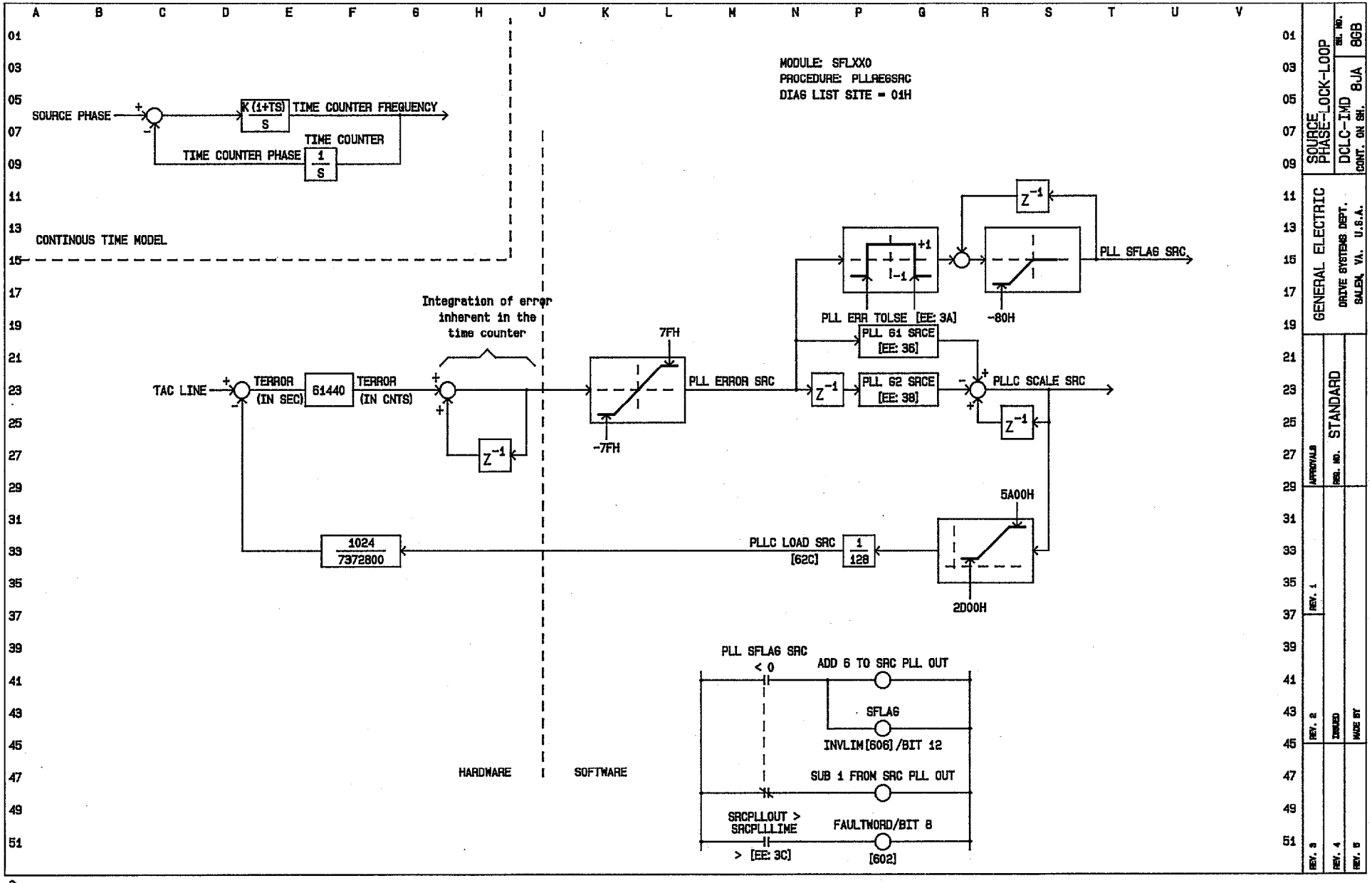


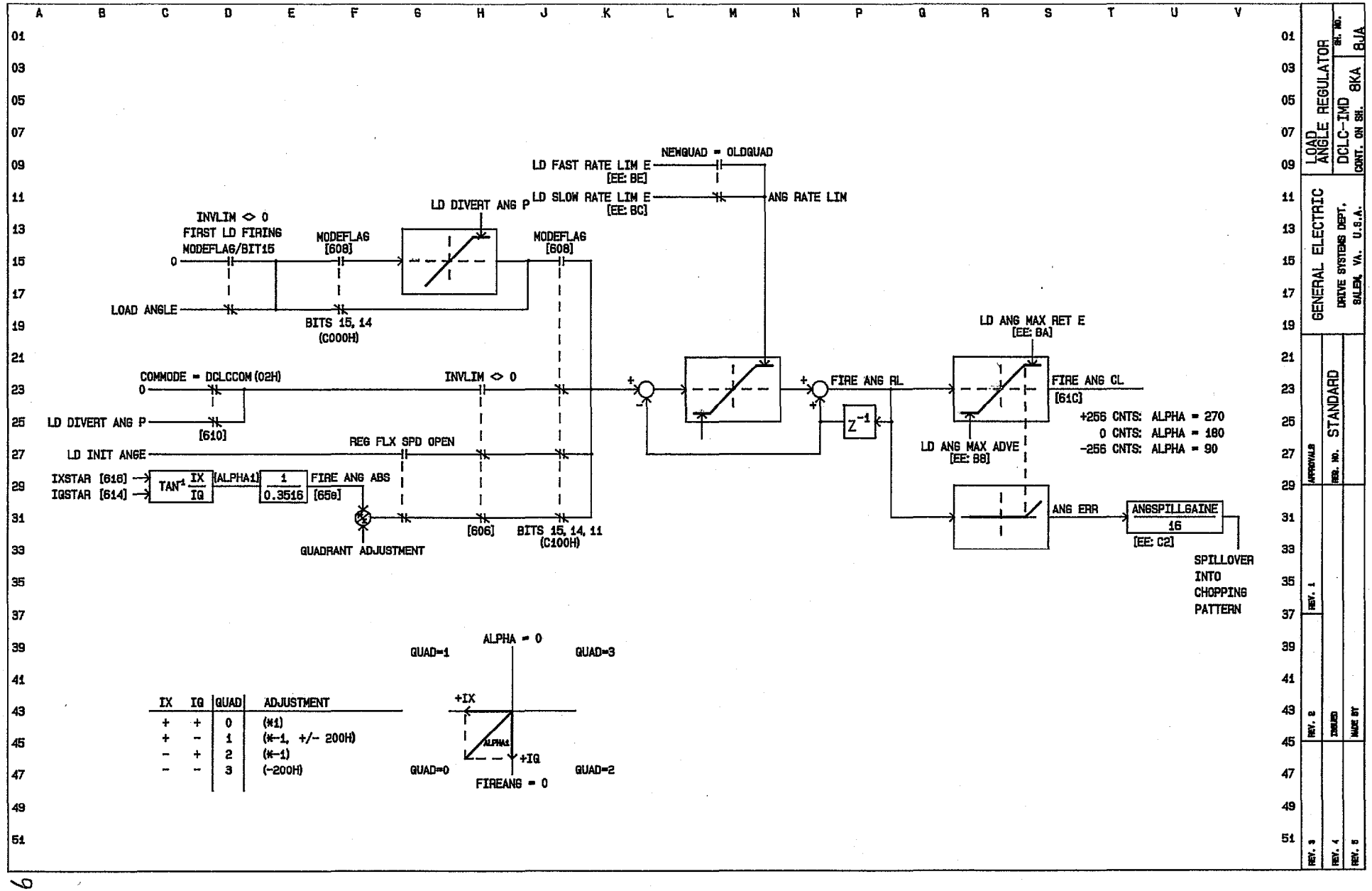
FIG. 3

TIME TO FIRE CALCULATION

1. THE TIMING FOR CELL FIRING IS ESTABLISHED BY THE SOURCE PHASE LOCK LOOP. IT ACTS TO MAINTAIN SYNCHRONIZM WITH THE BRIDGE VOLTAGE SUCH THAT 360 DEGREES IS DIVIDED INTO 1024 COUNTS (SEE FIG. 1). NOTE THAT THE REFERENCE PHASE IS A-N.
2. EACH CELL PAIR WHICH CONDUCTS ON A GIVEN L-L VOLTAGE IS ASSIGNED A VALUE CALLED "PH". FOR EXAMPLE, CELLS ASSOCIATED WITH A -B ARE CELLS NUMBERED 1 AND 6 (SEE FIG. 2). FROM THE PH TABLE THIS IS A PH OF 1. NOTE THAT THE ASSOCIATED OFFSET IS 55H COUNTS. THIS CORRESPONDS TO 30 DEGREES. THE DEFINITION OF OFFSET CORRESPONDS TO ALPHA = 0 DEGREES (FULL ADVANCE) FOR A GIVEN PH.
3. THE VALUE OF LINECOM CALCULATED BY THE CURRENT REGULATOR IS CONVERTED TO A FIRING ANGLE ACCORDING TO THE EQUATION ALPHA = ARCCOS(LINECOM). THIS IS IMPLEMENTED IN THE SOFTWARE WITH AN INDEXED LOOKUP TABLE.
4. THE VALUE OF ALPHA IS SUB FROM THE OFFSET FOR THE PARTICULAR PH BEING CALCULATED.
   THE RESULTANT IS TIME_TO_FIRE.
5. TIME TO GO (TIME SRC - TIME TO FIRE) IS THE DIFFERENCE BETWEEN PRESENT TIME AND TIME TO FIRE. THE SFTWRE ALLOWS FOR ROLLOVER (ANGLES > 360 DEGREES) AND TOO LATE TO FIRE.
6. THE RESULTING TIME TO GO (IN COUNTS) IS LOADED INTO THE FIRE COUNTER FOR THE CORRESPONDING NSFC. WHEN THE FIRE COUNTER IS DECREMENTED TO ZERO, THE FIRING INTERRUPT OCCURS AND THE NEW CELL MASK IS TRANSMITTED TO THE NSFC.

SOURCE PHASE-LOCK-LOOP
DCLC-IMD
CONT. ON SH. 8GB
SH. NO. 8GA
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
APPROVALS
REL. NO. STANDARD
7/24/86
REV. 1
REV. 2
REV. 3
REV. 4
REV. 5
ISSUED
MADE BY


MODULE: SFLXX0
PROCEDURE: PLLREGSRC
DIAG LIST SITE = 01H
SOURCE PHASE
K(1+TS)/S
TIME COUNTER FREQUENCY
TIME COUNTER
TIME COUNTER PHASE
1/S
CONTINOUS TIME MODEL
Integration of error inherent in the time counter
TAC LINE
TERROR (IN SEC)
61440
TERROR (IN CNTS)
7FH
-7FH
PLL ERROR SRC
PLL ERR TOLSE [EE: 3A]
+1
-1
-80H
PLL SFLAG SRC
PLL 61 SRCE [EE: 36]
PLL 62 SRCE [EE: 38]
PLLC SCALE SRC
5A00H
2D00H
PLLC LOAD SRC [62C]
1/128
1024/7372800
HARDWARE
SOFTWARE
PLL SFLAG SRC < 0
ADD 6 TO SRC PLL OUT
SFLAG
INVLIM[606]/BIT 12
SUB 1 FROM SRC PLL OUT
SRCPLLOUT > SRCPLLLIME
> [EE: 3C]
FAULTWORD/BIT 8
[602]
SOURCE PHASE-LOCK-LOOP
DCLC-IMD
8JA
8GB
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
STANDARD


NEWQUAD = OLDQUAD
LD FAST RATE LIM E
[EE: BE]
LD SLOW RATE LIM E
[EE: BC]
ANG RATE LIM
LD DIVERT ANG P
INVLIM <> 0
FIRST LD FIRING
MODEFLAG/BIT15
0
MODEFLAG
[608]
MODEFLAG
[608]
LOAD ANGLE
BITS 15, 14
(C000H)
LD ANG MAX RET E
[EE: BA]
COMMODE = DCLCCOM (02H)
0
INVLIM <> 0
FIRE ANG RL
FIRE ANG CL
[61C]
LD DIVERT ANG P
[610]
+256 CNTS: ALPHA = 270
0 CNTS: ALPHA = 180
-256 CNTS: ALPHA = 90
REG FLX SPD OPEN
LD ANG MAX ADVE
[EE: B8]
LD INIT ANGE
IXSTAR [616]
IQSTAR [614]
TAN-1 IX/IQ
[ALPHA1]
1/0.3516
FIRE ANG ABS
[65B]
ANG ERR
ANGSPILLGAINE
16
[EE: C2]
[606]
BITS 15, 14, 11
(C100H)
QUADRANT ADJUSTMENT
SPILLOVER
INTO
CHOPPING
PATTERN
ALPHA = 0
GUAD=1
GUAD=3
+IX
ALPHA1
+IQ
GUAD=0
GUAD=2
FIREANG = 0
IX IQ QUAD ADJUSTMENT
+ + 0 (*1)
+ - 1 (*-1, +/- 200H)
- + 2 (*-1)
- - 3 (-200H)
LOAD ANGLE REGULATOR
DCLC-IMD 8KA
CONT. ON SH. 8JA
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
STANDARD
APPROVALS
REV. 1
REV. 2
REV. 3
REV. 4
REV. 5
ISSUED
MADE BY

6

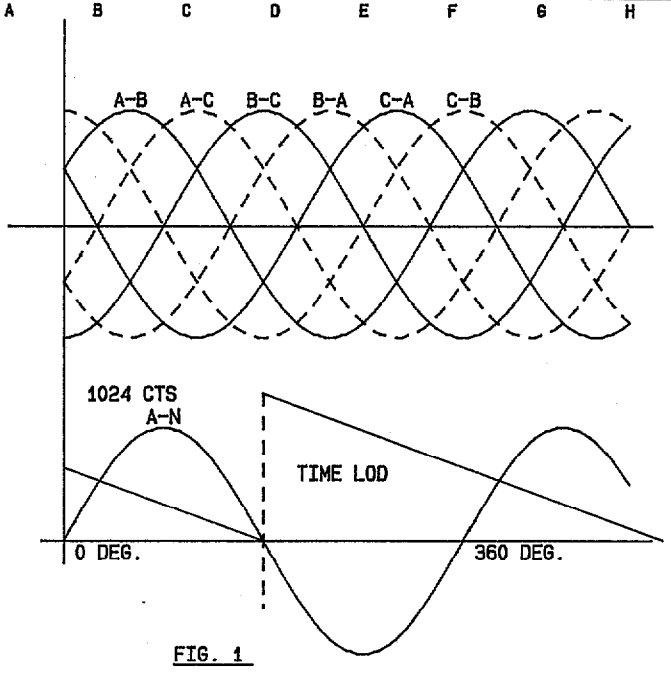


FIG. 1

PH TABLE		
L-L	PH	OFFSET (CTS)
A-B	1	3ABH
A-C	2	300H
B-C	3	255H
B-A	4	1ABH
C-A	5	100H
C-B	6	055H

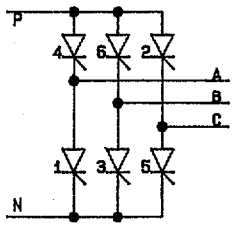


FIG. 2

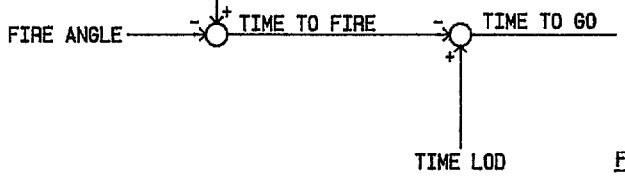


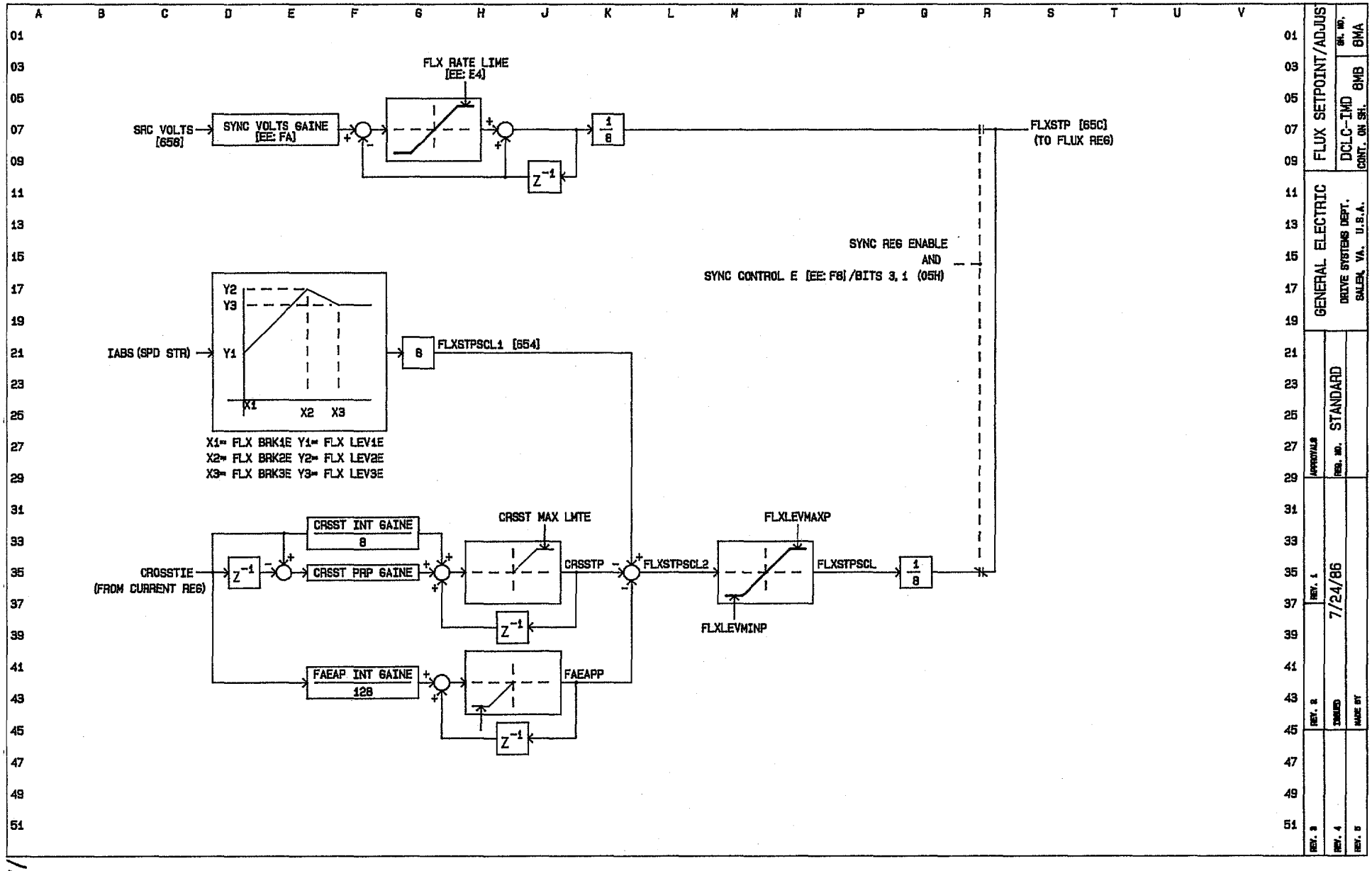
FIG. 3

TIME TO FIRE CALCULATION

1. THE TIMING FOR CELL FIRING IS ESTABLISHED BY THE LOAD PHASE LOCK LOOP. IT ACTS TO MAINTAIN SYNCHRONIZM WITH THE MOTOR VOLTAGE SUCH THAT 360 DEGREES IS DIVIDED INTO 1024 COUNTS (SEE FIG. 1). NOTE THAT THE REFERENCE PHASE IS A-N.
2. EACH CELL PAIR WHICH CONDUCTS ON A GIVEN L-L VOLTAGE IS ASSIGNED A VALUE CALLED "PH". FOR EXAMPLE, CELLS ASSOCIATED WITH A -B ARE CELLS NUMBERED 1 AND 6 (SEE FIG. 2). FROM THE PH TABLE THIS IS A PH OF 1. NOTE THAT THE ASSOCIATED OFFSET IS 55H COUNTS. THIS CORRESPONDS TO 30 DEGREES. THE DEFINITION OF OFFSET CORRESPONDS TO ALPHA = 180 DEGREES (FULL MOTOR) FOR A GIVEN PH
4. THE VALUE OF FIRE ANGLE IS SUB FROM THE OFFSET FOR THE PARTICULAR PH BEING CALCULATED THE RESULTANT IS TIME_TO_FIRE.
5. TIME TO GO (TIME LOD - TIME TO FIRE) IS THE DIFFERENCE BETWEEN PRESENT TIME AND TIME TO FIRE. THE SFTWRE ALLOWS FOR ROLLOVER (ANGLES > 360 DEGREES) AND TOO LATE TO FIRE.
6. THE RESULTING TIME TO GO (IN COUNTS) IS LOADED INTO THE FIRE COUNTER FOR THE CORRESPONDING NSFC. WHEN THE FIRE COUNTER IS DECREMENTED TO ZERO, THE FIRING INTERRUPT OCCURS AND THE NEW CELL MASK IS TRANSMITTED TO THE NSFC.

LOAD PHASE-LOCK-LOOP
DCLC-IMD CONT. ON SH. 8MA
SH. NO. 8KA
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
APPROVALS
REL. NO. STANDARD
REV. 1
7/24/86
REV. 2
ISSUED
MADE BY
REV. 3
REV. 4
REV. 5

01

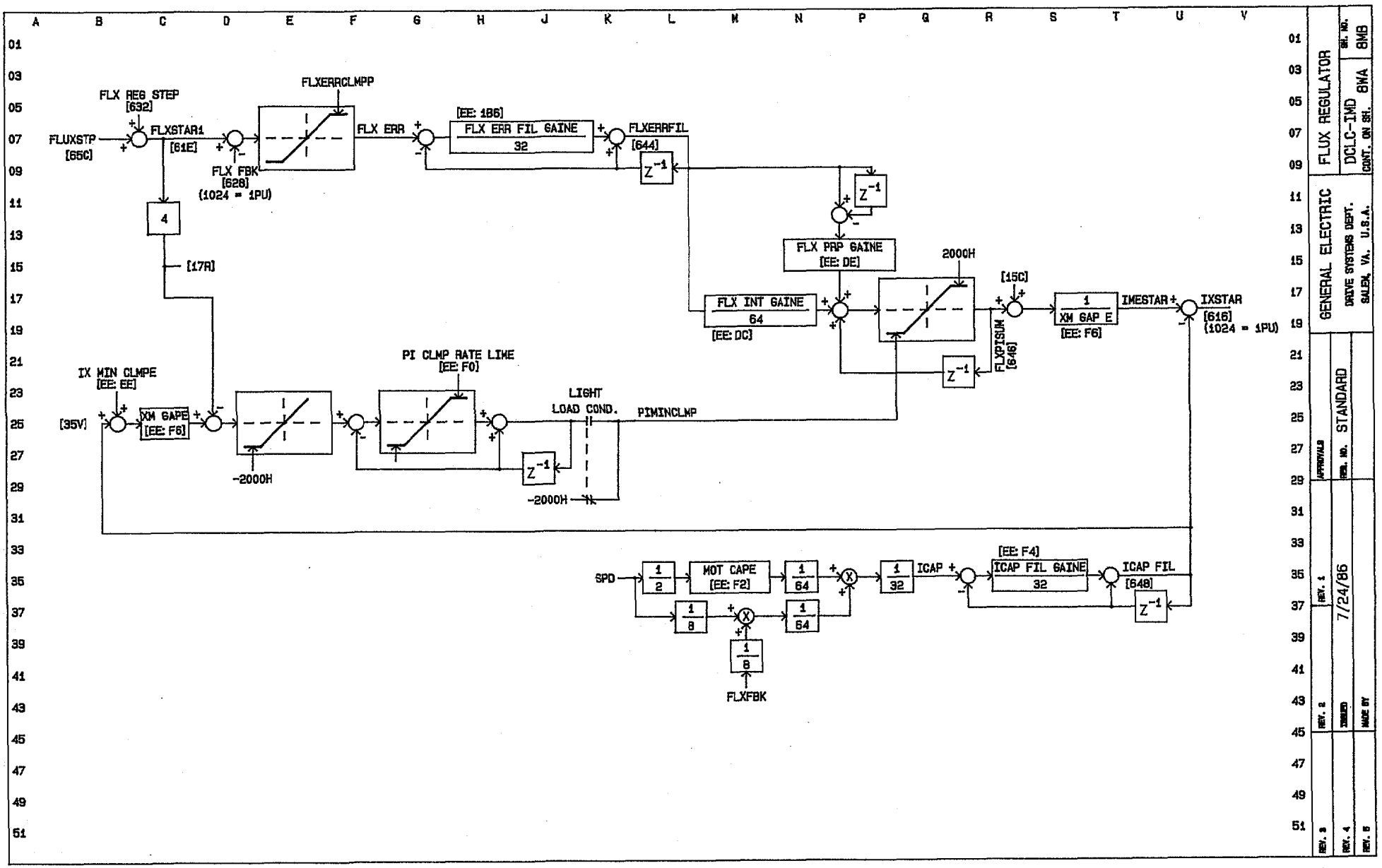


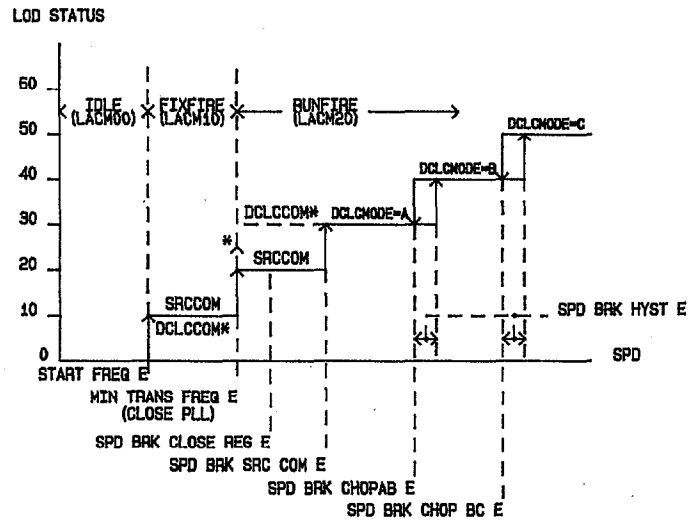
FLUX SETPOINT/ADJUS

DCLC-IMD SH. NO. 8MA
CONT. ON SH. 8MB

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SALEM, VA. U.S.A.

APPROVALS
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REV. 2 ISSUED
MADE BY
REV. 3
REV. 4
REV. 5

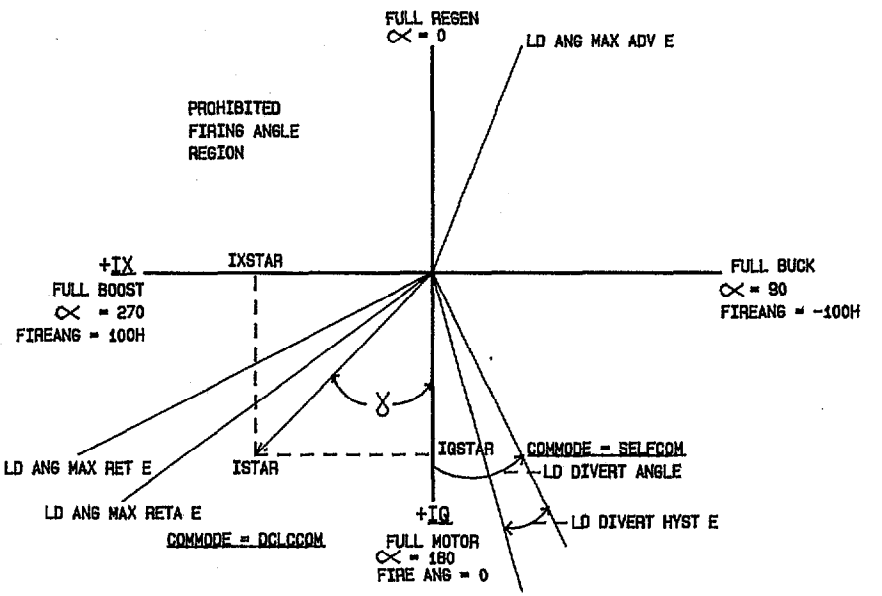

FLX REG STEP
[632]
FLUXSTP
[65C]
FLXSTAR1
[61E]
FLX FBK
[628]
(1024 = 1PU)
FLXERRCLMPP
FLX ERR
[EE: 186]
FLX ERR FIL GAINE
32
FLXERRFIL
[644]
Z-1
4
[17R]
FLX PRP GAINE
[EE: DE]
2000H
FLX INT GAINE
64
[EE: DC]
[15C]
1
XM GAP E
[EE: F6]
IMESTAR
IXSTAR
[616]
(1024 = 1PU)
FLXPISUM
[646]
IX MIN CLMPE
[EE: EE]
[35V]
XM GAPE
[EE: F6]
-2000H
PI CLMP RATE LIMI
[EE: F0]
LIGHT
LOAD COND.
PIMINCLMP
SPD
MOT CAPE
[EE: F2]
ICAP
[EE: F4]
ICAP FIL GAINE
ICAP FIL
[648]
FLXFBK
FLUX REGULATOR
DCLC-IMD
CONT. ON SH.
8MA
8MB
GENERAL ELECTRIC
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REV. 2
REV. 3
REV. 4
REV. 5
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DCLC MODE
A - 5th & 7th Harmonic Elimination
B - 5th Harmonic Elimination
C - NO Harmonic Elimination

* - SERIES/SHUNT DCLC CONFIGURATION

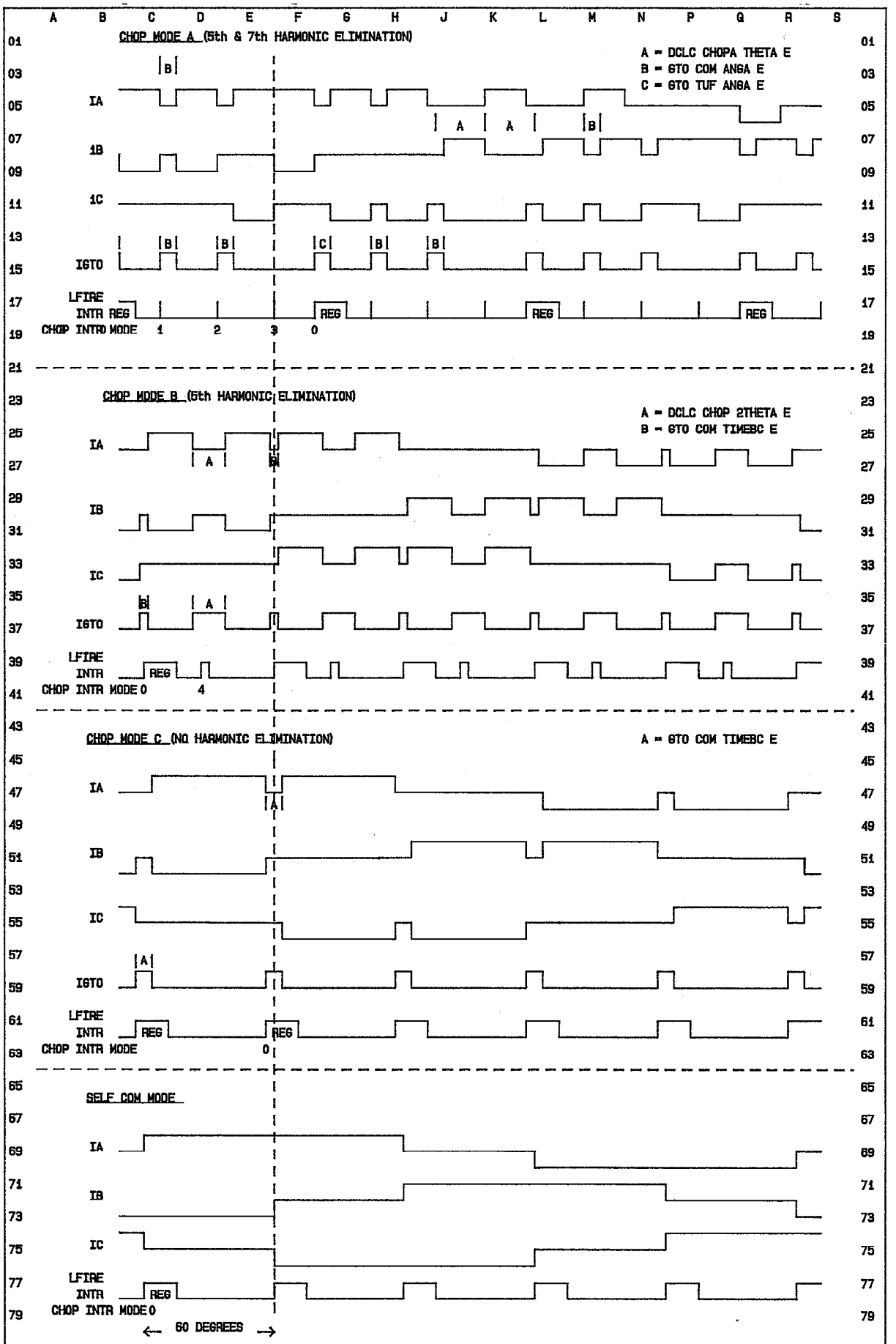
DCLC IMD STARTUP SEQUENCE



γ = FIREANG = TAN^{-1}(IQSTAR/IXSTAR)
1 CNT = 0.3515 DEGREE

INVERTER FIRING ANGLE REGIONS

STARTUP SEQUENCE & FIRING REGIONS
DCLC-IMD CONT. ON SH. 8WB
SH. NO. 8WA
GENERAL ELECTRIC
DRIVE SYSTEMS DEPT.
SALEM, VA. U.S.A.
APPROVALS
DES. NO. STANDARD
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REV. 2
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REV. 3
MADE BY
REV. 4
REV. 5



REV. 3	REV. 2	REV. 1	APPROVALS	GENERAL ELECTRIC	COMMUTATION MODES
REV. 4	ISSUED 7/25/86		REG. NO. STANDARD	DRIVE SYSTEMS DEPT. SALEM, VA. U.S.A.	DCLC-IMD CONT. ON SH. FNL. SH. NO. 8WB
REV. 5	MADE BY				



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